

COMPETITIVE PERSONAL MOTOR INSURANCE PRICING WITH SYNTHETIC QUOTE MARKETS: CONVERSION MODELLING AND PRICE OPTIMISATION

ACTUARIAL SCIENCE HONOURS RESEARCH REPORT

of

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Abstract

For an insurer, pricing the risk correctly is only half the problem. Competitive pricing requires insurers to balance expected claims costs with customer demand. However, the competing premiums and unsuccessful quotation outcomes needed to model customer behaviour are often unavailable in conventional insurance portfolio data. This dissertation develops a synthetic personal motor insurance framework to investigate whether competitive market-position information can improve conversion modelling and pricing decisions. A portfolio of 250,000 synthetic policies is generated across five annual cohorts, combining technical pricing, a simulated market with competing insurers, customer-choice modelling and price optimisation. Competitive information substantially improved conversion modelling. On the independent test sample, the selected Market-position GLM achieved a Spearman correlation of 0.843 with underlying conversion probabilities, compared with 0.668 for the Own-price GLM. A targeted pricing strategy selected using this model increased expected conversion from 17.79% to 22.03%, representing a relative improvement of 23.82%. This was achieved while keeping the aggregate quoted rate close to the technical-pricing benchmark and generating a 3.27% uplift in expected pre-expense underwriting profit. Structural sensitivity analysis found positive conversion and profit improvements in all ten examined scenarios, while ten independently regenerated experiments supported the stability of the main findings. As customer behaviour and competitor pricing were simulated, these results should not be interpreted as estimates of outcomes achievable in the Australian motor insurance market. The contribution is a reproducible framework for evaluating demand-aware insurance pricing when competitive quotation and conversion data are unavailable.

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Contents

Abstract i
Acknowledgements i
List of Figures iv
List of Tables v
1 Introduction 1
1.1 Motivation . . . 1
1.2 Research Questions . . . 2
1.3 Significance . . . 2
2 Literature Review 3
2.1 Technical Pricing, Demand and Competition . . . 3
2.1.1 Technical Adequacy and Commercial Pricing . . . 3
2.1.2 Competitive Choice Sets and Quote Conversion . . . 3
2.1.3 Strategic Pricing and Incomplete Market Observability . . . 4
2.2 Conversion Modelling for Demand-Aware Pricing . . . 4
2.2.1 Interpretable Benchmarks and Nonlinear Flexibility . . . 4
2.2.2 Price Endogeneity . . . 5
2.2.3 Prediction, Optimisation and Governance . . . 5
2.3 Synthetic Data Framework . . . 6
2.3.1 Synthetic Insurance Data and Incomplete Observability . . . 6
2.3.2 Internal Validation and Robustness of Pricing Results . . . 7
3 Synthetic Personal Motor Insurance Pricing Methodology 8
3.1 Inputs and Synthetic Portfolio Generation . . . 8
3.1.1 Data Structure . . . 9
3.1.2 Generation of Policy Characteristics . . . 10
3.2 Synthetic Claims and Underlying Expected Claims Cost . . . 11
3.2.1 Claim Frequency . . . 11
3.2.2 Claim Severity . . . 13
3.3 Technical Pricing . . . 14
3.4 Competitive Quote Market and Conversion Generation . . . 16
3.4.1 Quote-Market Construction . . . 16
3.4.2 Conversion Generation Process . . . 18
3.5 Conversion Modelling . . . 20
3.5.1 Model Specification . . . 20
3.5.2 Model Validation and Selection . . . 21
3.6 Pricing Optimisation . . . 22
3.6.1 Candidate Premiums and Predicted Conversion . . . 22
3.6.2 Pricing Objectives and Financial Constraints . . . 23
3.6.3 Uniform and Targeted Pricing Strategies . . . 25
3.6.4 Independent Evaluation . . . 26
3.7 Validation, Sensitivity and Uncertainty . . . 27
3.8 Computational Implementation . . . 29
4 Results 29
4.1 Technical Pricing and Synthetic Market Validation . . . 29
4.1.1 Technical Pricing . . . 29
4.1.2 Competitive Quote Market and Conversion Environment . . . 31
4.2 Conversion Model Performance . . . 32
4.2.1 Validation Fit and Model Selection . . . 32
4.2.2 Test Performance and Recovery of Underlying Conversion Probabilities . . . 33
4.3 Pricing Results . . . 34
4.3.1 Selection of the Targeted Pricing Structure . . . 34
4.3.2 Independent Test Evaluation . . . 35

4.4 Robustness and Uncertainty . . . 36
4.4.1 Structural Sensitivity . . . 36
4.4.2 Monte Carlo Regeneration . . . 37
4.4.3 Paired Bootstrap . . . 38
5 Discussion **38**
5.1 Competitive Information and Model Complexity . . . 39
5.2 From Predictive Performance to Pricing Decisions . . . 39
5.3 Interpretation of Targeted Repricing . . . 40
5.4 Robustness and Boundary Conditions . . . 41
5.5 Practical Implications . . . 41
5.6 Limitations . . . 42
6 Conclusion **43**
6.1 Further Work . . . 44
References **45**
A Document Conventions **48**
A.1 Acronyms and Abbreviations . . . 48
A.2 Summary of Notation . . . 48
B Calibration and Simulation Assumptions **50**
B.1 Empirical Inputs and Portfolio Calibration . . . 50
B.2 Baseline Competitor Assumptions . . . 52
B.3 Baseline Customer-Demand Assumptions . . . 53
B.4 Structural Sensitivity Assumptions . . . 54
C Supplementary Results and Model Specifications **56**
C.1 Complete Comparison of Targeting Variables . . . 56
C.2 Selected Vehicle-Value Segments and Multipliers . . . 56
C.3 Structural Sensitivity Results . . . 57
C.4 Monte Carlo Replication Detail . . . 57
C.5 Paired Test Bootstrap . . . 58
C.6 Model-Specification Register . . . 59
C.7 Interpretation of Selected Technical-Pricing Relativities . . . 60
D Computational Implementation and Reproducibility **61**
D.1 Module Structure . . . 61
D.2 Information Controls . . . 62
D.3 Code, Inputs and Reproducibility Requirements . . . 62

List of Figures

1 Overview of the synthetic motor insurance pricing framework. 8
2 Fitted versus underlying expected annual insurer claims cost for the 2025 cohort. 30
3 Initial premiums and expected purchase shares for the reference insurer and six simulated competitors. 31
4 Underlying expected and realised reference-insurer conversion by competitive quote rank. 32
5 Comparison of six conversion models with different information sets. 34
6 Additional expected purchases and underwriting profit from targeted repricing. 35
7 Structural sensitivity of targeted-pricing outcomes across ten simulated market, demand and pricing scenarios. 37
8 Results across ten independently regenerated synthetic datasets under fixed structural assumptions. . 38
C.1 Distribution of paired-bootstrap conversion uplift. 59
C.2 Fitted frequency and severity risk relativities from the technical-pricing GLMs. 61

List of Tables

1 Principal empirical calibration sources and their role in the synthetic portfolio . . . 9
2 Candidate conversion models and their explanatory information . . . 21
3 Pricing constraints applied to the targeted repricing . . . 24
4 Robustness procedures and sources of uncertainty . . . 27
5 Technical-pricing validation diagnostics on the independent 2025 cohort . . . 29
6 Baseline characteristics of the synthetic 2025 quote market . . . 31
7 Conversion-model validation performance and recovery of underlying conversion probabilities . . . 33
8 Leading admissible targeting structures on the validation sample . . . 34
9 Independent test sample comparison of technical and targeted pricing . . . 35
A.1 Acronyms and abbreviations . . . 48
A.2 Principal claims and technical-pricing notation . . . 48
A.3 Principal quote, demand and optimisation notation . . . 49
B.1 Empirical calibration register . . . 50
B.2 Selected scalar generation and claims assumptions . . . 51
B.3 Competitor stance and quote availability . . . 52
B.4 Competitor segment-appetite rules . . . 52
B.5 Parameters of the baseline demand DGP . . . 53
B.6 Baseline non-price insurer-choice utility adjustments . . . 54
B.7 Alternative customer-demand parameters . . . 54
B.8 Alternative market and pricing assumptions . . . 55
C.1 Validation comparison of 18 predefined eligible targeting bases . . . 56
C.2 Baseline validation-selected targeting rule . . . 56
C.3 Complete ten-case targeted-pricing sensitivity results . . . 57
C.4 Predictive stability across ten replications . . . 57
C.5 Independent-test pricing stability across ten replications . . . 58
C.6 Paired bootstrap estimates and 95% percentile intervals from the 10,000-profile independent test sample 58
C.7 Technical-pricing candidate specifications and selection . . . 59
C.8 Conversion-model information sets and flexibility . . . 60
D.1 Computational modules . . . 61

1 Introduction

1.1 Motivation

In personal motor insurance, setting a premium that adequately reflects expected claims costs does not necessarily ensure that a customer will purchase the policy. Technical pricing estimates the cost of insuring a particular risk and provides the basis for calculating the premium offered to the customer. Commercial pricing, however, must also consider the likelihood of purchase. Higher premiums may increase the underwriting profit earned per policy but discourage customers from purchasing, while lower premiums may attract additional business at the expense of underwriting adequacy. Actuarial pricing therefore requires consideration of both expected claims costs and customer demand (Spedicato et al., 2018; Taylor, 1986).

This relationship is particularly important when customers compare quotations from multiple insurers. Price-comparison websites allow customers to assess competing offers within a single search, making an insurer's competitive position potentially important to the purchasing decision. A premium may be attractive when competing insurers charge more but comparatively expensive when several cheaper alternatives are available. Measures such as price rank and relative price provide information about this competitive position that cannot be determined from the insurer's premium alone. Regulatory reviews have highlighted both the competitive benefits and potential shortcomings of price-comparison websites. The Australian Competition and Consumer Commission (2014) found that comparator websites can help consumers compare offers and encourage competition, but identified concerns about incomplete comparisons, commercial relationships and the transparency of rankings. Similarly, the Financial Conduct Authority (2014) found that an emphasis on quoted premiums could distract consumers from differences in coverage, excesses and policy terms. These findings establish the relevance of the available choice set to insurance purchasing decisions, while also showing that a customer's choice depends on more than price alone.

A principal challenge is that the information required to model competitive customer demand may not be available alongside conventional insurance portfolio data. Insurers generally maintain records of the policies they underwrite, including customer characteristics, premiums, exposure and claims experience. These records support technical pricing but may not contain competitors' quotations or information about customers who requested a quote without purchasing. Consequently, an insurer may be able to estimate expected claims costs without having the information needed to understand how its competitive position influences conversion.

Existing actuarial research recognises the importance of customer price responsiveness and competition in determining commercially effective premiums (Guven and McPhail, 2013; Guelman and Guillén, 2014; Spedicato et al., 2018). However, examining these relationships is difficult when competing quotations and unsuccessful conversion outcomes are unavailable. The research problem addressed by this dissertation is therefore how to construct and evaluate a competitive pricing framework when the information required to estimate customer demand cannot be directly observed.

To investigate this problem, the study develops a synthetic data framework for personal motor insurance, using public statistical data to inform selected policyholder and vehicle characteristics. The framework generates policy and claims records for technical-pricing estimation, followed by competing quotations and customer purchase outcomes for conversion modelling. The insurer whose pricing decisions are evaluated is referred to as the *reference insurer*. Competing premiums and customer outcomes are then generated for the same customer profiles, represented by quotations rather than separate synthetic claims portfolios.

The particular outcome studied here is the reference insurer's conversion, which occurs when the customer purchases from the reference insurer rather than choosing a competitor or not purchasing through the simulated market.

Because purchasing decisions are simulated, the probabilities underlying those decisions are known within the simulation. They are not provided to the fitted conversion models. The models instead learn from the customer characteristics, quoted premiums and simulated purchase outcomes. After a pricing strategy is selected, its performance can be checked against the underlying probabilities from the simulation. This allows the study to assess whether improvements predicted by the fitted conversion models match the specified assumptions under the simulated customer-choice mechanism.

1.2 Research Questions

This dissertation is guided by two research questions.

- Can a synthetic personal motor insurance framework, informed by Australian public data, generate a coherent quotation environment with the information required to approximate a comparison market?
- To what extent can competitive market-position information improve conversion modelling, and can the resulting demand model improve pricing decisions relative to a technical pricing benchmark?

The first question examines whether a synthetic dataset can establish consistent relationships between technical premiums, competing quotations, market position and customer conversion. The objective is to develop a controlled environment for investigating competitive pricing rather than to reproduce the actual pricing behaviour of Australian insurers and their customers.

The second question considers both predictive performance and pricing effectiveness. It examines whether competitive information improves conversion predictions beyond customer characteristics and the reference insurer's own premium, and whether the resulting models can support premium adjustments that increase expected conversion while maintaining specified financial requirements. This distinction is important because improved predictive accuracy does not necessarily lead to better pricing decisions.

1.3 Significance

The significance of this research lies in addressing the information gap between technical risk assessment and competitive pricing decisions. Although expected claims costs provide the actuarial basis for determining premiums, commercial outcomes also depend on whether customers accept the quotations offered. Where competitor prices and unsuccessful quotation outcomes are unavailable, the relationship between underwriting adequacy, competitive position and customer demand cannot readily be examined using conventional portfolio data.

The dissertation makes three methodological contributions. First, it develops a reproducible synthetic framework in which customer and vehicle characteristics, technical premiums, competing quotations and conversion outcomes are represented together. Competitive variables such as ranking and relative price are calculated directly from the generated quotations, ensuring that they reflect the available alternatives.

Second, the study distinguishes the value of additional competitive information from greater model complexity. Conversion models are compared using progressively richer information sets, followed by a comparison between a generalised linear model (GLM) and a generalised additive model (GAM) incorporating the same competitive information. This allows the study to investigate whether predictive improvements arise primarily from additional market information or from more flexible modelling methods.

Third, the framework evaluates conversion models through the pricing decisions they support. Premium adjustments are selected using estimated customer demand, subject to constraints designed to limit changes in the total premiums quoted across customers and preserve expected pre-expense underwriting profit, defined as expected premium revenue less expected claims costs. The selected strategy is then evaluated using the underlying simulated customer-choice mechanism, providing an assessment beyond conventional predictive accuracy.

The practical contribution is a transparent and reproducible framework for investigating demand-aware insurance pricing when complete competitive quotation data are unavailable. Structural sensitivity, Monte Carlo regeneration and bootstrap analysis are used to examine the stability of the findings. The framework therefore provides a basis for developing and testing competitive-pricing methods.

The dissertation proceeds as follows. Section 2 reviews the literature on technical pricing, competition, customer demand and synthetic data. Section 3 describes the simulation, modelling and optimisation methodology. Section 4 presents the results, including predictive comparisons, pricing outcomes and robustness analyses. Section 5 discusses their implications and limitations, while Section 6 concludes the study and identifies opportunities for further research.

2 Literature Review

2.1 Technical Pricing, Demand and Competition

2.1.1 Technical Adequacy and Commercial Pricing

Technical pricing forms the actuarial foundation of general insurance pricing. Its primary purpose is to estimate the expected cost of insuring a particular risk using policyholder risk characteristics. This estimate provides a starting point for determining the premium offered to a customer, typically with additional allowances required for expenses, capital and target returns. However, a premium that adequately covers expected costs does not necessarily represent the most commercially effective price, as the insurer must also consider the likelihood that the customer will purchase the policy.

The distinction between technical and commercial pricing has long been recognised in actuarial literature. Taylor (1986) examines underwriting strategy in a competitive insurance environment, demonstrating the importance of considering market conditions alongside the insurer's own risk and financial objectives. More recent research by Spedicato et al. (2018) incorporates customer-demand modelling into pricing optimisation, recognising that commercial outcomes depend on both the premium charged and the probability that a policy is purchased.

This creates a trade-off between premiums and customer conversion. Increasing premiums may improve the contribution earned on each policy sold but discourage purchases, while lower premiums may attract additional customers at the expense of contribution per policy. Guven and McPhail (2013) examine price elasticity in insurance pricing, which describes how customer demand responds proportionately to changes in premiums. Understanding this relationship is important because changes in price may affect both the volume of business written and the financial contribution generated.

Guelman and Guillén (2014) further demonstrate the importance of distinguishing customer price responsiveness from other factors influencing purchasing behaviour. Their research highlights that customers receiving different premiums may also differ in risk characteristics and purchasing preferences, making the effect of price differences difficult to isolate from observed insurance data.

For the present dissertation, technical pricing establishes the risk-based benchmark against which alternative premiums are evaluated. Conversion modelling then provides the additional information needed to examine how pricing adjustments may affect customer demand.

2.1.2 Competitive Choice Sets and Quote Conversion

Customer conversion refers to the acceptance of an insurance quotation, usually represented as a binary outcome indicating whether the customer purchases the offered policy. In a competitive insurance market, customer conversion may depend on both the quoted premium and the alternative offers available. The collection of offers a customer can choose between is referred to as the *choice set*. An insurer's attractiveness within this set may depend on its premium relative to competing quotations, rather than its price in isolation.

Price-comparison websites provide a relevant example of this environment. They allow customers to compare premiums from multiple insurers within a single search, increasing price transparency and making differences between available offers easier to identify. The Australian Competition and Consumer Commission (2014) examines the influence of comparator websites on competition and consumer outcomes, while the Financial Conduct Authority (2014) investigates their role in the United Kingdom's general insurance market. These investigations establish the relevance of comparison platforms to insurance distribution, although they do not directly quantify the effect of individual competitive measures on conversion.

The importance of competitive position can be illustrated by considering an insurer offering a premium of $1,000. This may be the cheapest available quotation for one customer but substantially more expensive than the alternatives available to another. Although the insurer's premium is unchanged, its competitive position differs between the two customers. Measures such as price rank, which indicates the position of an offer when quotations are ordered from cheapest to most expensive, and relative price, which compares the offered premium with a competing price, can capture this difference.

These considerations motivate the inclusion of competitive market-position variables in conversion models. By generating alternative insurer quotations, the present study can examine whether information about the reference insurer's

position within each customer's choice set improves conversion prediction beyond customer characteristics and the reference insurer's own premium.

2.1.3 Strategic Pricing and Incomplete Market Observability

Actuarial research has examined how insurer pricing decisions interact with customer demand and competitive conditions. Taylor (1986) establishes an early framework for underwriting strategy under competition, while Emms et al. (2007) investigate optimal premiums when business volume responds to an insurer's price relative to the market. Pantelous and Passalidou (2013) similarly consider premium optimisation within a competitive insurance environment.

Further research incorporates strategic interactions between insurers. Game-theoretic models consider how one insurer's pricing decisions may influence, and be influenced by, the actions of competitors. Boonen et al. (2018) investigate these interactions in general insurance markets, while Mourdoukoutas et al. (2021) examine pricing in a competitive stochastic insurance setting. These approaches recognise that competing insurers may respond to changes in market conditions rather than maintaining fixed prices.

Modelling strategic competition requires assumptions about competing insurers' pricing objectives and how they respond to changes in one another's premiums. This information may be difficult to obtain because insurers cannot generally observe their competitors' internal pricing models. This dissertation therefore holds competing quotations fixed while the reference insurer evaluates alternative premiums. The approach isolates the effect of changing the reference insurer's competitive position, but cannot assess how competitors might respond to its pricing decisions.

Incomplete market information creates a related practical difficulty. Conventional insurance portfolio data may contain sufficient information to estimate expected claims costs but lack competitor quotations and outcomes for customers who did not purchase. The information required for technical pricing may therefore be available even when the information needed to model competitive demand is not. Constructing a synthetic quote market allows these components to be examined together under explicitly defined assumptions.

2.2 Conversion Modelling for Demand-Aware Pricing

2.2.1 Interpretable Benchmarks and Nonlinear Flexibility

Conversion modelling estimates the probability that a customer purchases an insurance policy using their characteristics, the quoted premium and other available information. Logistic regression provides a suitable starting point because conversion is a binary outcome and the model produces estimated purchase probabilities. As a generalised linear model (GLM), logistic regression also offers a relatively interpretable way of examining relationships between explanatory variables and customer purchasing behaviour.

However, customer responses to premiums and competitive position may be nonlinear. For example, moving from the second-cheapest quotation to the cheapest may have a different effect on conversion than moving between two relatively expensive quotations. A conventional logistic GLM assumes that its predictors have linear effects on the odds of conversion unless additional terms are introduced. Nonlinear relationships can be represented using transformations or splines, while generalised additive models (GAMs) allow selected predictors to have flexible, smooth effects estimated from the data.

Existing insurance research demonstrates the potential value of flexible modelling methods. Spedicato et al. (2018) compare conventional GLMs with machine-learning approaches in insurance price optimisation, illustrating how modelling choices can influence the resulting pricing decisions. Henckaerts et al. (2021) examine flexible methods for capturing complex relationships in insurance tariff modelling, while Yeo et al. (2001) investigate nonlinear customer responses to premium changes in motor insurance retention. Although retention differs from new-business conversion, these studies support examining whether flexible model structures can represent customer responses more effectively.

These studies demonstrate the potential value of flexible modelling methods, but greater complexity does not necessarily improve predictions or pricing decisions. More informative predictors may offer greater benefits than additional flexibility applied to a limited information set. To examine this distinction, the dissertation first compares logistic GLMs using progressively richer information, from customer and technical-pricing characteristics to own-price and competitive market-position variables. It then compares the Market-position GLM with a GAM using the same broad competitive information. This separates the benefit of additional competitive information from the incremental effect of nonlinear flexibility.

2.2.2 Price Endogeneity

Price endogeneity arises when observed premiums are associated with other factors that also influence customer purchasing behaviour. This is a confounding problem: insurers set prices partly on customer risk characteristics, some of which may also be associated with purchasing preferences. Consequently, a comparison of different customers' observed premiums does not necessarily estimate how the same customer would respond to a different premium (Guelman and Guillén, 2014).

Consider two hypothetical customers. A 22-year-old driving a sports car receives a premium of $1,800 and has a conversion probability of 15%, while a 45-year-old driving a family sedan receives a premium of $900 and has a conversion probability of 30%. Although the customer offered the higher premium is less likely to purchase, the difference cannot be attributed to price alone. Age, vehicle type, income, purchasing preferences and other characteristics may also influence conversion.

The relevant question for pricing is how the same customer's probability of purchasing would change if their premium changed while other conditions remained fixed. For example, reducing the first customer's premium from $1,800 to $1,600 might increase their conversion probability from 15% to 22%. These probabilities are illustrative, but the comparison demonstrates the customer response that a pricing model aims to estimate.

In observed insurance data, only the purchasing outcome associated with the premium actually offered is generally available. The same customer's decision at an alternative premium cannot be observed simultaneously. This makes it difficult to distinguish the effect of changing price from differences between customers. Guelman and Guillén (2014) address this identification problem when estimating price elasticity in automobile insurance using causal-inference methods. Similar difficulties arise in wider demand research, where observed prices and purchasing decisions may both depend on underlying consumer preferences and market conditions (Berry et al., 1995).

The synthetic framework provides a controlled environment for examining how changes in premiums affect customer purchasing decisions. Synthetic policy and claims records are generated to estimate expected claims costs and establish a risk-based pricing reference. The same customer characteristics are then used to construct a competitive quotation environment, in which customers receive an offer from the reference insurer alongside quotations from other simulated competitors.

Rather than generating separate policy and claims portfolios for each competitor, their quotations are calculated using the common risk-based reference, with adjustments representing differences in general pricing approach, customer-group preferences and quotation-level variation. This allows customers to face different competing offers without requiring separate technical-pricing models for every insurer.

Simulated purchasing outcomes are then used to fit conversion models, which estimate how the probability of purchasing from the reference insurer varies with its premium and competitive position. These predictions inform the selection of premium adjustment.

The approach is deliberately narrower than modelling the complete insurance market. It allows the effect of changes to an insurer's premium to be examined under controlled assumptions, but does not represent differences arising from competitors' independently estimated claims risks or their strategic responses to repricing.

2.2.3 Prediction, Optimisation and Governance

Accurate conversion predictions do not necessarily lead to effective pricing decisions. A model may identify customers who are more likely to purchase insurance without accurately predicting how their purchasing probabilities change when premiums are adjusted. Although measures such as log loss assess the accuracy of predicted probabilities (Gneiting and Raftery, 2007), they do not directly measure the financial consequences of decisions made using those predictions. This distinction is recognised where improvements in predictive accuracy do not necessarily produce better optimisation outcomes (Elmachtoub and Grigas, 2022).

Mayorga and Torres (2017) demonstrate how customer price responsiveness can inform motor insurance premium selection. They estimated customers' probabilities of renewing their policies using a generalised additive model, incorporating renewal premiums and other customer characteristics. Alternative premium increases were then evaluated using the fitted renewal probabilities. Their optimisation sought to maximise expected financial margin while maintaining customer retention above a specified minimum and restricting the size of premium increases. This allowed different premium adjustments to be selected for customer groups according to their estimated price sensitivity and financial value.

A related application by Spedicato et al. (2018) examines new-business conversion. Their framework combines estimated claims costs with conversion probabilities predicted at alternative quoted premiums. For each candidate price, the expected underwriting result depends on both the contribution earned if the customer purchases and the probability of that purchase occurring. These studies illustrate how customer-demand models can be used alongside technical-pricing models to assess alternative premiums, rather than relying on estimated claims costs alone.

The present dissertation seeks to maximise the expected number of new-business conversions while satisfying specified financial safeguards. These include retaining at least the benchmark expected pre-expense underwriting profit and maintaining the portfolio expected claims ratio below its assumed ceiling. Supplementary uniform-pricing strategies instead maximise expected pre-expense underwriting profit, subject to conversion and financial requirements.

Expected pre-expense underwriting profit is defined as the premium expected to be collected from policies sold, less their fitted expected claims costs, with both amounts weighted by customers' purchasing probabilities. This simplified measure excludes acquisition, operating, reinsurance and capital costs and therefore does not represent net insurer profit. **Expected profit uplift** refers to the improvement in this amount relative to the technical-pricing benchmark, rather than the amount itself. These measures and the corresponding optimisation constraints are defined formally in Section 3.6.

Pricing decisions must also consider governance and customer outcomes. Models that assign different premiums across customer segments may produce results that are difficult to explain or justify, particularly where differences reflect sensitive characteristics or customer behaviour. The Australian Human Rights Commission and Actuaries Institute (2022) highlight the importance of avoiding unjustified discrimination in insurance pricing and underwriting, while the Australian Securities and Investments Commission (2024) examines governance challenges associated with artificial intelligence.

These considerations support using interpretable modelling methods, transparent assumptions and explicit financial constraints. The present study evaluates whether competitive information can improve pricing decisions within the simulated environment, but does not establish whether the resulting premiums would be appropriate for implementation. This would require additional assessment of expenses, regulation, fairness and customer outcomes.

2.3 Synthetic Data Framework

2.3.1 Synthetic Insurance Data and Incomplete Observability

Synthetic data consist of artificially generated observations designed to reproduce specified characteristics or statistical relationships. They are increasingly used where access to detailed information is restricted by privacy, confidentiality or commercial sensitivity. El Emam et al. (2020) discuss the generation of synthetic data for analytical applications, while Nowok et al. (2016) demonstrate methods for creating datasets that preserve selected statistical properties of observed information.

In actuarial research, synthetic data can support the development and evaluation of modelling methods without requiring access to confidential policyholder or claims records. Avanzi et al. (2021), for example, developed SynthETIC, an individual-claims simulator that generates insurance claims experience under controlled assumptions. Such approaches allow researchers to examine model performance under known conditions and systematically modify the relationships governing simulated outcomes.

The motivation for using synthetic data in this dissertation extends beyond confidentiality. Conventional insurer portfolio records contain information about policies written, including policyholder characteristics, premiums and claims experience. These records support technical pricing but may not contain competing quotations or purchasing outcomes for customers who received a quotation without buying a policy. Without this information, the relationship between an insurer's competitive position and customer conversion cannot be readily examined.

The study therefore constructs the required datasets in sequence. First, publicly available Australian data inform the selection and distributions of personal motor insurance characteristics, including driver age, geographic location and vehicle attributes. Synthetic policy and claims records are then generated to support the estimation of expected claims costs. Next, the fitted technical premiums provide a pricing reference for generating quotations from the reference insurer and other simulated competitors. Customer purchasing decisions are subsequently generated using a specified choice model, producing the outcomes required for conversion modelling.

Although selected policy characteristics are informed by Australian data, the competing premiums and customer purchasing relationships are generated using assumptions rather than observed market behaviour. The resulting

framework therefore provides a controlled environment for examining competitive pricing, but does not establish that the simulated conversion rates or price responses are representative of Australian motor insurance customers.

An important advantage of this approach is that the underlying customer-choice probabilities are known because they are specified by the simulation. These probabilities are withheld when the conversion models are fitted and premium adjustments are selected. Once a strategy has been chosen, its performance can be evaluated using the underlying probabilities rather than relying solely on the fitted model's predictions. This allows the study to assess whether its modelling and pricing decisions recover the relationships built into the simulated environment.

2.3.2 Internal Validation and Robustness of Pricing Results

The usefulness of synthetic data depends on the purpose for which they are generated. When synthetic data are intended to represent an observed population, their quality can be assessed by comparing characteristics such as distributions and relationships between variables. However, this dissertation uses synthetic data primarily to investigate conversion modelling and pricing decisions under specified market conditions. The relevant question is therefore whether the conclusions remain reliable when the assumptions or simulated observations change. This distinction is consistent with the need to evaluate synthetic data according to their intended application and limitations (Jordon et al., 2024; Jones et al., 2025).

Three complementary methods are used to examine the stability of the results. First, structural sensitivity analysis evaluates alternative assumptions about competing quotations, customer price sensitivity and pricing restrictions. By changing these assumptions and repeating the modelling and pricing procedures, the study examines whether the improvements observed under the baseline scenario depend on particular market conditions. Sensitivity analysis is an established approach for assessing how model conclusions depend on uncertain inputs (Saltelli et al., 2008).

Second, Monte Carlo regeneration repeats the synthetic experiment using different random seeds. Each replication generates new observations, refits the relevant models and selects a new pricing strategy. Comparing results across these independently generated datasets helps determine whether the findings are consistent or depend heavily on the particular random sample used in the original experiment. This follows the general principle of Monte Carlo simulation, which uses repeated random sampling to examine variation in model outcomes (Robert and Casella, 2004).

Finally, paired bootstrap analysis examines uncertainty associated with the composition of the independent test sample. The same customer observations are resampled for both the technical-pricing benchmark and the selected targeted strategy, preserving the direct comparison between them. The pricing strategy remains fixed throughout this procedure. Repeated resampling is then used to estimate the uncertainty surrounding the improvement in expected conversion and expected pre-expense underwriting profit (Efron and Tibshirani, 1993).

Together, these procedures examine three different sources of uncertainty: the assumptions used to construct the synthetic market, random variation between independently generated datasets, and the customers included in the final test sample.

3 Synthetic Personal Motor Insurance Pricing Methodology

The methodology begins with synthetic annual policy and claims records for one reference insurer. These records provide the data needed to estimate expected claims costs. Using only the 2025 customer profiles, a second stage generates the reference insurer's quotation, competing quotations and simulated purchase outcomes. These conversion observations are used to compare fitted demand models and select premium adjustments. The final pricing strategies are evaluated on an independent sample using the assumed customer-choice mechanism. Figure 1 shows how the datasets, fitted models and evaluation stages are connected.

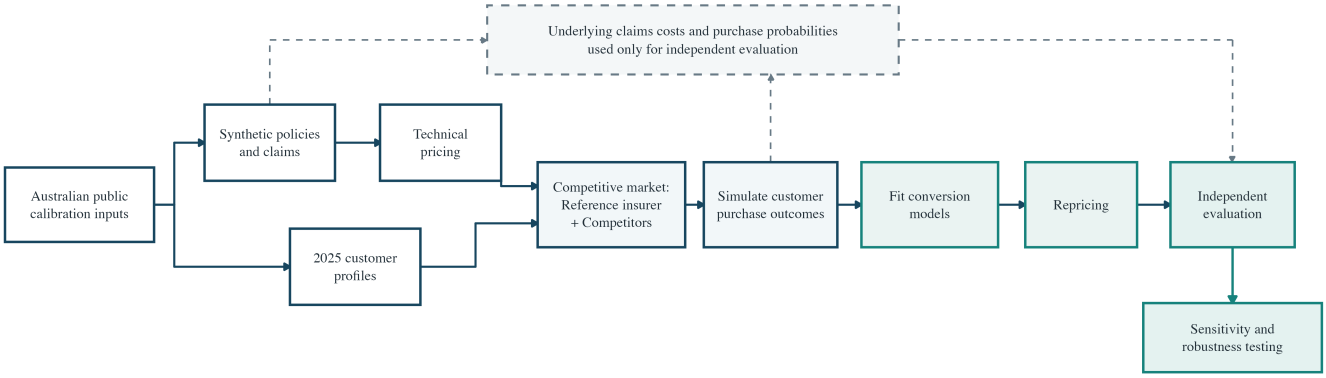


Figure 1: Overview of the synthetic motor insurance pricing framework. Synthetic policy and claims records are generated using Australian-calibrated inputs and used to estimate technical premiums. These premiums provide the basis for competing quotations and simulated purchasing decisions. Conversion models are then fitted to inform constrained premium adjustments, which are evaluated using the underlying simulated customer-choice mechanism. The dashed information channel identifies quantities withheld during model fitting and pricing selection. Structural sensitivity, Monte Carlo regeneration and paired bootstrap analysis assess the robustness of the findings.

3.1 Inputs and Synthetic Portfolio Generation

The first stage generates synthetic policy and claims data for a reference insurer offering personal comprehensive motor insurance. Publicly available Australian statistics are used to inform the distributions of selected driver, geographic and vehicle characteristics. For example, vehicle registration records inform the types and ages of vehicles represented in the portfolio, while licence-holder statistics inform the distribution of driver ages.

This use of published information is referred to as *empirical calibration*. It involves using observed statistics to determine selected inputs to the simulation, rather than estimating every characteristic from an actual insurer's policy records. Where suitable information is unavailable, additional assumptions are introduced and documented separately. The resulting portfolio is therefore informed by Australian data but is not intended to reproduce an observed insurer's portfolio.

The synthetic data are generated in two main stages. The first constructs policy and claims records for the reference insurer. Module 00 prepares the empirical calibration inputs, including vehicle registration distributions, driver characteristics and geographic classifications. Module 01 uses these inputs to generate individual policy records containing customer, vehicle and underwriting characteristics, together with simulated claims experience. These records provide the data used to estimate the reference insurer's technical premiums in Section 3.3.

The second stage constructs the competitive quotation and conversion data using the 2025 customer profiles. Following technical-pricing estimation, Module 03 generates quotations for the reference insurer and six simulated competitors. Module 04 then generates customer purchasing outcomes using a specified behavioural model. These additional records allow the study to examine how the reference insurer's premium and competitive position relate to the probability of purchase.

Table 1 summarises the principal sources informing the first stage of data generation. The corresponding calibration assumptions and their implementation are documented in Appendix B.

Table 1: Principal empirical calibration sources and their role in the synthetic portfolio

Component	Principal source	Role in simulation
Vehicle registrations and fleet composition	Australian Bureau of Statistics (2022) and Bureau of Infrastructure and Transport Research Economics (2025b)	State, manufacture-year and make/model distributions
Driver population	Bureau of Infrastructure and Transport Research Economics (2025a)	Driver-age distribution
Geography and remoteness	Bureau of Infrastructure and Transport Research Economics (2025b) and Australian Bureau of Statistics (2021a)	State and remoteness distributions
Vehicle segment and motive power	Australian Government (2026) and Federal Chamber of Automotive Industries (2024)	Vehicle classification and technology mix
Vehicle specifications	CarExpert (2026)	Engine, power, torque, transmission and weight
Vehicle values	Royal Automobile Club of Queensland (2025)	Reference prices and depreciation
Vehicle usage	Australian Bureau of Statistics (2020) and Australian Bureau of Statistics (2021b)	Annual-distance benchmark and motive-power usage relativities
Historical monetary values	Australian Bureau of Statistics (2026)	Adjustment of vehicle values and claims costs to a common price basis
Additional underwriting characteristics	Simulation assumptions	Excess, parking, financing and other policy attributes

3.1.1 Data Structure

The reference insurer portfolio contains 250,000 synthetic personal comprehensive motor insurance records across five annual cohorts from 2021 to 2025, with 50,000 records generated for each year. Each record represents an independent policy risk with one year of exposure, rather than a policyholder followed across multiple years. The multi-year structure provides historical claims experience for technical-pricing development, while reserving the final cohort for independent evaluation.

The sample size is a modelling choice intended to increase the number of purchasing outcomes and the precision of the planned estimation and evaluation procedures, while keeping repeated simulation and model fitting feasible. It is not intended to reproduce the size of an Australian insurer's portfolio. The principal model selection results are also compared with a 50,000-policy experimental run; this comparison evaluates stability across two simulation sizes rather than establishing a statistically optimal sample size. Equal numbers of records are generated for each year to maintain a consistent experimental structure, rather than to represent observed annual business volumes.

Renewals, cancellations and mid-term policy changes are excluded because the study focuses on new-business quotation decisions over a single annual coverage period. Including these additional processes would require separate assumptions about policy duration, renewal behaviour and exposure adjustments, which are outside the scope of the present experiment.

The cohorts serve different purposes in technical-pricing development. Claims experience from 2021–2023 is used to fit candidate frequency and severity models. The 2024 cohort is used to compare their predictive performance and select the preferred specifications. Following selection, the models are subsequently refitted using the combined 2021–2024 experience and evaluated on the independent 2025 cohort.

The preceding cohorts therefore provide claims experience for estimating risk-based premiums, whereas the 2025 cohort provides a controlled cross-sectional market in which alternative quotations and purchasing probabilities can be compared.

Historical conversion rates would require additional assumptions about the evolution of competitor prices, distribution arrangements and customer preferences over time. Those assumptions are not needed to test the narrower question of whether access to market-position variables improves prediction and supports repricing within a fixed quote market. The 2025 conversion dataset still has separate training, validation and independent test samples; it is not fitted and evaluated on the same observations.

A longitudinal simulation of conversion histories, followed by projection to a later cohort, would be a valuable extension. It could test the model's predictive performance when applied to data from a different time period and the value of previous conversion experience, but it addresses a different question from the controlled cross-sectional comparison undertaken here. Accordingly, the results should not be presented as evidence that historical conversion information is unimportant in practice.

3.1.2 Generation of Policy Characteristics

The portfolio includes geographic, driver, vehicle and underwriting characteristics. These variables are generated in a defined sequence to preserve relevant dependencies. For example, licence experience depends on driver age, vehicle age depends on the manufacture year and policy year, and physical vehicle specifications are assigned using compatible reference records.

Geographic and driver characteristics: State is assigned using annual passenger-vehicle registration distributions published by Bureau of Infrastructure and Transport Research Economics (2025b) and Australian Bureau of Statistics (2021a). Remoteness is subsequently generated conditional on State using BITRE vehicle-location information and ABS geographic classifications. The five remoteness categories are Major Cities, Inner Regional, Outer Regional, Remote and Very Remote. Where postal areas overlap multiple remoteness categories, Census dwelling counts are used to allocate registrations between them (Australian Bureau of Statistics, 2022).

Policies are assigned to approximately 450 synthetic localities within the State and remoteness categories. Policies sharing a locality receive a common adjustment to expected claim frequency, representing geographical risk differences not captured by their observable characteristics. This approach is motivated by Gschlößl and Czado (2007), who demonstrate the importance of incorporating spatial effects when modelling motor insurance claims. The synthetic localities and the magnitude of their risk adjustments are specified simulation assumptions rather than estimates derived from observed Australian postcode claims experience.

Driver ages are generated using national licence-holder distributions published in the BITRE *Australian Infrastructure and Transport Statistics Yearbook 2025* (Bureau of Infrastructure and Transport Research Economics, 2025a). Ages are sampled within the corresponding age bands, and licence duration is derived from the simulated driver age and an assumed age at first licensing.

Vehicle characteristics:

Vehicle manufacture year is sampled from the relevant policy-year and State registration distribution. Vehicle age is then calculated as the difference between the policy year and manufacture year, with the simulation restricted to vehicles no more than 30 years old. Vehicle make, model, segment and motive power are assigned using the corresponding registration distributions and reference classifications.

Vehicle classifications are informed by the Australian Government's Green Vehicle Guide and FCAI segmentation criteria (Australian Government, 2026; Federal Chamber of Automotive Industries, 2024). Additional vehicle specifications, including engine capacity, cylinder count, power, torque, transmission and unladen weight, are assigned using CarExpert vehicle records (CarExpert, 2026).

Vehicle values: Insured vehicle values are estimated using RACQ's 2025 vehicle-running-cost data, which provides a reference for new-vehicle prices and estimated values after five years (Royal Automobile Club of Queensland, 2025).

The RACQ reference data provide indicative new vehicle prices and five-year residual value for selected models. To assign values at other ages, the simulation extrapolates each residual value exponentially from the five-year point. A numerical floor of \$3,000 is then imposed to prevent the simulated value of an older vehicle from approaching zero, while a ceiling prevents depreciation from producing a value above the reference new-vehicle price. The chosen floor and the extrapolation outside the observed five-year point are pragmatic simulation assumptions, not an insurer's empirically estimated depreciation function.

Let B_i denote the price assigned to a new vehicle in 2025, r_i the proportion of that price remaining after five years and a_i the vehicle's age. The estimated vehicle value on the 2025 price basis is

$$V_i = \min\{B_i, \max(3000, B_i r_i^{a_i/5})\}.$$

Historical monetary values are adjusted using ABS price indices. The motor-vehicle price index is used to express vehicle values and write-off amounts at their corresponding policy-year price levels, while the motor-vehicle maintenance and repair index is used for partial claims. Claims costs are subsequently converted to a common 2025 price basis for technical-pricing estimation. This prevents historical changes in vehicle and repair prices from being treated as differences in underlying claims risk (Australian Bureau of Statistics, 2026).

Annual usage and additional underwriting characteristics: Annual kilometres are generated using a lognormal distribution to ensure positive distances while allowing some vehicles to travel substantially further than others. The lognormal form and its coefficient of variation of 0.45 are specified simulation assumptions, rather than distributions estimated from Australian population-level mileage records. The evidence of Charlton et al. (2019) is used below to motivate age-related variation in driving patterns, not the lognormal distribution itself.

The simulated annual distances are subsequently scaled to an average of approximately 11,100 kilometres per passenger vehicle, consistent with the Australian Bureau of Statistics' Survey of Motor Vehicle Use for 2019–20 (Australian Bureau of Statistics, 2020). The use of this historical average as a common calibration target across the 2021–2025 cohorts is a simplifying assumption.

Additional adjustments are applied for driver age, remoteness and motive power to introduce differences in expected vehicle usage. Age-related variation is motivated by research documenting changes in driving distance across older age groups (Charlton et al., 2019). Geographic adjustments reflect differences in transport accessibility and reliance on private vehicles in regional and remote Australia (Australian Institute of Health and Welfare, 2005). Motive-power adjustments are informed by the ABS experimental fuel-specific usage estimates (Australian Bureau of Statistics, 2021b).

The precise age and remoteness multipliers are specified simulation assumptions. For example, drivers aged 24 or younger receive an age multiplier of 1.10, compared with 1.00 for drivers aged 25–59. A 19-year-old and a 40-year-old living in the Remote category both receive a remoteness multiplier of 1.24. Holding other characteristics and the initial simulated mileage constant, the younger driver's unscaled annual distance is therefore 10% higher. The resulting distances are recalibrated to the cohort-level mileage target.

Additional underwriting characteristics include voluntary excess, overnight parking, vehicle financing, modifications, pre-existing damage and regular-driver status. Voluntary excess is included because it determines the policyholder's contribution towards an insured claim and consequently affects the insurer's expected claim payments. Australian insurers offer different excess levels, and publicly available insurer and comparison-site information illustrates how the selected excess can affect the premium (Allianz Australia Insurance Limited, 2015; Compare the Market, 2026).

The simulation assigns excess amounts of $400, $600, $800, $1,000 and $1,500. These values span a range illustrated in published Australian insurance information, but the probabilities assigned to each amount are modelling assumptions. The simulated excess is deducted from gross claim amounts when determining insurer payments.

The completed portfolio provides the characteristics used to simulate claims and estimate technical premiums.

3.2 Synthetic Claims and Underlying Expected Claims Cost

3.2.1 Claim Frequency

Reported claim counts are generated using a Poisson distribution, a conventional starting point for modelling insurance claim frequency over a fixed exposure period (Frees, 2010; Ohlsson and Johansson, 2010). The distribution represents the number of claims occurring during the policy year, with each policy's expected frequency determined by its risk characteristics.

For policy i, the reported claim count is generated as

$$N_i \mid \lambda_i \sim \text{Poisson}(\lambda_i),$$

where N_i is the number of reported claims and λ_i is the underlying expected annual claim frequency.

Expected frequency is constructed using multiplicative risk relativities, consistent with the risk-classification approach used in Dionne and Vanasse (1989) and Denuit et al. (2007). This structure allows each characteristic to increase or decrease expected claim frequency relative to a reference level. Driver age, licence experience and annual kilometres represent observable risk characteristics, while geographic and policy-specific random effects capture additional differences between otherwise similar policies.

The precise relationships and coefficient values are specified for the simulation. Their purpose is to generate interpretable differences in underlying risk that the fitted technical-pricing models must subsequently estimate.

Expected frequency is therefore specified as

$$\lambda_i = c \times R_i^A \times R_i^L \times R_i^K \times G_{\ell(i)} \times U_i,$$

where c is a common frequency scaling factor, R_i^A, R_i^L and R_i^K represent the adjustments for driver age, licence experience and annual kilometres, respectively. The remaining terms introduce geographic and individual variation in claims risk.

The observable risk adjustments are specified as

$$\begin{aligned} R_i^A &= \min\left[2.20, \exp\{0.00055(A_i - 50)^2\}\right], \\ R_i^L &= \exp[-0.020 \min(L_i, 15)], \\ R_i^K &= \left(\frac{K_i}{11{,}100}\right)^{0.55}, \end{aligned}$$

where A_i is driver age, L_i is years licensed and K_i is annual kilometres.

The selected relativities introduce three different relationships with expected claims frequency. The age function assumes a U-shaped pattern, with the lowest age-related frequency around 50 years and higher frequency among younger and older drivers. This is motivated by evidence that crash involvement varies by driver age (Ryan et al., 1998). The precise curvature and maximum relativity of 2.20 are simulation assumptions.

The licence-experience function assumes that expected frequency decreases as driving experience increases, with the reduction limited to the first 15 years. The kilometre function assumes that greater annual driving distance increases claim frequency, reflecting greater exposure to road use. This relationship is motivated by Ferreira and Minikel (2012), who demonstrate that annual mileage is informative for automobile insurance risk. The exponent of 0.55 is assumed, allowing claim frequency to increase less than proportionately with mileage. The licence-experience coefficient and the corresponding caps are also specified assumptions.

Two additional random effects introduce differences in claim frequency that are not explained by the observable risk characteristics. The geographic effect $G_{\ell(i)}$ is shared by policies within the same synthetic locality, representing common conditions that may affect claims risk, such as local road environments or exposure to hazards. The policy-specific effect represents residual differences between otherwise similar drivers. The use of such effects is motivated by mixed-Poisson approaches to actuarial risk classification, which allow unobserved heterogeneity to affect claim frequency (Dionne and Vanasse, 1989; Denuit et al., 2007).

The overall claims frequency scale c is calibrated to an annual claimant rate of 14%, informed by the Insurance Council of Australia (2025), which reports that approximately one in seven motor insurance policyholders makes a claim each year. This target concerns the proportion of policies with at least one reported claim, rather than the average number of claims per policy.

$$\frac{1}{n_{2025}} \sum_{i:\,\text{year}_i=2025} [1 - \exp(-\lambda_i)] = 0.14.$$

Under the Poisson distribution, the probability of at least one claim is one minus the probability of zero claims. The common frequency scaling factor is therefore selected so that the average of these probabilities across the 2025 cohort equals 14%.

The same underlying frequency is used to simulate current-year claims and five preceding years of claims history. These counts are generated independently conditional on the policy's expected frequency, allowing prior claim counts

and consecutive claim-free years to reflect underlying risk differences while retaining random variation in reported experience.

3.2.2 Claim Severity

A separate gross claim amount is generated for each reported claim. Two types of claims are distinguished: **write-off claims**, where a vehicle is written off, and **partial claims**, where the vehicle is repaired. This distinction is motivated by the different financial consequences of vehicle replacement and repair. The Insurance Council of Australia (2025) reports that write-offs account for approximately 25% of motor insurance claims expenditure.

Each reported claim is assigned to one of these categories using a Bernoulli random variable.

A common write-off probability is applied across policies as a simplifying assumption. The probability is calibrated so write-off claims account for the intended proportion of simulated claims expenditure, not the proportion of claims classified as write-offs.

Let T_{ij} indicate whether claim j for policy i is a write-off:

$$T_{ij} \sim \text{Bernoulli}(p_{WO}),$$

where p_{WO} is the probability that a reported claim results in a write-off.

For write-off claims, the gross claim amount is set equal to the simulated vehicle value. Partial-claim amounts are instead generated using a lognormal distribution (Klugman et al., 2012; Omari et al., 2018). This ensures positive claim amounts and allows a right-skewed distribution, with many relatively small claims and fewer large claims.

$$X_{ij} \mid T_{ij} = 0 \sim \text{Lognormal}(\eta_i, \sigma_X^2),$$

where η_i and σ_X determine the location and variability of partial claim amounts on the logarithmic scale. The coefficient of variation for partial claims is fixed at 1.00, providing substantial variation in simulated claim amounts. This value is a modelling assumption. The distribution's location parameter is adjusted to achieve the specified expected partial claim amount for each policy.

Expected partial-claim severity is also adjusted according to vehicle value:

$$E(X_{ij} \mid T_{ij} = 0) \propto \left(\frac{V_i}{V_0}\right)^{0.35},$$

where V_i is the simulated vehicle value on the 2025 price basis and V_0 is the reference vehicle value.

This reflects the assumption that damage to more valuable vehicles may involve higher repair costs, although repair costs need not increase in direct proportion to vehicle value. A power function is used because it provides a positive and interpretable severity relativity while allowing a less-than-proportional increase.

The exponent of 0.35 is specified for the simulation. For example, doubling vehicle value increases expected partial-claim severity by approximately 27%, holding all other components constant. The relationship therefore introduces a moderate vehicle-value effect without assuming that twice the vehicle value produces twice the repair cost.

The insurer payment is determined after deducting the policyholder's voluntary excess:

$$Y_{ij} = \max(X_{ij} - e_i, 0),$$

where X_{ij} represents the gross claim amount and e_i is the excess.

Under an ordinary per-claim deductible, the insurer pays the amount of the gross claim exceeding the excess, with a minimum payment of zero (Klugman et al., 2012). Consequently, a reported claim can produce no insurer payment if its gross amount does not exceed the excess.

This distinction is retained throughout the study. Reported claims are used to describe the underlying claims-generation process, whereas the technical-pricing models estimate frequency and the average payment of claims where there is an insurer payment, referred to as **paid-claims**.

The severity simulation is calibrated using two industry-level reference values reported by the Insurance Council of Australia (2025): an average motor claim size of approximately \$5,202 and total losses accounting for approximately 25% of claims expenditure. These values provide calibration targets rather than a complete empirical distribution of individual claim amounts.

The write-off probability and partial-claim severity scale are determined jointly so that the simulated payments approximate these targets after allowing for voluntary excesses. The average claim amount is calculated across all reported claims, including those resulting in no insurer payment. The procedure therefore reproduces selected aggregate characteristics of the published claims information without claiming to reproduce the full distribution of Australian motor insurance claims.

Historical claim amounts are generated at the monetary values applicable to their policy year and then converted to the common 2025 price basis used for technical pricing.

The underlying expected annual claims cost for each policy is therefore

$$\mu_i = \lambda_i s_i,$$

where λ_i is the expected annual frequency of reported claims and s_i is the expected insurer payment per reported claim, including the effect of voluntary excesses and claims resulting in zero payment.

Unlike realised claims experience, the resulting expected claims cost is calculated directly from the specified simulation relationships. It is retained as an underlying reference for evaluating the fitted technical-pricing models.

3.3 Technical Pricing

The reference insurer's technical-pricing model is developed from the synthetic policy and claims experience. A frequency–severity approach is used to estimate annual insurer claims costs, with separate generalised linear models (GLMs) for claim frequency and average claim severity (Ohlsson and Johansson, 2010; Frees, 2010).

Although the underlying simulation generates all reported claims, technical pricing is based on claims resulting in a payment by the insurer after applying the policyholder's excess. Claims fully absorbed by the excess are excluded from the paid-claim count because they produce no insurer expenditure.

Frequency therefore measures the expected number of paid claims per policy year, while severity measures the expected insurer payment conditional on a payment. This definition ensures that multiplying the two fitted components provides an estimate of the insurer's expected annual claims expenditure, rather than gross losses before deductibles.

Claim frequency: The number of paid claims is modelled using a quasi-Poisson GLM with a logarithmic link. This specification is suitable for non-negative claim counts, while the logarithmic link ensures positive fitted frequencies and allows risk characteristics to have multiplicative effects (Ohlsson and Johansson, 2010).

The quasi-Poisson variance specification also accommodates differences between the observed variability of claim counts and that implied by a standard Poisson model. This flexibility is relevant because the synthetic claims process contains geographic and policy-specific risk effects that are not fully observed by the fitted model (Ver Hoef and Boveng, 2007).

Let C_i denote the number of paid-claims for policy i,

$$\log \mathbb{E}(C_i \mid X_i) = \beta_0 + X_i^\top \beta + \log E_i,$$

where X_i contains its observed rating characteristics, β represents the estimated regression coefficients and E_i its policy exposure.

The fitted model is specified using a log link and an exposure offset. All simulated policies have one year of exposure, so E_i equals one and the offset is zero.

The dispersion parameter measures variation relative to the Poisson mean–variance relationship.

$$\mathrm{Var}(C_i \mid X_i) = \phi \mathbb{E}(C_i \mid X_i),$$

where ϕ is the dispersion parameter. A dispersion estimate close to one indicates limited evidence of additional dispersion, while larger values indicate greater variability.

Claim severity: Average paid-claim severity is estimated using a Gamma GLM with a logarithmic link. The Gamma specification accommodates strictly positive, right-skewed payment amounts, while allowing variance to increase with expected claim severity (de Jong and Heller, 2008; Ohlsson and Johansson, 2010).

$$\log E(S_i \mid X_i) = \gamma_0 + X_i^\top \gamma,$$

where S_i is the average insurer payment per paid claim and γ contains the estimated severity coefficients.

The severity model uses only policies with at least one insurer payment. Policies without paid claims are retained in frequency modelling but excluded from severity estimation, since their average paid-claim severity is undefined.

Model specification and selection: Four candidate specifications are evaluated separately for frequency and severity: Benchmark, Flexible, Flexible + motive power, and Flexible + physical vehicle specifications. The benchmark models use a smaller set of risk characteristics, while the flexible models incorporate natural cubic splines and additional vehicle attributes. Splines allow the relationship between continuous characteristics and expected claims outcomes to vary nonlinearly without prescribing a single functional shape (Hastie et al., 2009). The complete candidate formulas are presented in Appendix C.

The candidates are fitted using claims experience from 2021–2023 and compared using the independent 2024 validation cohort. The frequency specification is selected by the lowest mean Poisson deviance, while the severity specification is selected by the lowest claim-count-weighted mean Gamma deviance. These measures evaluate how closely the fitted outcomes agree with observed claims experience on data not used to fit the candidate models, with lower values indicating better predictive performance (McCullagh and Nelder, 1989; Ohlsson and Johansson, 2010).

In the experiment, the selected frequency specification was Flexible + physical vehicle specifications. It includes natural cubic splines for driver age, licence years and logarithmic annual kilometres, together with prior claims history, State, remoteness, vehicle segment, motive power, transmission and physical vehicle specifications. Its fitted log mean can be represented as

$$\begin{aligned}\log \hat{\lambda}_i = \hat{\beta}_0 &+ f_1(A_i) + f_2(L_i) + f_3(\log K_i) \\ &+ b_{\text{claims}(i)} + b_{\text{state}(i)} + b_{\text{remoteness}(i)} \\ &+ b_{\text{segment}(i)} + b_{\text{motive}(i)} + b_{\text{transmission}(i)} \\ &+ \sum_{j \in \mathcal{V}} f_j(Z_{ij}) + b_{\text{cylinders}(i)} + \log E_i.\end{aligned}$$

Here, f denotes fitted natural cubic spline functions, b denotes estimated categorical effects and $\mathcal{V}$ denotes the continuous physical specifications: engine capacity, power, torque and unladen weight. The categorical terms are shorthand for estimated factor effects rather than unweighted additions to the log mean.

The selected severity specification was the Benchmark model, which includes logarithmic vehicle value, voluntary excess and vehicle segment without a vehicle-age spline:

$$\log \hat{s}_i = \hat{\gamma}_0 + \hat{\gamma}_V \log V_i + c_{\text{excess}(i)} + c_{\text{segment}(i)}.$$

Both formulas describe the specifications selected using the 2024 validation cohort; their coefficients are re-estimated using the combined 2021–2024 data.

Following model selection, both models are refitted using the combined 2021–2024 development data. Aggregate balancing adjustments are then applied to reconcile predicted paid-claim counts and claim payments with their corresponding observed totals in this period. Frequency predictions are scaled by the ratio of observed to predicted paid claims, while severity predictions are adjusted using the ratio of observed to predicted aggregate payments among policies with paid claims. These adjustments are calculated without reference to the 2025 test cohort and do not change the models' relative risk rankings.

The resulting factors are approximately 1.0000 for frequency and 1.0018 for severity, indicating that the fitted models already closely reproduce their respective development-period totals. Balancing therefore serves as an explicit aggregate reconciliation rather than a material adjustment to the estimated claims costs.

Calculation of the technical premium: The fitted expected claims cost provides the pure risk premium for each policy i calculated as

$$\hat{\mu}_i = \hat{\lambda}_i \times \hat{s}_i,$$

where $\hat{\lambda}_i$ is the fitted annual frequency of paid claims and $\hat{s}_i$ is the fitted average insurer payment per paid claim.

For the pricing experiment, a uniform loading of 30% is applied to construct a benchmark quoted premium, denoted by

$$\pi_i = 1.30 \times \hat{\mu}_i.$$

This loading provides a simplified allowance above expected insurer claims expenditure, without separately modelling acquisition expenses, operating costs or capital requirements. The 30% value is an assumed parameter of the simulation and is not presented as an observed Australian market loading.

Applying the same loading to all policies preserves their relative fitted claims-cost differences. It also produces an indicative expected claims ratio of 76.92% on the fitted claims-cost basis, calculated as 1/1.30. The resulting loaded benchmark is subsequently used to generate the initial reference-insurer quotation, competing quotations and candidate premium adjustments.

3.4 Competitive Quote Market and Conversion Generation

The 2025 cohort is used to construct a synthetic competitive quotation market in which each customer receives an offer from the reference insurer and quotations from up to six simulated competitors. The purpose is to examine how a customer's decision to purchase from the reference insurer depends not only on its quoted premium but also on the prices offered by alternative insurers.

The process consists of two stages. First, quotations are generated for the reference insurer and competing insurers, and the resulting competitive-position measures are calculated. Second, customer purchasing decisions are simulated using a two-stage behavioural model. Together, these stages produce the quotation characteristics and binary conversion outcomes required for subsequent modelling and pricing optimisation.

3.4.1 Quote-Market Construction

Reference insurer premium: The technical-pricing model provides a risk-based benchmark for each customer, but the price quoted to a customer may not equal that benchmark.

To examine the relationship between quoted premiums and purchasing behaviour, the reference insurer's initial quotation is generated by applying a randomly selected commercial multiplier to its fitted pricing benchmark.

$$P_i = \pi_i \omega_i, \qquad \omega_i \sim \text{Uniform}(0.95, 1.05)$$

Here, π_i represents the reference insurer's fitted pricing benchmark from Section 3.3, P_i is its initial quotation, and ω_i is a commercial multiplier drawn independently from a uniform distribution between 0.95 and 1.05. The multiplier allows the quoted premium to vary by up to 5% above or below the benchmark without changing the customer's fitted expected claims cost.

This additional variation is introduced for conversion-model development rather than to correct or improve the technical-pricing model. In particular, it allows customers with otherwise comparable risk characteristics to receive different commercial pricing adjustments, creating information from which the conversion models can learn the relationship between relative quoted price and simulated purchasing behaviour. The 5% range is a chosen experimental assumption and is not estimated from observed insurer discounts.

The initial multiplier ω_i generates price variation in the synthetic training market, whereas the multiplier m_i introduced in Section 3.6 represents a proposed pricing decision selected by the optimiser. Both operate relative to the same fitted benchmark, but serve different purposes.

Competing insurer premiums: Six stylised competitors are introduced to represent different pricing approaches, including one value-oriented insurer, two mainstream insurers, one premium-brand insurer and two specialist insurers. Six is a deliberate simulation choice: it provides variation in price rank, relative premiums and available quotations while keeping the competing choice set and its assumptions manageable. It is not intended to match the number of major Australian insurers or the complete range of offers on a comparison website. The study does not separately test whether changing the number of potential competitors to ten or twenty would alter its conclusions; the market-structure sensitivity scenarios instead modify competitor pricing and quote availability. Separate claims portfolios and technical-pricing models are not estimated for these competitors. Instead, all competing quotations are generated from the same fitted pricing benchmark for a given customer, with insurer-specific adjustments creating differences between the offers.

This design reflects the study's focus on the reference insurer's pricing decisions rather than the independent underwriting performance of every competitor. Allowing different insurers to quote different premiums for a common customer is consistent with the competitive nature of insurance pricing, in which insurers may adopt different pricing strategies and customer-group preferences (Taylor, 1986; Mourdoukoutas et al., 2025).

A log-multiplicative formula is used to construct the competing premiums Q_{ik} . For competitor k and customer i,

$$Q_{ik} = \pi_i \exp(\alpha_k + \delta_{ik} + \varepsilon_{ik}),$$

It combines three components: an insurer-specific pricing adjustment α_k , a customer-segment adjustment δ_{ik}, and random quotation variation ε_{ik}.

The exponential transformation ensures that the resulting premium multipliers are positive. The formula and its parameter values are specified as part of the synthetic experiment rather than estimated from observed competing quotations.

Random quotation variation is generated from a normal distribution with mean zero and standard deviation 0.045 on the logarithmic price scale. After exponentiation, the competing-premium multipliers are restricted to between 0.75 and 1.35 times the fitted pricing benchmark. These bounds prevent extreme simulated quotations while retaining differences between insurers. Both the dispersion and the bounds are specified simulation assumptions.

Competitors also have different probabilities of returning a quotation, ranging from 0.90 to 0.98 in the baseline configuration. This allows some insurers to be absent from a customer's available choice set. A minimum of two competing quotations is enforced so that relative price and market-position measures can be calculated for every customer. Each profile therefore contains the reference insurer's quotation and between two and six competing offers. The availability probabilities and minimum-quotation rule are design assumptions, not observed Australian aggregator statistics.

Competitive market position: The reference insurer's position is calculated from the quotations available to each customer.

Two principal measures are used: relative price and price rank. These capture different aspects of competition. Relative price measures how much more or less the reference insurer charges than the cheapest competitor, whereas price rank indicates how many available insurers offer a lower premium. Both are relevant to examining customer purchasing decisions in a quotation-comparison environment (Financial Conduct Authority, 2014).

Let $\mathcal{K}_i$ denote the set of competitors providing a quotation to customer i. The cheapest competing premium is

$$Q_i^{\min} = \min_{k \in \mathcal{K}_i} Q_{ik}.$$

The reference insurer's relative price is then

$$RP_i = \frac{P_i}{Q_i^{\min}}.$$

For example, a relative price of 1.10 means that the reference insurer is 10% more expensive than the cheapest available competitor.

The reference insurer's rank within the available quotations is calculated as

$$R_i = 1 + \sum_{k \in \mathcal{K}_i} \mathbb{I}(Q_{ik} < P_i),$$

where $\mathbb{I}(Q_{ik} < P_i)$ equals one when competitor k offers a lower premium than the reference insurer and zero otherwise. A rank of one indicates that no competing quotation is cheaper, while a rank of three indicates that two competing insurers offer lower premiums. Quotations with identical premiums share the same rank under this definition.

Both measures are calculated directly from the generated quotations rather than assigned independently.

Importantly, these measures can change when the reference insurer adjusts its premium, even if all competing quotations remain unchanged. This provides the connection between quotation generation and the pricing optimisation: each proposed premium can be evaluated using the competitive position it would create.

3.4.2 Conversion Generation Process

Customer conversion is generated using a two-stage purchasing mechanism. First, a customer decides whether to purchase a policy through the simulated quotation market. If a purchase occurs, the customer then selects one insurer from the available quotations. Separating these decisions allows the simulation to distinguish overall purchasing activity from the reference insurer's ability to attract customers within the competitive choice set.

The mechanism combines a logistic market-participation model with a conditional multinomial-logit insurer-choice model. Logistic and random-utility choice models are established approaches for representing discrete purchasing decisions (Train, 2009). The particular explanatory variables, parameter values and calibration targets used here are specified assumptions of the synthetic experiment.

Stage 1 — Market participation: The first stage determines the probability that a customer purchases insurance through the simulated comparison market. This probability is modelled using a logistic function, which maps a linear combination of customer characteristics and market conditions to a value between zero and one (Train, 2009).

The probability that customer i purchases insurance from any insurer in the available choice set is

$$q_i = \text{logit}^{-1}\left(\theta_0 + Z_i^\top \theta\right),$$

where q_i is the probability of purchasing through the market, Z_i contains the characteristics influencing the purchasing decision and θ represents their effects.

Market participation depends on the median premium among the customer's available quotations relative to the overall market median, together with driver age and previous claims. Higher relative market premiums reduce the assumed probability of purchase, while the customer characteristics introduce additional variation.

The baseline market-price sensitivity coefficient is set to 0.75. Age effects and claims-history adjustments are specified separately, with the participation intercept calibrated so that the mean predicted purchasing probability across the 2025 cohort equals 78%. This is a simulation target rather than an estimate of observed Australian quotation-market purchasing behaviour.

Stage 2 — Insurer choice: Conditional on purchasing insurance through the market, the customer selects one insurer from the available quotations. This decision is represented using a random-utility framework, in which each insurer's offer is assigned an attractiveness measure, or utility. Multinomial-logit choice probabilities are then calculated from these utilities (Train, 2009).

Utility depends on several factors. Customers generally receive lower utility from quotations that are expensive relative to competing offers, but this effect varies according to their individual price sensitivity. The model also allows insurers to differ in their non-price attractiveness, recognising that customers may value attributes other than the lowest premium. Additional rank and price-threshold terms introduce stronger responses when quotations become materially less competitive.

For insurer h in customer i's choice set, utility is specified as

$$\begin{aligned} U_{ih} = B_{ih} + \tau_{ih} - \beta_i \log(r_{ih}) \\ + H(R_{ih}) - T_P(r_{ih}) - T_R(R_{ih}) \\ + \kappa I(h = \text{ref}). \end{aligned}$$

The terms represent different influences on customer choice:

- U_{ih} defines the Utility of insurer h's quotation to customer i
- B_{ih} captures the customer's non-price preference for an insurer.
- τ_{ih} introduces additional random variation in preferences between customers and insurers.
- β_i measures customer-specific price sensitivity.
- r_{ih} compares an insurer's premium with the cheapest offer in the complete choice set.
- $H(R_{ih})$ provides additional preference for higher ranked quotations.
- $T_P(r_{ih})$ penalises quotations that exceed a specified relative-price threshold.
- $T_R(R_{ih})$ penalises quotations falling below a specified rank threshold.
- κ adjusts the reference insurer's utility to achieve the intended average conversion rate.

These behavioural relationships are specified for the simulation rather than estimated from observed consumer choices. The additional terms allow customer conversion to respond to both the size of price differences and the insurer's position within the available quotation set. Their baseline parameter values are provided in Appendix B.

Each utility is transformed into a conditional insurer-choice probability through the multinomial-logit function:

$$w_{ih} = \frac{\exp U_{ih}}{\sum_{\ell \in \mathcal{J}_i} \exp U_{i\ell}},$$

where $\mathcal{J}_i$ is the complete set of available insurers, including the reference insurer.

The resulting probabilities sum to one across all available insurers. An insurer with higher utility receives a higher conditional probability of selection, although it is not guaranteed to be chosen.

The conditional probabilities depend on quoted premiums because both relative price and rank enter the utility function. Consequently, reducing the reference insurer's premium may increase its utility and conditional selection probability while reducing the probabilities assigned to competing offers. The magnitude of the change depends on the customer's price sensitivity, the competing quotations and the remaining preference components.

The reference insurer's conversion probability is obtained by multiplying the probability of purchasing through the simulated market by the conditional probability of selecting the reference insurer:

$$p_i = q_i \times w_i,$$

where w_i is the conditional probability of selecting the reference insurer after deciding to purchase through the market.

This gives the unconditional probability that a customer accepts the reference insurer's quotation.

The reference insurer's preference adjustment is calibrated so that the average expected conversion probability across the 2025 cohort equals 18%. This differs from the 78% market-participation target because customers who purchase through the market may select any available insurer. Both targets are assumptions used to establish the baseline synthetic purchasing environment rather than observed Australian conversion statistics.

Generation of conversion outcomes: Customer purchase decisions are simulated from the calculated probabilities. Each customer first receives a market-purchase outcome using their participation probability. Where a purchase occurs, the selected insurer is drawn according to the conditional insurer-choice probabilities. The reference insurer's conversion outcome equals one when the customer purchases its policy and zero otherwise.

A zero outcome therefore includes customers who select another insurer and customers who do not purchase through the simulated market. This procedure generates 50,000 binary reference-insurer conversion observations for the 2025 cohort. The underlying probabilities are retained separately for evaluating predictions and pricing decisions, but are not provided as response values when fitting the conversion models.

Consider the following hypothetical customer:

Quantity	Initial quotation	Proposed repricing
Fitted pricing benchmark	\$1,000	\$1,000
Reference insurer's quotation	\$1,020	\$950
Cheapest competing quotation	\$950	\$950
Other competing quotations	\$990, \$1,030	\$990, \$1,030
Reference insurer's rank	3	1 (tied cheapest)
Relative price	1.074	1.000

Initially, the reference insurer quotes \$1,020 and two competitors are cheaper.

Reducing its quotation to \$950 changes its competitive position, despite the fitted claims cost remaining unchanged.

Those updated market-position values are used by the customer-choice mechanism and, subsequently, by the fitted conversion model to calculate the probability of purchasing from the reference insurer.

The connection between pricing and conversion can therefore be summarised as follows. The technical-pricing model establishes a customer-specific pricing reference, from which the reference insurer's quoted premium is constructed. That premium determines its price relative to competing insurers and its position within the available quotations. These measures enter the customer-choice mechanism, influencing the probability that the customer selects the reference insurer. During optimisation, alternative premiums are evaluated by recalculating the affected competitive-position measures and predicting the resulting conversion probabilities.

3.5 Conversion Modelling

Following the generation of customer purchasing outcomes, conversion models are fitted to estimate the probability that a customer purchases insurance from the reference insurer. Unlike the underlying two-stage choice mechanism, these models use only observable customer and vehicle characteristics, quoted premiums and competitive market information. The simulated purchase probabilities and underlying preference parameters are excluded from model estimation.

Six candidate models are compared to determine whether incorporating competing quotations improves conversion predictions beyond those obtained from customer characteristics and the reference insurer's own premium. This comparison is important because the purpose of the modelling stage is not simply to predict which customers purchase insurance, but to estimate how purchasing probabilities change when the reference insurer adjusts its premium.

3.5.1 Model Specification

Five logistic generalised linear models (GLMs) and one generalised additive model (GAM) are considered. Logistic regression provides an interpretable benchmark for modelling binary purchasing outcomes, while the GAM allows additional nonlinear relationships between conversion and the explanatory variables (Spedicato et al., 2018; Wood, 2017).

For customer i, the logistic GLM takes the general form

$$\text{logit}(\hat{p}_i) = \beta_0 + Z_i^\top \beta,$$

where $\hat{p}_i$ is the estimated probability of purchasing from the reference insurer, Z_i contains the explanatory variables available to the model and β represents their estimated coefficients. The candidate specifications differ in the information included in Z_i, as summarised in Table 2.

Table 2: Candidate conversion models and their explanatory information

Model	Information included
Technical-only GLM	Customer and vehicle characteristics, claims history and estimated technical premium.
Own-price GLM	Technical-only variables plus the reference insurer's quoted premium relative to technical premium
Rank GLM	Own-price variables plus the reference insurer's rank percentile
Relative-price GLM	Own-price variables plus its premium relative to the cheapest competitor
Market-position GLM	Combines rank and relative price with competitor-price variation and quote-availability measures
Market-position GAM	Uses the same broad competitive information as the Market-position GLM, with additional nonlinear flexibility

The Technical-only GLM provides the initial benchmark using customer characteristics and estimated claims risk. The Own-price GLM then introduces the reference insurer's commercial price adjustment, measured by

$$OP_i = \log\left(\frac{P_i}{\pi_i}\right),$$

where P_i is the reference insurer's initial quoted premium and π_i is its fitted technical premium. This measures the quoted premium relative to the risk-based benchmark, separating the commercial pricing adjustment from differences in expected claims costs between customers.

The Rank and Relative-price GLMs separately examine two aspects of competition. The Rank GLM considers where the reference insurer appears among available quotations, while the Relative-price GLM measures how much its premium differs from the cheapest competitor. Comparing these models with the Own-price GLM helps establish whether competitive position provides additional predictive information beyond the insurer's own pricing adjustment.

The Market-position GLM combines both measures with further information about the available quotations. These include the premium gap above the cheapest competitor, the reference insurer's premium relative to the market median, competitor-price dispersion and the number of competing quotations. The excess-price-gap feature is zero until the reference quotation is more than 8% above the cheapest competitor, and then measures the excess above that threshold. The 8% threshold is a predefined modelling choice introduced to represent a possible nonlinear competitive disadvantage, not an estimate from observed Australian quotation data. The full model specifications are provided in Appendix C.

Natural cubic splines are used for selected continuous predictors in the logistic GLMs to accommodate nonlinear effects. The Market-position GAM extends this flexibility through penalised cubic regression smooths, with smoothing parameters estimated using restricted maximum likelihood (Wood, 2017). The comparison between the full GLM and GAM therefore assesses whether greater flexibility materially improves conversion predictions when both models have access to broadly the same competitive information.

3.5.2 Model Validation and Selection

Conversion modelling uses the 50,000 synthetic quotation profiles generated for the 2025 cohort. These are divided into 30,000 training observations, 10,000 validation observations and 10,000 independent test observations. All candidate models are initially fitted using the training sample, while the validation sample is used to compare predictive performance and select between the two full market-position specifications.

The primary model-selection measure is binary log loss:

$$LL = -\frac{1}{n}\sum_{i=1}^{n}\left[D_i \log(\hat{p}_i) + (1 - D_i)\log(1 - \hat{p}_i)\right],$$

where D_i is the simulated binary conversion outcome and $\hat{p}_i$ is the fitted conversion probability.

Lower log loss indicates more accurate probability predictions, with larger penalties assigned to confident predictions that are incorrect. Log loss is a strictly proper scoring rule, making it suitable for assessing probabilistic predictions (Gneiting and Raftery, 2007). This is particularly important because the subsequent optimisation uses predicted purchase probabilities rather than binary classifications.

Model performance is also assessed using the Brier score, area under the receiver operating characteristic curve (AUC) and probability-calibration measures. AUC evaluates the ability to distinguish customers who convert from those who do not, while calibration examines whether predicted probabilities are consistent with observed conversion frequencies. Calibration is particularly relevant when predicted probabilities are subsequently used to evaluate alternative premiums (Van Calster et al., 2019).

The Market-position GLM is retained as the default full-market specification because it provides a comparatively interpretable model. The GAM is selected only when it reduces validation log loss by at least 0.5% relative to the GLM and achieves a calibration slope between 0.70 and 1.30. A calibration slope near one indicates that the variation in predicted probabilities is approximately consistent with observed outcomes, whereas a substantially lower slope generally indicates overly extreme predictions. The improvement threshold and acceptable calibration interval are prespecified modelling criteria. They require the additional complexity of the GAM to provide a measurable predictive benefit without substantially compromising calibration.

Following model selection, all six candidate specifications are refitted using the combined training and validation observations and evaluated on the independent test sample. Because the underlying conversion probabilities are known within the simulation, the fitted probabilities can also be compared with their generating values using mean absolute error, root mean squared error and Spearman rank correlation. These additional measures assess how closely the models recover the underlying purchasing probabilities and their ordering across customers. The generating probabilities are used only for evaluation and are not supplied to the fitted models during development.

The subsequent pricing optimisation follows a related but distinct procedure. Models fitted on the training sample are used to predict conversion under alternative premiums, with probability recalibration and pricing-strategy selection performed using validation observations. Once the pricing rules have been selected, the models refitted on the combined training and validation sample are used to evaluate the selected strategies on the independent test sample.

Although the validation sample contributes to both probability recalibration and pricing selection, the test sample remains separate from those decisions. This allows the final evaluation to assess whether the selected pricing strategies generalise beyond the observations used during their development.

3.6 Pricing Optimisation

The final modelling stage uses the estimated conversion probabilities to select premiums for the reference insurer. The technical-pricing model establishes the premium required to cover estimated claims costs with the specified loading, while the conversion models estimate how the probability of purchase varies with the premium offered and its position relative to competing quotations.

Pricing optimisation combines these estimates to identify premium adjustments that improve expected customer conversion while maintaining specified financial requirements. This approach follows the demand-aware pricing framework examined by Spedicato et al. (2018) and Mayorga and Torres (2017), in which estimated claims costs and customer purchasing probabilities jointly inform premium decisions.

Four strategies are evaluated. The first retains the technical premium for every customer and serves as the benchmark. The second and third apply a single premium multiplier across the portfolio, using either the Own-price GLM or the selected Market-position conversion model. The fourth allows premiums to vary between customer segments using the selected Market-position model. The uniform strategies examine the effect of changing the overall pricing level, while the targeted strategy examines whether different adjustments across customer groups can improve conversion without materially changing the aggregate quoted premium level.

3.6.1 Candidate Premiums and Predicted Conversion

The starting point is the fitted technical premium from Section 3.3. For customer i, the estimated annual claims cost is denoted by $\hat{\mu}_i$, and the technical premium is $\pi_i = 1.30\hat{\mu}_i$, incorporating the assumed 30% loading. The pricing exercise considers alternative premiums by applying a multiplier to this benchmark:

$$P_i(m_i) = m_i \pi_i, \qquad m_i \in \{0.95, 0.96, \ldots, 1.05\}. \tag{1}$$

Here, m_i is the multiplier being considered for customer i. A value of 1.00 retains the technical premium, while 0.95 and 1.05 represent reductions and increases of 5%, respectively. The eleven candidate multipliers provide a finite set of pricing alternatives, allowing their expected outcomes to be compared before a decision is made.

These multipliers should be distinguished from the initial commercial adjustments introduced in Section 3.4. The original quotations were generated by applying independently simulated multipliers to technical premiums, providing the price variation used to generate and model customer purchasing outcomes.

In contrast, the multipliers in Equation 1 are decision variables: the optimisation selects among them to recommend new premiums. Both use the same $\pm 5\%$ range, limiting the proposed changes to the scale of price variation represented in the simulated quote data.

For each candidate multiplier, the reference insurer's quotation is replaced by $P_i(m_i)$ while the competing insurers' quotations remain unchanged. The reference insurer's competitive rank, relative price, price gap and other price-dependent variables are then recalculated. These updated values are supplied to the fitted conversion model to obtain

$$\hat{p}_i(P_i(m_i)) = \widehat{\Pr}\left(D_i = 1 \mid Z_i(P_i(m_i))\right),$$

where $D_i = 1$ indicates that the customer purchases insurance from the reference insurer, and $Z_i(P_i(m_i))$ contains the customer characteristics and price-dependent predictors evaluated at the proposed premium. The fitted model therefore estimates conversion separately for each alternative price, rather than applying a fixed conversion probability to every proposed premium.

For example, consider a customer with a technical premium of \$1,300 and a cheapest competing quotation of \$1,250. At the technical premium, the reference insurer is 4% more expensive than that competitor. Applying a multiplier of 0.95 reduces its premium to \$1,235, making it cheaper than the previously lowest competing quotation. This changes the reference insurer's competitive position and consequently the conversion probability predicted by the Market-position model. The optimiser assesses whether the expected improvement in conversion justifies the lower premium received if the customer purchases.

Before these predicted probabilities are used for pricing decisions, they are recalibrated using the validation sample. The conversion models initially fitted on the training observations produce probabilities at the original simulated quotations. A logistic recalibration model is estimated by relating validation conversion outcomes to their original predicted probabilities:

$$\operatorname{logit}\left(\hat{p}_i^{\text{cal}}\right) = a + b \operatorname{logit}(\hat{p}_i),$$

where a adjusts the overall level of predicted conversion and b adjusts the spread of probabilities between customers.

The estimated coefficients are applied to predictions under each candidate premium. Recalibration is conducted separately for the Own-price GLM and the selected Market-position model because their probability predictions may differ in calibration.

The recalibrated probability $\hat{p}_i^{\text{cal}}(P_i(m_i))$ is used in all subsequent pricing calculations. This procedure assumes that the recalibration relationship estimated at the original quotations remains applicable over the permitted range of premium adjustments. No underlying simulated purchasing probabilities or test-sample conversion outcomes are used to recalibrate the fitted models.

3.6.2 Pricing Objectives and Financial Constraints

The premium-selection problem involves a trade-off. Reducing a premium may improve its competitiveness and increase the probability of purchase, but it also reduces the premium received from each policy written. Increasing a premium may improve the expected profit from an accepted policy while reducing the number of customers who purchase. The optimisation therefore assesses both expected conversion and financial outcomes for each candidate price.

For a set of n quotations, the predicted number of conversions under a proposed pricing strategy is

$$\hat{N} = \sum_{i=1}^{n} \hat{p}_i^{\text{cal}}(P_i(m_i)).$$

The corresponding expected conversion rate is $\hat{N}/n$. Each customer's contribution to this total is their predicted probability of purchasing the reference insurer's policy, so the calculation allows different customers to respond differently to the proposed premiums.

Financial performance is assessed using expected pre-expense underwriting profit, defined for a set of n quotations as:

$$\hat{M} = \sum_{i=1}^{n} \hat{p}_i^{\text{cal}}(P_i(m_i)) \left[P_i(m_i) - \hat{\mu}_i\right].$$

The difference between the proposed premium and fitted expected claims cost represents the pre-expense underwriting amount earned if the customer purchases the policy. Multiplying this amount by the probability of purchase gives the expected pre-expense underwriting profit attributable to that quotation. Summing across customers provides the portfolio measure.

The two uniform-pricing strategies maximise expected pre-expense underwriting profit while satisfying their conversion and financial requirements. The targeted strategy instead maximises expected conversions, subject to preserving benchmark expected profit and satisfying additional constraints. An increase in the financial measure relative to technical pricing is referred to as **expected profit uplift**, while the ratio of strategy profit to benchmark profit is referred to as **expected profit retention.**

The targeted strategy is subject to five principal financial and commercial restrictions, in addition to the permitted range of individual premium multipliers. These are summarised in Table 3.

Table 3: Pricing constraints applied to the targeted repricing

Constraint	Requirement	Purpose
Premium adjustment	0.95–1.05	Limit the size of individual premium changes
Aggregate quoted rate level	99.5%–100.5%	Maintain approximately unchanged total quoted premiums
Expected profit retention (all customers)	At least 100%	Preserve expected pre-expense underwriting profit
Expected profit retention (each customer group)	At least 90%	Limit expected profit deterioration within individual customer groups
Expected claims ratio	At most 80%	Maintain claims adequacy
Expected conversion	At least 100% of benchmark	Require expected conversion to be no lower than the technical benchmark

The aggregate quoted rate level compares the total premiums offered under the proposed strategy with those offered at technical rates:

$$RL = \frac{\sum_i P_i(m_i)}{\sum_i \pi_i}, \qquad 0.995 \leq RL \leq 1.005.$$

A rate level of 1.00 indicates that total quoted premiums are unchanged. The permitted deviation of 0.5% allows small differences arising from the discrete candidate multipliers while preventing the targeted strategy from achieving higher conversion simply by reducing premiums across the portfolio. Importantly, this measure includes every quotation regardless of whether the customer purchases. It is a measure of the aggregate pricing level, not expected premium revenue.

The expected profit retention requirements compare expected pre-expense underwriting profit with the technical-pricing benchmark. At portfolio level, the proposed strategy must retain at least 100% of the benchmark amount. Within each selected customer segment, expected pre-expense underwriting profit under its chosen multiplier must

remain at least 90% of the amount obtained by pricing that segment at technical rates. This provides flexibility to reduce premiums for some customer groups, provided the required portfolio-wide expected profit is maintained.

Claims adequacy is assessed through the expected claims ratio:

$$\widehat{CR} = \frac{\sum_i \hat{p}_i^{\text{cal}}(P_i(m_i))\hat{\mu}_i}{\sum_i \hat{p}_i^{\text{cal}}(P_i(m_i))P_i(m_i)} \leq 0.80.$$

Both expected claims costs and written premiums are weighted by the probability of purchase, so the ratio reflects the expected composition of policies written rather than all quotations offered. The 80% ceiling is an assumed commercial constraint, not a regulatory requirement. With the 30% technical loading used in Section 3.3, the fitted technical-pricing benchmark implies a claims ratio of approximately $1/1.30 = 76.92\%$. The higher ceiling allows some deterioration in claims adequacy while preventing excessive premium reductions.

These restrictions are imposed when selecting a strategy using validation data. They express the study's chosen commercial risk appetite rather than claiming that the resulting premiums would satisfy every financial, regulatory or customer-treatment requirement faced by an operating insurer.

3.6.3 Uniform and Targeted Pricing Strategies

The strategies differ in how many premium multipliers the insurer can select and which conversion model informs those decisions.

For uniform repricing, every quotation receives the same multiplier. The eleven alternatives in Equation 1 are evaluated separately using the Own-price GLM and the selected Market-position model. For each model, the selected multiplier maximises expected pre-expense underwriting profit, provided expected conversion remains at least 95% of the model's technical-pricing benchmark, expected portfolio profit does not decrease, and the expected claims ratio remains at or below 80%.

The uniform strategies are not subject to the narrow aggregate rate-level restriction imposed on targeted pricing. They are included to establish how the premium recommendations differ when the insurer uses only its own price information compared with the fuller competitive information set, while maintaining the same portfolio-wide adjustment for every customer.

Targeted repricing allows the multiplier to vary across customer groups. Rather than assigning a separate premium adjustment to every individual quotation, the portfolio is divided using one observable characteristic at a time. Candidate segmentation variables include driver age and experience, previous claims, state, remoteness, vehicle characteristics, annual kilometres and voluntary excess. Each provides a different way to group customers who may have different risk characteristics or responses to changes in competitive pricing.

Continuous variables are generally divided into quartiles using cut-offs determined from the validation sample, while categorical variables retain their relevant groups. At least 50 validation observations are required within each segment. These restrictions reduce the reliance on small groups and avoid constructing highly detailed pricing cells from combinations of multiple characteristics.

Each candidate segmentation structure is optimised separately. Within a structure, all customers in segment g receive the same multiplier m, but different segments may receive different multipliers. No pricing direction is prescribed in advance. For example, higher-value vehicles are not automatically assigned higher premiums; the optimiser evaluates the available premium adjustments according to their predicted conversion and financial outcomes.

The selection problem is formulated as a binary linear programme. Define $x_{gm} = 1$ if multiplier m is selected for segment g, and zero otherwise. Exactly one multiplier must be selected for each segment:

$$\sum_{m \in \mathcal{M}} x_{gm} = 1, \qquad x_{gm} \in \{0, 1\},$$

where $\mathcal{M} = \{0.95, 0.96, \ldots, 1.05\}$ is the set of permitted multipliers.

For each segment and multiplier, the fitted Market-position model is used to calculate the expected number of conversions:

$$\hat{N}_{gm} = \sum_{i \in g} \hat{p}_i^{\text{cal}}(m\pi_i).$$

The targeted optimisation then selects the combination of segment multipliers that maximises total expected conversions:

$$\max_{\mathbf{x}} \sum_{g} \sum_{m \in \mathcal{M}} x_{gm} \hat{N}_{gm}.$$

The corresponding expected pre-expense underwriting profit, quoted premium and probability-weighted claims measures are calculated in advance for every segment–multiplier combination. This allows the requirements in Table 3 to be imposed as linear restrictions on the binary selection variables. Candidate multipliers that fail the minimum segment expected profit requirement are excluded before optimisation.

The problem is solved using binary linear programming through the R package `lpSolve`. The solver selects the combination of multipliers providing the highest predicted conversion while satisfying all the specified constraints simultaneously. This is important because the best multiplier for a segment in isolation may not be feasible when the insurer must maintain aggregate quoted rates, portfolio expected profit and claims adequacy.

The procedure is repeated for each eligible segmentation variable. The preferred segmentation structure is selected according to its validation-sample predicted conversion rate. When competing structures produce expected conversion rates within 0.05 percentage points of the highest result, preference is given to the structure containing fewer segments. This avoids adopting a more complicated segmentation structure when its additional predicted conversion benefit is negligible.

The resulting strategy specifies both the characteristic used for targeting and the multiplier assigned to each group. These decisions are fixed before being applied to the independent test sample.

3.6.4 Independent Evaluation

All pricing strategies are selected using the 10,000-observation conversion-validation sample. The Own-price and Market-position development models are fitted using the training observations, while the validation observations determine probability recalibration, uniform multipliers and the targeted segmentation structure. The final models refitted using the combined training and validation data are subsequently used to produce predictions on the independent 10,000-observation test sample. Neither test conversion outcomes nor the underlying simulated purchasing probabilities are used to select pricing decisions.

The final evaluation distinguishes between the conversion probabilities predicted by the fitted models and those implied by the known customer-choice mechanism from Section 3.4. The former determine which premium adjustments are recommended; the latter are used to assess those decisions after selection.

For each test customer and selected strategy s, the proposed reference insurer premium is inserted into the original simulated comparison market, with competitor quotations held unchanged. The two-stage demand mechanism is then recalculated, including market participation, competitive ranks, relative prices and conditional insurer-choice probabilities. The resulting reference insurer conversion probability is

$$p_i^{\text{DGP}}(s) = q_i(s)\, w_i(s),$$

where $q_i(s)$ is the simulated probability of purchasing through the comparison market and $w_i(s)$ is the conditional probability of choosing the reference insurer under strategy s.

This provides a test of whether the pricing decisions selected using fitted conversion models improve outcomes under the specified customer-choice mechanism, rather than merely appearing beneficial according to the models used to select them.

For each strategy, the mean expected conversion rate is

$$\bar{p}_s = \frac{1}{n} \sum_{i=1}^{n} p_i^{\text{DGP}}(s).$$

Let $\bar{p}_0$ denote expected conversion under the technical-pricing benchmark. Absolute conversion uplift is calculated as $\Delta p_s = \bar{p}_s - \bar{p}_0$, expressed in percentage points, while relative uplift is

$$\Delta p_s^{\text{rel}} = \frac{\bar{p}_s - \bar{p}_0}{\bar{p}_0}.$$

These distinguish the change in conversion rate from the proportional improvement over the benchmark.

Expected pre-expense underwriting profit is also recalculated using the underlying simulated purchasing probabilities:

$$M_s^{\text{DGP}} = \sum_{i=1}^{n} p_i^{\text{DGP}}(s) \left[P_i(s) - \hat{\mu}_i \right].$$

The evaluation retains fitted expected claims costs because the exercise is intended to assess pricing decisions supported by the technical-pricing model. The underlying simulation is used to independently evaluate demand response, not to replace estimated actuarial costs with unobserved claims-cost parameters.

Final comparisons consider expected conversion, absolute and relative conversion uplift, expected profit retention, aggregate quoted rate level and expected claims ratio. The results establish whether the pricing recommendations remain effective when assessed against the specified synthetic purchasing behaviour and whether the financial restrictions imposed during validation remain satisfied on the held-out test sample.

The evaluation is independent of pricing selection, but remains conditional on the assumed customer-choice mechanism. Consequently, improvements observed in this experiment demonstrate the value of market-position information within the constructed competitive environment, rather than establishing that equivalent improvements would necessarily occur in an observed Australian motor insurance market.

3.7 Validation, Sensitivity and Uncertainty

Additional validation and robustness procedures assess whether the principal findings remain consistent under alternative assumptions, independently generated synthetic datasets and changes in the composition of the final test sample. These procedures address different sources of uncertainty and provide complementary assessments of the simulation framework (Kleijnen, 2015).

Validation and information controls: The modelling framework maintains a separation between model development and final evaluation. The 2025 cohort is excluded from technical-pricing model estimation. For conversion modelling and pricing optimisation, the independent test sample is withheld from model specification, probability recalibration and strategy selection.

The underlying expected claims costs and customer-choice probabilities are also stored separately from the observations used to fit models and select premiums. They are used for subsequent evaluation, rather than being supplied directly to the fitted models.

Validation checks are performed throughout the computational framework to assess the consistency of generated records, model predictions and pricing calculations. In particular, competitive-position variables are recalculated for each candidate premium to ensure that predicted conversion reflects the revised quotation environment.

Robustness analysis: Three complementary procedures are used to examine different sources of uncertainty, as summarised in Table 4.

Table 4: Robustness procedures and sources of uncertainty

Procedure	What changes?	Purpose
Structural sensitivity analysis	Market, demand and pricing assumptions	Assess dependence on the specified modelling assumptions
Monte Carlo regeneration	Randomly generated portfolios, claims and conversion outcomes	Assess stability across independent synthetic datasets
Paired test-sample bootstrap	Composition of the held-out test sample	Quantify sampling uncertainty in the final pricing comparison

Structural sensitivity analysis: Ten cases are evaluated, comprising the baseline and nine alternatives covering competitive conditions, customer demand and pricing constraints. The analysis varies one major group of assumptions at a time to examine how changes in the simulated environment affect model performance and pricing recommendations.

Market scenarios examine stronger or weaker competition and greater variation in competing premiums. Demand scenarios vary customer price sensitivity, ranking preferences and nonlinear price thresholds. Pricing scenarios alter the permitted range of premium adjustments or impose a stricter claims-ratio ceiling.

Each scenario changes a related group of assumptions while retaining the remaining baseline structure. The detailed parameter changes are documented in Appendix B. This design allows changes in the results to be attributed more clearly to the assumption group being examined, although it does not provide a comprehensive assessment of interactions between different assumption groups (Saltelli and Annoni, 2010).

Only the affected stages of the simulation are rerun. Market scenarios regenerate competing quotations and subsequent modelling stages. Demand scenarios retain the original quotations but regenerate customer-choice outcomes and subsequent models. Pricing scenarios retain the fitted market and demand structure while repeating optimisation under revised constraints.

The underlying portfolio and technical-pricing models remain unchanged across these scenarios. Where possible, common random-number seeds are retained to reduce simulation variation unrelated to the assumptions being examined. This follows the common-random-numbers approach used to improve the comparability of simulation experiments (Kleijnen, 1988).

Each scenario is evaluated according to whether competitive market-position information improves conversion modelling beyond the Own-price GLM and whether targeted repricing increases expected conversion while preserving at least the benchmark expected pre-expense underwriting profit.

Monte Carlo regeneration: This procedure examines whether the findings depend on the particular random observations generated in the baseline experiment. Repeated simulation under fixed structural assumptions allows the variability of model selection and pricing outcomes across independently generated datasets to be assessed (Kleijnen, 2015).

The complete simulation and modelling process is repeated using ten independent sets of random-number seeds. Empirical calibration inputs and structural assumptions remain fixed, while new portfolios, claims experience, competing quotations and conversion outcomes are generated.

Technical-pricing and conversion models are refitted, and model and pricing-strategy selection are repeated using the same procedures as in the baseline analysis.

This allows predictive performance, selected model specifications, targeting structures and pricing outcomes to be compared across independently generated datasets. Given the limited number of replications, the analysis is interpreted as a stability assessment rather than a precise estimate of simulation uncertainty.

Paired bootstrap analysis: The final procedure examines uncertainty associated with the particular customers included in the 10,000-profile test sample. A paired nonparametric bootstrap is conducted using 5,000 samples drawn with replacement, following the resampling approach described by Efron and Tibshirani (1993).

For each resample, the technical-pricing benchmark and selected targeted strategy are evaluated using the same customer profiles. Keeping the observations paired preserves the relationship between each customer's expected outcomes under the two strategies, allowing uncertainty in their difference to be assessed.

The fitted models and selected pricing decisions remain fixed throughout resampling. Each bootstrap sample is used to recalculate expected conversion uplift, expected profit retention, expected claims ratio and aggregate quoted rate level.

Two-sided 95% bootstrap percentile intervals are calculated using the 2.5th and 97.5th percentiles of the resulting distributions (Efron and Tibshirani, 1993). These intervals describe uncertainty associated with the composition of the held-out test sample, conditional on the fitted models, selected pricing strategy and specified simulation assumptions.

Together, the three procedures examine sensitivity to modelling assumptions, variation across independently generated datasets and uncertainty associated with the composition of the final test sample.

3.8 Computational Implementation

The analysis is implemented in R through a sequence of R Markdown modules. This structure allows the simulation, statistical modelling and pricing optimisation procedures to be executed reproducibly, with the underlying code and analytical outputs documented together. Reproducible computational workflows are particularly important in simulation-based research because they allow results to be traced to the assumptions, data and procedures used to produce them (Peng, 2011; Xie et al., 2018).

The accompanying computational materials and appendices document the modelling specifications, simulation assumptions, required calibration inputs and supplementary results.

4 Results

This section presents the results of the synthetic personal motor insurance framework. First, the technical-pricing models and competitive quote market are examined to assess whether the simulation produces a coherent environment for analysing customer conversion. The predictive performance of the conversion models is then compared to determine the value of incorporating competitive market-position information. Finally, pricing optimisation evaluates whether these improvements support more effective premium decisions, while structural sensitivity, Monte Carlo regeneration and bootstrap analysis assess the stability of the findings.

4.1 Technical Pricing and Synthetic Market Validation

4.1.1 Technical Pricing

The synthetic portfolio comprised 250,000 policies and generated 38,947 reported claims, of which 36,464 resulted in insurer payments. The remaining 2,483 claims were fully covered by policyholder excesses and therefore did not contribute to the paid-claim frequency or severity used for technical pricing.

Validation on the 2024 cohort favoured a quasi-Poisson GLM for paid-claim frequency and a Gamma GLM for paid-claim severity, both using logarithmic link functions. The selected frequency specification incorporated nonlinear effects of driver age, licence experience and annual kilometres, together with claims history, geography, vehicle segment, motive power and physical vehicle specifications. The physical variables included engine capacity, cylinders, power, torque, transmission and unladen weight. In contrast, the selected severity model was the simpler benchmark specification, using logarithmic vehicle value, voluntary excess and vehicle segment.

The selected specifications were refitted using the combined 2021–2024 experience and evaluated on the independent 2025 cohort. Table 5 summarises their validation diagnostics; the complete model definitions are given in Section 3.3 and Appendix C.

Table 5: Technical-pricing validation diagnostics on the independent 2025 cohort

Diagnostic	Result
Frequency dispersion	1.030
Severity dispersion	1.196
2025 frequency O/E	1.009
2025 severity O/E	0.981
2025 claims-cost O/E	1.001
Underlying claims-cost Spearman correlation	0.807
Mean fitted-to-underlying claims-cost ratio	0.991
Claims-cost RMSE	$440.03
Claims-cost MAE	$278.06

Observed-to-expected (O/E) ratios compare actual claims experience with model predictions. A ratio of one indicates aggregate agreement; values below or above one indicate that the model overpredicted or underpredicted realised outcomes, respectively. Dispersion estimates describe the variability remaining in the fitted models, while

Spearman correlation assesses how well fitted claims costs preserve the ranking of policies according to their underlying expected risk. Mean absolute error (MAE) measures the average absolute difference between fitted and underlying expected claims costs, whereas root mean squared error (RMSE) places greater weight on larger errors.

The fitted models produced frequency and severity dispersion estimates of 1.030 and 1.196, respectively. On the independent 2025 cohort, the claims-cost observed-to-expected (O/E) ratio was 1.001. This means that the total claims expenditure was almost equal to the amount predicted for the 2025 policies. Figure 2 shows that fitted and underlying expected claims costs also had a Spearman correlation of 0.807, indicating that the model generally assigned higher expected costs to higher-risk policies. The model did not have access to the underlying claims-cost relationships when it was fitted.

The mean fitted claims cost was approximately 0.9% below its underlying expected value. Predictions for particular policies were less precise: the RMSE was \$440.03 and the MAE was \$278.06. Although the predicted total was close to the actual claims expenditure, the model did not estimate every policy's expected claims cost accurately. Adding physical vehicle specifications reduced frequency validation deviance by only approximately 0.006% relative to the motive-power model.

Technical-pricing candidates were selected by the lowest validation deviance, without a separate requirement for a meaningful improvement over a simpler model. By contrast, the conversion GAM had to improve validation log loss by at least 0.5% to be selected over the Market-position GLM. This difference in the selection rules is a limitation of the experimental design. In particular, the frequency model's 0.006% improvement should not be treated as robust evidence that the additional physical specifications improve prediction.

Across all five annual cohorts, fitted expected claims cost averaged \$791.43 per policy. Applying the 30% loading produced a mean technical premium of \$1,028.85. The 2025 cohort used for the competitive quote market instead had a mean technical premium of \$1,041.48. The fitted premiums provided a risk-differentiated pricing benchmark while retaining the estimation uncertainty that arises when technical premiums are developed from observed claims experience rather than known underlying costs.

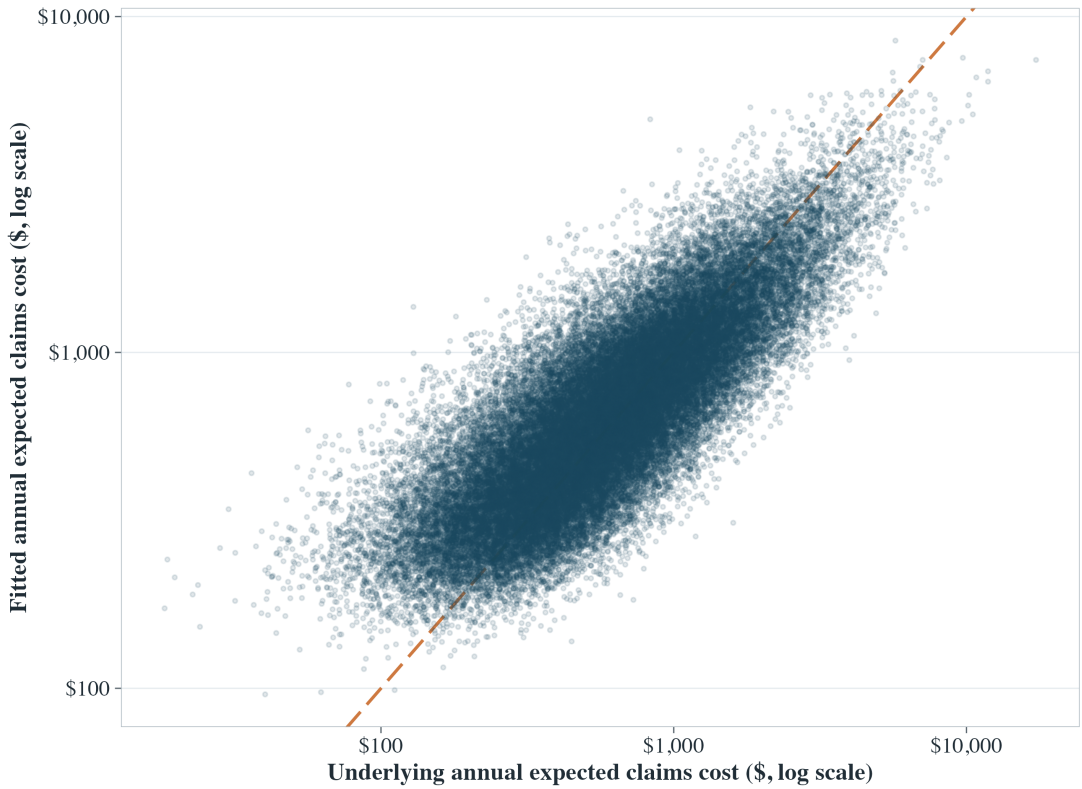


Figure 2: Fitted versus underlying expected annual insurer claims cost for the 2025 cohort. Each point represents a policy; both axes use logarithmic scales and the dashed diagonal represents equality between fitted and underlying expected cost. Deviations from the line indicate estimation error, while rank recovery is summarised in Table 5. The underlying claim costs were withheld during model fitting.

4.1.2 Competitive Quote Market and Conversion Environment

The 2025 cohort was used to construct the competitive quote market. Each customer received between two and six competing quotations, with an average of 5.63 competitors returning an offer. Table 6 summarises the resulting market characteristics.

Table 6: Baseline characteristics of the synthetic 2025 quote market

Quote-market measure	Result
Customer profiles	50,000
Mean number of returned competitors	5.63
Mean technical premium	$1,041.48
Mean reference insurer quote	$1,041.58
Mean competing insurer quote	$1,046.92
Median reference insurer rank	4
Mean reference insurer rank	3.75
Reference insurer cheapest	4.83%
Reference insurer within 5% of cheapest	30.07%
Median relative price to cheapest	1.0754
Mean gap above cheapest	7.74%

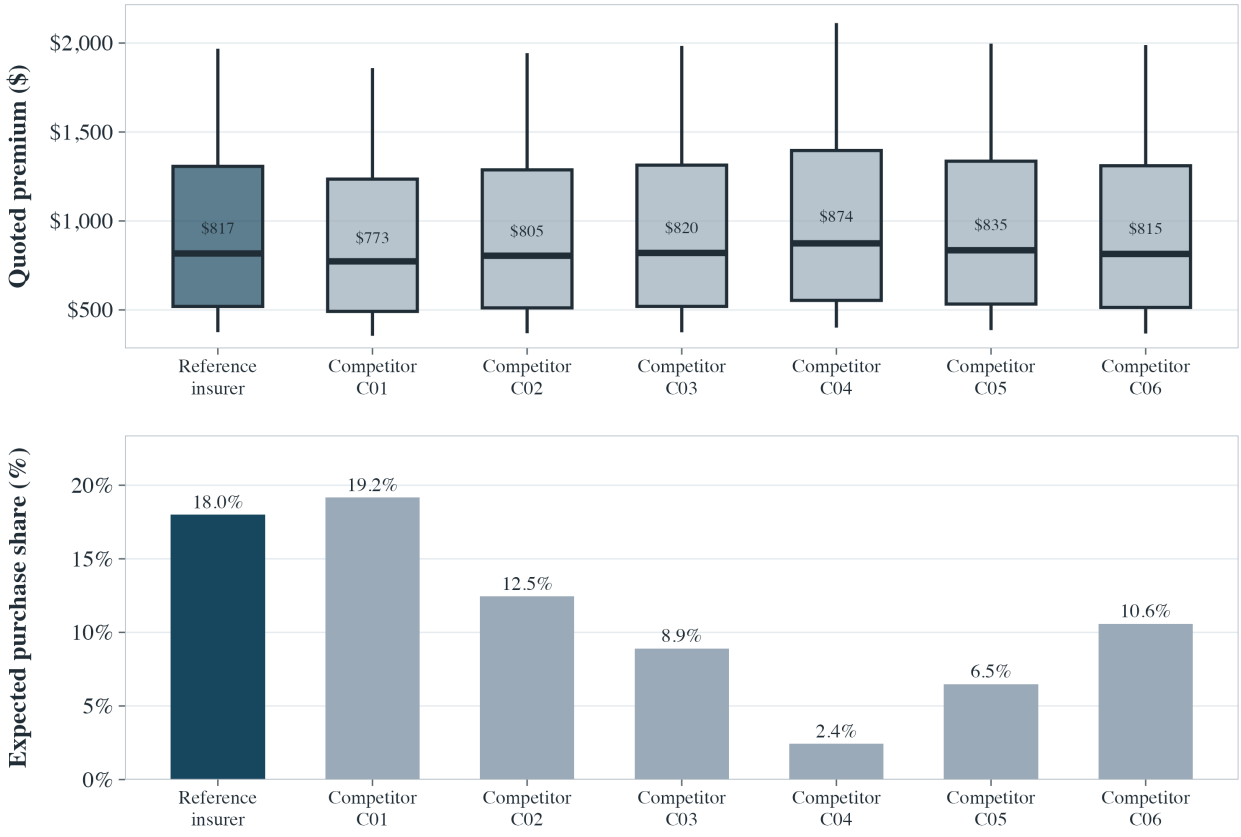


Figure 3: Initial quoted premiums and expected purchase shares for the reference insurer and six simulated competitors. The upper panel shows the distribution of returned quotations, with boxes representing the interquartile range (25th–75th percentiles), horizontal lines indicating the median, and whiskers extending from the 10th to the 90th percentiles. Median premiums are labelled within each box. The lower panel shows each insurer's expected purchases as a proportion of all 50,000 customer enquiries, calculated using market-participation and conditional insurer-choice probabilities. The expected purchase shares sum to the overall market-participation probability rather than 100%.

The reference insurer's mean premium was similar to the average competing premium, but its offer was rarely the cheapest for an individual customer. Its median rank was fourth, and it was the cheapest insurer in only 4.83% of enquiries. Its median relative price was 1.0754, meaning that the reference insurer typically quoted 7.54% more than the cheapest available competitor. Thus, similar average premiums across insurers do not imply that their quotations are equally competitive for individual customers.

The simulated purchasing outcomes were close to their predefined targets. The average probability of buying from any insurer in the simulated market was 78.00%, while 78.28% of customers actually purchased in the generated outcomes. For the reference insurer, the average underlying conversion probability was 18.00%, compared with 18.11% of customers who actually purchased. The small differences arise because purchasing outcomes were randomly generated from the specified probabilities.

Expected reference-insurer conversion decreased consistently as its quotation became less competitive. For customers ranked first, expected conversion was 37.94%, while lower-ranked quotations had substantially smaller purchasing probabilities. Spearman correlations between underlying conversion probability and quotation rank or relative price were −0.716 and −0.722, respectively. These relationships are consequences of the specified customer-choice mechanism and provide patterns for the fitted models to recover.

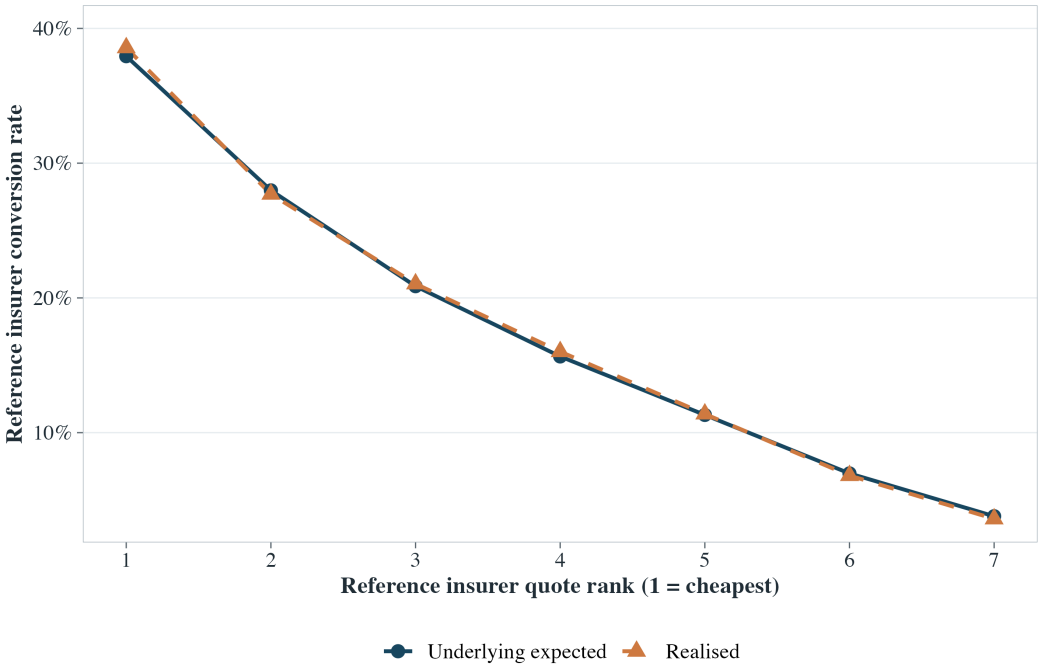


Figure 4: Underlying expected and realised reference-insurer conversion by initial quotation rank in the 2025 synthetic market. Rank one denotes the cheapest quotation. The expected series uses the generating purchase probabilities, while the realised series uses binary simulated outcomes. Differences reflect purchase-outcome randomness, and the rank relationship is generated by the assumed customer-choice mechanism.

4.2 Conversion Model Performance

4.2.1 Validation Fit and Model Selection

Six conversion models were fitted using 30,000 training profiles and compared on the 10,000-profile validation sample. The primary comparison used binary log loss, with AUC and calibration measures providing complementary evidence. Table 7 also reports Spearman correlations with withheld underlying conversion probabilities on the independent test sample. These correlations were calculated only for evaluation and did not enter model selection.

Table 7: Conversion-model validation performance and recovery of underlying conversion probabilities

Model	Validation log loss	Validation AUC	Validation calibration slope	Test underlying Spearman
Technical-only GLM	0.4758	0.5408	0.843	0.187
Own-price GLM	0.4600	0.6387	0.984	0.668
Rank GLM	0.4556	0.6529	0.965	0.744
Relative-price GLM	0.4517	0.6635	1.007	0.782
Market-position GLM	0.4461	0.6750	0.999	0.843
Market-position GAM	0.4449	0.6751	1.015	0.850

The Technical-only GLM achieved validation log loss of 0.4758 and AUC of 0.541, showing that claims-risk and customer characteristics alone provided limited information about which customers would buy. Adding the reference insurer's own commercial price adjustment reduced log loss to 0.4600 and increased AUC to 0.639. Including quotation rank or relative price improved predictions further. In this simulated market, knowing how a premium compared with competing offers was more informative than knowing the reference insurer's price adjustment alone.

The Market-position GLM achieved validation log loss of 0.4461, AUC of 0.675 and a calibration slope of 0.999. Its log loss was approximately 3.02% lower than that of the Own-price GLM. The Market-position GAM performed slightly better on log loss (0.4449), but this was only a 0.27% improvement over the Market-position GLM. Under the predefined selection rule, the GAM needed an improvement of at least 0.5% and acceptable calibration to justify its added complexity. Although the GAM met the calibration requirement, it did not meet the improvement threshold. The Market-position GLM was therefore selected for the baseline experiment.

The improvement from adding competitive information was materially larger than the incremental advantage from the GAM. Within the simulated environment, the information contained in competing quotations contributed more consistently to predictive performance than additional nonlinear modelling flexibility.

4.2.2 Test Performance and Recovery of Underlying Conversion Probabilities

After selection, the Market-position GLM was refitted using the combined training and validation samples and evaluated on 10,000 independent test profiles. Its test log loss was 0.4382, AUC was 0.6714 and calibration slope was 0.969. The model predicted an average conversion rate of 18.11%, compared with 17.77% of customers who purchased in the simulated test outcomes. The underlying average probability used to generate those outcomes was 17.94%. The three figures differ because one is a model prediction, one is a realised simulated outcome and one is the known generating probability. Their closeness indicates reasonable aggregate calibration, although individual predictions still contain errors.

The highest predicted probability decile recorded a realised conversion rate of 34.4%, compared with 17.77% across the test sample. Customers ranked in this decile were therefore approximately 1.94 times as likely to convert as the average customer in the test sample. Expected calibration error was 0.0192, summarising the average difference between predicted and realised conversion across probability groups.

Because the experiment is synthetic, the fitted conversion probabilities can be compared with the probabilities used to generate customer purchases. On the independent test sample, the Market-position GLM had a probability MAE of 0.0453 and RMSE of 0.0598, corresponding to errors of 4.53 and 5.98 percentage points, respectively. Its Spearman correlation with the underlying probabilities was 0.843, compared with 0.668 for the Own-price GLM. This means that the Market-position model ranked customers more closely to their true simulated likelihood of purchase. The GAM had a slightly higher correlation of 0.850. However, these correlations were calculated using the underlying simulated conversion probabilities, which were reserved for independent evaluation rather than model selection. The GLM was therefore retained based on the predefined validation criteria.

These results show that incorporating competitive market-position information helped the models recover the price and rank effects specified in the simulated customer-choice mechanism.

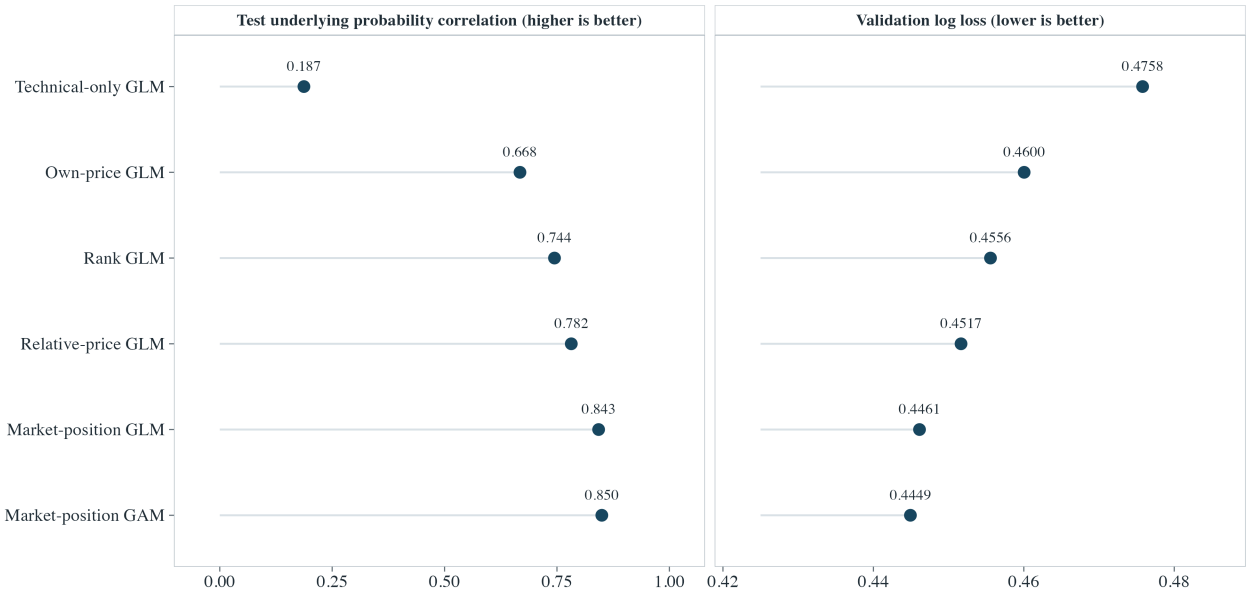


Figure 5: Comparison of six conversion models with different information sets. The left panel shows independent-test Spearman correlation with withheld underlying conversion probabilities (higher is better); the right shows validation binary log loss (lower is better). Rank and Relative-price GLMs are parallel extensions of the Own-price GLM; the Market-position GLM combines competitive predictors, while the GAM adds flexibility. Model selection uses validation performance, not independent-test correlation.

4.3 Pricing Results

4.3.1 Selection of the Targeted Pricing Structure

Eighteen predefined customer characteristics were tested as possible bases for targeted pricing. For each one, the optimiser selected premium adjustments that increased predicted conversion while satisfying the financial constraints in Section 3.6. **Relative conversion uplift** measures the percentage increase in expected purchases compared with technical pricing. **Expected profit retention** measures expected pre-expense underwriting profit as a percentage of the technical-pricing benchmark: 100% means unchanged profit, and 102% means 2% higher profit. Table 8 shows the five highest-conversion strategies that satisfied the constraints.

Table 8: Leading admissible targeting structures on the validation sample

Targeting variable	Relative conversion uplift	Expected profit retention
Vehicle-value band	22.58%	102.21%
Vehicle-age band	18.85%	102.34%
Vehicle segment	16.44%	102.65%
Unladen-weight band	11.94%	102.82%
Power band	11.70%	102.15%

Note: Results use fitted conversion probabilities and expected claims costs on the validation sample relative to technical pricing.

Vehicle-value band achieved the largest admissible conversion improvement, at 22.58% with profit retention of 102.21%. Although other segments produced higher profit retention, vehicle-value band was selected because the optimisation prioritised expected conversion once its financial requirements were satisfied.

The selected strategy divided customers into four vehicle-value quartiles and assigned multipliers of 0.96, 0.96, 1.01 and 1.01, respectively. Customers in the lower two quartiles were therefore offered premiums 4% below their technical premiums, while those in the upper two quartiles were offered premiums 1% above technical rates. Although the optimiser could choose four different multipliers, it selected only two distinct adjustments: one for the lower half and one for the upper half of vehicle values.

On the validation sample, total quoted premiums were 99.54% of the technical-pricing benchmark, expected profit retention was 102.21% and the expected claims ratio was 77.94%. The lowest expected profit retention among the selected quartiles was 90.12%, only slightly above the required 90%. This suggests that the requirement for each quartile to retain at least 90% of its benchmark expected profit limited the premium adjustments. These results show that vehicle-value grouping was the best of the tested options within this synthetic experiment; they do not show that vehicle value itself causes differences in customer price sensitivity.

4.3.2 Independent Test Evaluation

The validation-selected segmentation and multipliers were applied to the independent 10,000-profile test sample. Performance was evaluated using the underlying simulated customer-choice probabilities, which were withheld during model estimation and price selection.

With all customers quoted their technical premiums, expected conversion was 17.79%. The selected targeted strategy increased this to 22.03%. The difference is 4.24 percentage points, or a 23.82% increase relative to the original conversion rate. Across 10,000 independent test enquiries, this represents approximately 424 additional expected policies. Table 9 compares the remaining pricing and financial outcomes.

Table 9: Independent test sample comparison of technical and targeted pricing

Measure	Technical benchmark	Targeted strategy
Aggregate quoted rate level	100.00%	99.53%
Underlying expected conversion	17.79%	22.03%
Relative conversion uplift	—	23.82%
Expected policies written	1,778.8	2,202.5
Expected pre-expense underwriting profit	$399,158	$412,210
Expected profit retention	100.00%	103.27%
Expected claims ratio	76.92%	77.94%



Figure 6: Independent-test comparison of the technical-pricing benchmark and validation-selected targeted strategy across 10,000 simulated customer enquiries. The chart shows expected policies written and expected pre-expense underwriting profit, calculated using the withheld customer-choice probabilities. Targeted repricing adds approximately 424 expected policies and $13.1 thousand in expected pre-expense underwriting profit. Competing quotations are held fixed; profit excludes acquisition, operating, reinsurance and capital costs.

Three conversion rates are used for different purposes. The technical-pricing benchmark assumes that every customer is quoted their fitted technical premium, giving an expected conversion rate of 17.79%. The original quotations include randomly generated commercial adjustments, so their underlying expected conversion rate is 17.94%. The realised conversion rate of 17.77% is the proportion of customers who actually purchased in the simulated outcomes at those original quotations. Only the first of these is the benchmark for the repricing comparison.

The aggregate quoted rate level was 99.53%, meaning that the sum of premiums offered to all customers was approximately 0.47% lower than under technical pricing. Expected pre-expense underwriting profit rose from $399,158 to $412,210, a 3.27% improvement, or approximately $13.1 thousand. This measure weights premiums and fitted claims costs by the simulated probabilities of purchase. It excludes separately modelled operating, acquisition, reinsurance and capital costs, so it should not be interpreted as net insurer profit.

The expected claims ratio increased from 76.92% to 77.94%, but remained below the specified 80% ceiling. In other words, claims expenditure accounted for a slightly larger share of expected written premiums, even though expected profit increased. The two lower vehicle-value quartiles, which received 4% discounts, generated approximately 530.5 additional expected policies and $37.2 thousand in extra expected pre-expense underwriting profit. The two higher quartiles, which received 1% increases, generated approximately 106.7 fewer expected policies and $24.2 thousand less expected profit. The net gain was 423.8 expected policies and $13,053 in expected profit (approximately $13.1 thousand). The group figures are rounded separately, so their displayed values may not sum exactly to the net result.

For comparison, both uniform strategies selected a multiplier of 0.97, reducing every premium by 3%. On the independent test sample, both increased expected conversion by 40.40% relative to technical pricing and achieved expected profit retention of 121.55%. However, their total quoted rate level was 97%, and they were chosen to maximise expected profit without the targeted strategy's narrow 99.5%–100.5% rate-level constraint. Their larger improvements therefore should not be taken as evidence that uniform pricing is better under the same requirements: the strategies solve different optimisation problems.

4.4 Robustness and Uncertainty

4.4.1 Structural Sensitivity

Structural sensitivity analysis examined whether the principal findings remained consistent under alternative competitive, customer-demand and pricing assumptions. Ten predefined cases were evaluated, including the baseline. Competitive market-position information improved conversion prediction and recovery of underlying customer-choice probabilities relative to the Own-price GLM in all ten cases. Targeted repricing also increased expected conversion while retaining at least benchmark expected pre-expense underwriting profit in every case.

As shown in Figure 7, relative conversion uplift ranged from 6.51% under low price sensitivity to 32.15% under stronger price-threshold effects. Expected profit retention ranged from 100.32% to 110.67%. All ten scenarios improved expected profit, but the low-price-sensitivity scenario increased it by only 0.32%. This shows that repricing can produce much smaller financial gains when customers respond less strongly to premium changes. The agent-mediated demand scenario gave 28.75% relative conversion uplift and 109.47% profit retention, but this is the outcome of an assumed demand setting, not evidence about actual automated purchasing.

Tightening the expected claims-ratio ceiling to 78% did not change the selected baseline strategy because it already satisfied this requirement. The wide-pricing stress changed both the permitted range and the candidate-price grid, with relative conversion uplift of 8.96% and profit retention of 102.53%. This result should not be attributed solely to wider bounds because the grid resolution also changed.

Vehicle-value band remained the preferred targeting variable in all ten cases. The Market-position GLM was selected in eight cases and the GAM in two, indicating that the preferred segmentation basis was stable across these assumed scenarios even though the selected model family and premium multipliers varied.

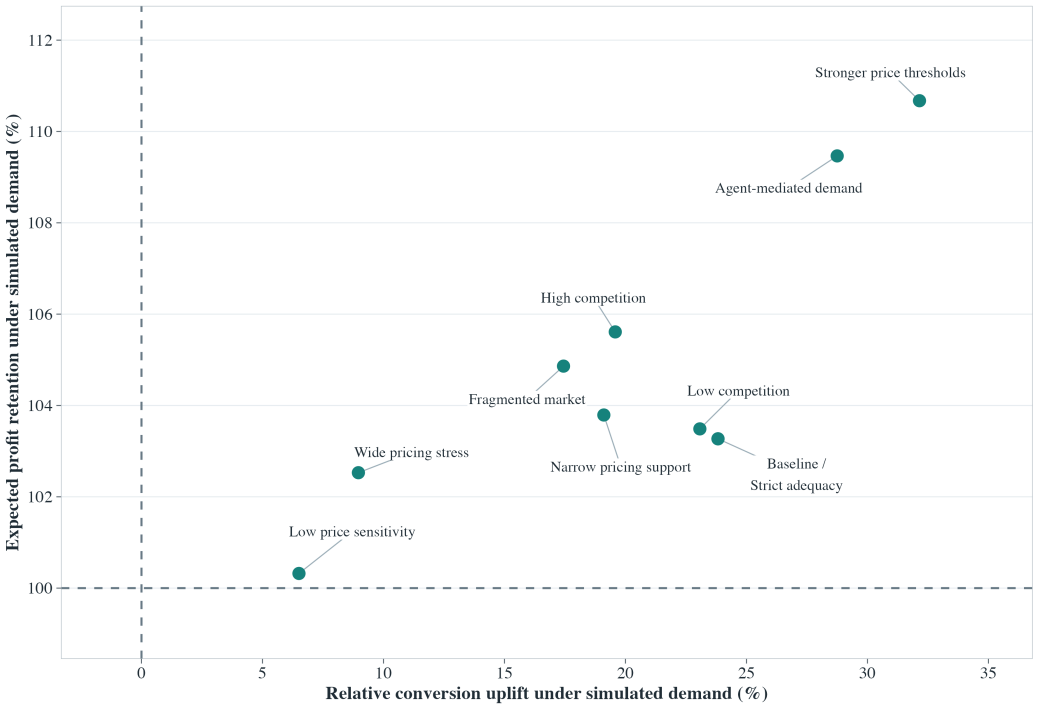


Figure 7: Structural sensitivity of targeted-pricing outcomes across ten simulated market, demand and pricing scenarios. Horizontal position indicates relative conversion uplift; vertical position indicates expected profit retention, both evaluated using withheld simulation demand probabilities. The horizontal dashed line at 100% denotes unchanged expected pre-expense underwriting profit; all ten cases lie above this threshold. Baseline and Strict adequacy coincide because they produce identical plotted outcomes. Wide-pricing stress changes both the search bounds and candidate-price grid.

4.4.2 Monte Carlo Regeneration

The complete simulation and modelling process was independently regenerated ten times to assess whether the results depended on the particular random observations in the baseline experiment. Empirical calibration inputs and structural assumptions remained fixed, while new portfolios, claims experience, competing quotations and conversion outcomes were generated for each replication.

The selected Market-position model outperformed the Own-price GLM in all ten replications. Improvements in independent test log loss ranged from 2.29% to 3.13%, averaging 2.73%. Recovery of underlying conversion probabilities also improved consistently, with a mean Spearman correlation gain of 0.177. These findings indicate that the predictive value of competitive information was not specific to the initial generated dataset.

Targeted repricing produced positive expected conversion and expected profit improvements in every replication. Relative conversion uplift ranged from 15.12% to 25.00%, averaging 22.93%, while expected profit retention ranged from 102.67% to 103.90%, with a mean of 103.10%. Vehicle-value band and the Market-position GLM were selected in all ten replications, although the technical-pricing specifications and individual premium multipliers varied.

All ten strategies kept total quoted premiums within 99.5%–100.5% of the technical-pricing benchmark on the validation samples used to select them. In one independent test sample, the resulting rate level was 99.47%, slightly below the lower bound. This is possible because the mix of test customers differs from the mix used for selection; the constraint was not imposed again after seeing the test data. Expected profit remained above the benchmark in all ten test samples.

An earlier exploratory experiment used 50,000 policies, including 10,000 quotation profiles and 2,000 independent test profiles. In that experiment, the Market-position GAM improved validation log loss by 0.51% relative to the corresponding GLM, just above the 0.5% selection threshold. In the 250,000-policy experiment, the improvement was 0.27%, and the Market-position GLM was selected in all ten Monte Carlo replications. Because the two experiments generated different observations, this comparison does not isolate the effect of sample size. It shows that the value of competitive information remained evident in the larger experiment, while the extra flexibility of the GAM was not consistently selected.

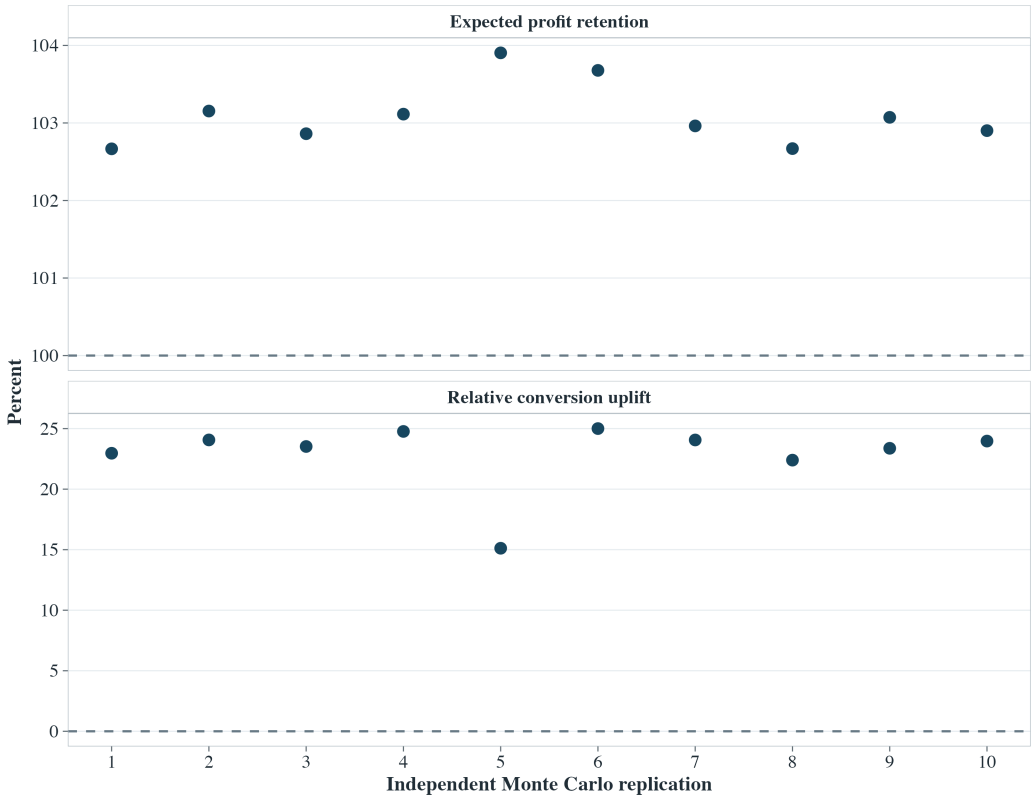


Figure 8: Results across ten independently regenerated synthetic datasets under fixed structural assumptions. Each point represents a complete modelling and repricing repetition. The panels show expected profit retention and relative conversion uplift under the generating choice mechanism; the Market-position GLM was selected in all ten replications. Dashed reference lines denote no profit change (100%) and zero conversion improvement. The figure assesses simulation stability, not real-market forecast uncertainty.

4.4.3 Paired Bootstrap

The paired bootstrap drew 5,000 samples with replacement from the 10,000 independent test profiles. Each resample used the same customers for the technical benchmark and the fixed targeted strategy, allowing a direct comparison. The estimated absolute conversion improvement was 4.24 percentage points, with a 95% percentile interval of 4.10–4.38 percentage points. Relative conversion uplift was 23.82% (95% interval: 23.03%–24.62%), and expected profit retention was 103.27% (95% interval: 102.81%–103.72%). Every resample showed positive expected conversion uplift and expected profit above the benchmark.

The targeted aggregate quoted rate level was 99.53%, with a 95% bootstrap interval of 99.48%–99.57%. Although the point estimate satisfied the 99.5% lower bound used in validation, some resamples fell below it. The selected strategy therefore has limited headroom against this constraint when the customer mix changes. These intervals describe uncertainty from resampling the test customers with the models, premiums and simulated demand mechanism held fixed. They do not measure uncertainty about actual Australian purchasing behaviour or about which strategy a new sample would select. The complete results are in Appendix C.

5 Discussion

The results show that knowing how the reference insurer's premium compares with competing quotations improves conversion predictions in the simulated market. Adding this information was more consistently useful than making the statistical model more complex. Targeted repricing also increased expected conversion and expected pre-expense underwriting profit in the baseline experiment. However, the sensitivity results show that better predictions do not guarantee large financial gains, particularly when customers respond less strongly to price changes. This section explains these findings, their practical implications and their limitations.

5.1 Competitive Information and Model Complexity

A central finding is that incorporating competitive market-position information provided greater and more consistent improvements in conversion prediction than introducing additional nonlinear modelling flexibility. The Own-price GLM outperformed the Technical-only GLM, demonstrating the value of incorporating the reference insurer's quoted premium alongside customer characteristics and technical risk. Further improvements occurred when relative price, quotation rank and other competitive-position measures were included.

This finding is consistent with the distinction between technical and commercial insurance pricing discussed by Taylor (1986), Guelman and Guillén (2014) and Spedicato et al. (2018). Technical pricing estimates the expected cost of insuring a customer, whereas commercial pricing must also account for the likelihood of purchase. In a comparison market, that likelihood may depend not only on the premium offered but also on the alternatives available. Consequently, the same premium can represent a favourable or unfavourable offer depending on competing quotations. Competitive-position variables provide information about this relationship that cannot be obtained from the reference insurer's premium alone.

The GLM–GAM comparison shows why a small improvement in predictive accuracy does not always justify a more complex model. The GAM had slightly lower validation log loss, but its improvement over the Market-position GLM was only 0.27%, below the predefined 0.5% threshold. The GLM was therefore selected for the baseline experiment, and it was also selected independently in all ten Monte Carlo replications. Under these selection rules, the extra flexibility of the GAM did not provide a large enough benefit to outweigh its complexity.

By comparison, adding competitive information improved prediction in every structural sensitivity case and every Monte Carlo replication. In this simulation, knowing the relative prices and ranks of competing quotations provided more consistent value than allowing a model to fit more complex patterns. A simpler model may therefore be preferable when it gives nearly the same predictions and is easier to interpret and validate.

Nevertheless, these results do not establish that GLMs are generally preferable to GAMs. Their relative performance depends on the relationships governing customer behaviour and the information available for estimation. In particular, competitive effects were built into the simulated customer-choice mechanism, making models that incorporate competitive information well suited to recovering those relationships.

5.2 From Predictive Performance to Pricing Decisions

A conversion model is useful for pricing only if it helps the insurer choose better premiums. A model can predict which customers are likely to buy at their original quotations without accurately predicting how their behaviour would change if those premiums were increased or reduced. Better predictive scores therefore do not automatically mean better pricing decisions.

This distinction is particularly important in competitive markets because changing a premium also changes its position relative to other offers. For example, a modest premium reduction may move the reference insurer from the second-cheapest to the cheapest quotation, potentially producing a larger response than an equivalent reduction that leaves its competitive rank unchanged. Evaluating alternative premiums therefore requires competitive-position variables to be recalculated rather than held fixed.

For each proposed premium, the framework recalculated the reference insurer's relative price, quotation rank and other measures of competition. The fitted conversion model then predicted demand at that premium and guided the pricing decision. After the strategy was chosen, it was assessed on the independent test sample using the purchase probabilities defined by the simulation. This distinction matters because a strategy that looks favourable according to fitted predictions might not perform as well under the process used to generate customer behaviour.

The targeted strategy increased expected conversion by 23.82% relative to technical pricing and expected pre-expense underwriting profit by 3.27%. These improvements were also present when the chosen premiums were evaluated against the underlying simulated purchase probabilities, rather than only the predictions used to select them. This supports the effectiveness of the strategy under the assumed demand relationships.

This ability to evaluate alternative prices is a methodological advantage of the synthetic framework. In observed insurance data, purchasing outcomes are generally available only for premiums actually offered, making it difficult to determine how the same customer would have responded to another price. The simulation provides the underlying probabilities needed to make these comparisons directly. Its value therefore lies in supporting controlled evaluation

of pricing methods, consistent with the emphasis on assessing synthetic data according to their intended analytical purpose (Jordon et al., 2024; Jones et al., 2025).

5.3 Interpretation of Targeted Repricing

Pricing involves a trade-off between how much the insurer earns on each policy and how many customers buy. Reducing a premium makes an offer more attractive, but lowers the amount received if that customer purchases. The discount improves total expected pre-expense underwriting profit only when the resulting increase in purchases is large enough to offset the smaller amount earned per policy.

The uniform-pricing strategies illustrated this trade-off by producing larger conversion improvements after reducing every quoted premium by 3%. However, these strategies were optimised using a different objective and were not subject to the narrow aggregate quoted rate-level restriction applied to targeted repricing. Their results therefore provide a useful comparison, but not a directly equivalent alternative under the same constraints.

The targeted strategy had a different objective: increase expected purchases while keeping the total premiums quoted across customers close to the technical-pricing benchmark and meeting the financial constraints. It used vehicle-value quartiles to choose premiums rather than applying the same reduction to everyone. The lower two quartiles received 4% discounts and the upper two received 1% increases.

Vehicle-value band was selected because it gave the largest predicted conversion increase among the eighteen eligible segmentation variables while satisfying the financial constraints. This does not mean that vehicle value directly causes price sensitivity. Rather, within the simulation, grouping customers by vehicle value produced a useful combination of risk, competing prices and predicted purchasing responses.

The validation results show why the optimiser chose different adjustments for lower- and higher-value vehicles. Premiums fell by 4% in each of the two lower-value quartiles. The model predicted approximately 54.32% and 53.84% more purchases in these groups, while expected profit increased to 127.43% and 127.54% of the respective technical-pricing benchmarks. In the two higher-value quartiles, premiums rose by 1%. Expected purchases fell to 86.44% and 86.31% of benchmark levels, and expected profit fell to 90.16% and 90.12%. The gains in the discounted groups were therefore partly offset by losses in the groups receiving increases.

Although the optimiser considered four vehicle-value quartiles, it selected only two distinct multipliers. This effectively divided customers into the lower and upper halves of the vehicle-value distribution. A separate two-group split was not evaluated as a candidate; testing whether that simpler structure produces comparable results would be a useful extension.

Both discounted quartiles had expected claims ratios of 80.13% on the validation sample, slightly above the 80% limit. This did not breach the optimisation rules because the limit applied to the combined results for all customers, not separately to each quartile. Each quartile was instead required to retain at least 90% of its benchmark expected profit. The result shows why insurers should examine the financial outcomes for each group even when the combined results meet the requirements.

These outcomes arise because a premium adjustment affects both the profit earned on each accepted policy and the number of customers expected to purchase. In the simulated market, responses to relative prices, quotation rankings and price thresholds can be strong enough for a discount to increase total expected profit.

The overall pricing results illustrate this trade-off. Following targeted repricing, the expected claims ratio increased from 76.92% to 77.94%, indicating that expected claims expenditure represented a slightly greater proportion of expected written premium. Nevertheless, the increase in expected conversion generated sufficient additional business to raise total expected pre-expense underwriting profit.

The three main measures answer different questions. Expected conversion measures how many customers are likely to purchase. The expected claims ratio measures how much of the premium earned from those policies is needed to cover fitted claims costs. Expected pre-expense underwriting profit measures the resulting premium income less fitted expected claims expenditure, allowing for the probability that each customer buys. A strategy can improve one measure while worsening another, so they must be assessed together.

Vehicle-value segmentation was selected in all ten structural sensitivity cases and all ten independent Monte Carlo replications, although the exact multipliers varied. This indicates that the preferred segmentation basis was consistent across the examined settings, although the exact premium adjustments remained sensitive to market and demand assumptions. In an operational setting, the suitability of the selected segments would require continued evaluation as customer behaviour, claims experience and competitive conditions change.

5.4 Robustness and Boundary Conditions

The structural sensitivity and Monte Carlo results qualify the baseline findings by examining changes in market conditions and random simulation outcomes. The most consistent result was the predictive value of competitive information: Market-position models outperformed the Own-price GLM in all ten sensitivity cases and all ten independent replications. This indicates that the predictive improvement was not confined to the original synthetic dataset or to a particular set of simulated observations.

Targeted repricing increased expected conversion and preserved at least benchmark expected pre-expense underwriting profit in all ten structural sensitivity scenarios. The size of the improvement, however, depended on the assumed demand response. When customers were less sensitive to prices, relative conversion uplift was only 6.51% and expected profit retention was 100.32%, meaning expected profit increased by just 0.32%. This shows that more accurate conversion modelling can support profitable repricing without guaranteeing a substantial financial gain.

The Monte Carlo results similarly showed favourable pricing outcomes in all ten independently generated portfolios, with mean relative conversion uplift of 22.93% and mean expected profit retention of 103.10%. Vehicle-value band was selected in every case. The Market-position GLM was also selected in all ten replications, although the fitted technical-pricing formulas and precise segment multipliers varied.

In one Monte Carlo replication, the test-sample aggregate quoted rate level was 99.47%, slightly below the 99.5% lower limit. The selected strategy had met the constraint on its validation sample, where the pricing decision was made. It was then applied unchanged to a different mix of customers in the independent test sample. The test result therefore shows that the same pricing strategy can change the total premiums quoted relative to technical pricing when the mix of customers changes. It does not indicate an error in the optimisation procedure.

The paired bootstrap tested how the findings changed when different combinations of the same 10,000 test profiles were sampled. Across 5,000 resamples, the 95% percentile interval for absolute expected conversion uplift was 4.10–4.38 percentage points, while expected profit retention was 102.81%–103.72%. Both intervals were favourable. The interval for total quoted premiums, however, was 99.48%–99.57% of the benchmark, crossing the 99.5% lower bound. This shows that the financial gains were stable when the test customers were resampled, but the total premiums offered could fall just below the permitted range. The resamples held the pricing rule and assumed customer behaviour fixed, so they do not measure uncertainty in actual Australian market demand.

5.5 Practical Implications

The principal practical implication is that conversion modelling should complement, rather than replace, technical pricing. Expected claims costs provide the actuarial basis for assessing premium adequacy, while conversion models estimate the likelihood that customers will accept the premiums offered. Considering both components allows pricing decisions to account for customer acquisition and expected pre-expense underwriting profit.

For insurers operating in comparison-based markets, competitor quotations may provide useful information beyond that contained in their own prices. Measures such as relative price and quotation rank describe the alternatives available to individual customers and may help explain differences in purchasing behaviour. However, obtaining these benefits in practice would depend on the availability, reliability and representativeness of competitive quotation data.

The modelling results also suggest that improving the information available to a conversion model may be as important as adopting more sophisticated statistical techniques. In the simulated environment, a Market-position GLM captured most of the predictive improvement achieved by the GAM, while offering a simpler modelling structure. This supports beginning with interpretable approaches and introducing greater flexibility when it produces meaningful improvements in predictive performance or pricing outcomes.

The optimisation results further emphasise that customer acquisition should not be evaluated in isolation. Premium reductions may attract additional business while weakening claims adequacy or reducing profit within particular customer groups. Practical pricing decisions would therefore require clear financial limits and checks on both the combined results and the outcomes for each customer group.

Governance and customer fairness are also important considerations. A statistically effective segmentation characteristic is not necessarily appropriate for insurance pricing, particularly if its use produces unjustified differences between customer groups. Any operational application would require assessment of regulatory obligations, explainability, discrimination risks and customer outcomes, consistent with the considerations discussed by the Australian Human Rights Commission and Actuaries Institute (2022).

Finally, the study demonstrates how synthetic experimentation can support the development of pricing methods when proprietary competitor quotations and conversion outcomes are unavailable. The framework allows researchers to test modelling choices, optimisation constraints and evaluation procedures before applying them to observed insurance data. Its practical value is therefore methodological: it provides a controlled and reproducible approach for investigating demand-aware pricing, rather than premium adjustments that insurers could adopt directly.

5.6 Limitations

The principal limitation is that customer purchasing behaviour is simulated rather than observed. The customer-choice mechanism explicitly incorporates responses to relative price, quotation ranking and competitive thresholds. The improvements achieved by Market-position models therefore demonstrate their ability to recover relationships built into the simulation. They do not establish that Australian motor insurance customers exhibit the same responses, and the reported conversion improvements should not be interpreted as empirical estimates of benefits available in the Australian market.

The study also cannot determine how much of the 3.27% expected profit improvement was caused specifically by using competitor information. The Market-position GLM predicted conversion more accurately than the Own-price GLM, but the study did not compare targeted pricing strategies based on both models under the same objective and constraints. It therefore shows that the selected Market-position model supported better pricing than the technical benchmark, not the additional profit that would be earned solely from access to competitor quotations.

Technical-pricing models were selected using the lowest validation deviance, while the conversion GAM needed at least a 0.5% improvement over the corresponding GLM. This difference in the treatment of model complexity is a limitation of the experimental design. In particular, the chosen frequency specification improved validation deviance by only approximately 0.006% over the motive-power alternative. Such a small difference should not be interpreted as robust evidence that the additional physical vehicle specifications improve prediction.

The synthetic competitive market is also simplified. Competing insurers generate premiums from a common fitted technical-pricing reference rather than separate claims histories and independently estimated risk models. Consequently, the experiment does not fully represent differences in insurers' assessments of the same risk or the adverse-selection effects that those differences could create. Actual insurers may also differ in underwriting objectives, expenses and target returns. Competing quotations are held fixed during repricing, so the framework does not capture strategic responses to the reference insurer's decisions.

Although Australian public data inform the synthetic portfolio, the generated policies do not represent an observed insurer's portfolio. Some demographic, vehicle and geographic characteristics are calibrated to published statistics, while other underwriting relationships and risk adjustments rely on modelling assumptions. The technical-pricing results consequently demonstrate the recovery of expected claims costs within the constructed environment, rather than establishing the accuracy of the assumed risk relationships in an operational insurance market.

The scope of the pricing experiment introduces further limitations. Premium adjustments are restricted to relatively small changes around technical pricing, and the underlying customer-choice mechanism is assumed to remain stable when these changes are applied. The findings therefore provide limited evidence about larger premium adjustments or markets in which customer behaviour changes substantially over time.

The analysis also focuses on new-business conversion rather than renewal retention. Existing policyholders may respond differently to premium changes because of switching costs, familiarity with their insurer and other behavioural factors. Similarly, the customer-choice model represents differences in premiums and stylised insurer preferences without explicitly modelling variation in coverage, service quality, reputation or optional benefits.

The financial evaluation is restricted to expected pre-expense underwriting profit. Although the technical premium includes a general 30% loading, the optimisation does not separately model operating expenses, acquisition costs, reinsurance, capital charges, taxation or investment income. The reported profit improvements therefore do not represent changes in net insurer profitability or establish the solvency implications of the selected strategy.

The independent pricing evaluation used the simulation's underlying purchasing probabilities, but calculated expected claims costs using the fitted technical-pricing models rather than the known claims-cost parameters. As a result, the expected profit estimates can still be affected by technical-pricing error, even though the conversion response is evaluated against the generating mechanism.

Finally, the robustness analyses cannot eliminate uncertainty arising from unexamined market conditions. Structural sensitivity evaluates a predefined set of alternative assumptions, but does not comprehensively examine their interactions. The ten Monte Carlo replications provide a useful assessment of stability across independently generated datasets, although they are insufficient for precise estimation of simulation uncertainty. The paired bootstrap evaluates variation across customer profiles while holding model selection, pricing decisions and the underlying demand relationships fixed.

These procedures strengthen confidence in the internal reliability of the findings but do not establish their external validity. Applying the framework to an operational insurance market would require observed quotation and conversion data to validate customer-demand relationships, empirical assessment of competitor pricing and a more comprehensive financial and governance evaluation.

6 Conclusion

This dissertation examined whether a synthetic quote market could help evaluate conversion models and pricing decisions when actual competitor quotations and unsuccessful purchase outcomes are unavailable. The study generated 250,000 synthetic personal motor policies across 2021–2025. The 50,000 customer profiles from 2025 were used to simulate quotations from the reference insurer and six competitors, followed by customer purchasing outcomes. Technical pricing established the claims-cost benchmark; conversion models predicted purchasing probabilities; and constrained optimisation selected alternative premiums. The selected decisions were then tested against purchasing probabilities withheld during model development.

The first research question asked whether a coherent competitive quotation environment could be generated synthetically. The simulation produced consistent links between policy risks, technical premiums, competing quotations, price rankings and purchasing outcomes. Average expected market participation was 78.00%, compared with 78.28% of customers purchasing in the simulated outcomes. For the reference insurer, expected and realised conversion were 18.00% and 18.11%, respectively. Separately, technical-pricing validation showed that fitted claims costs preserved much of the underlying risk ranking, with a Spearman correlation of 0.807. These results show that the framework produced the intended relationships and closely met its calibration targets.

The second research question asked whether competitor information improved conversion prediction and supported better pricing decisions. Including the reference insurer's own quotation improved predictions over using risk characteristics alone. Adding competitor prices, relative price and quotation rank improved them further. On the independent test sample, the selected Market-position GLM achieved a Spearman correlation of 0.843 with the underlying conversion probabilities, compared with 0.668 for the Own-price GLM. The GAM was slightly more accurate on this measure, but its gain in validation log loss was below the predefined threshold for choosing the more complex model. On the validation sample, adding competitive information reduced log loss by 3.02% relative to the Own-price GLM, compared with an additional 0.27% reduction from the GAM's greater flexibility. The gain from competitor information was therefore much larger than the gain from the GAM's extra flexibility.

Using the selected Market-position GLM, targeted repricing increased expected conversion from 17.79% to 22.03%. This is an increase of 4.24 percentage points, or 23.82% relative to the technical-pricing benchmark. Across 10,000 independent test enquiries, the strategy produced approximately 424 additional expected policies and 3.27% more expected pre-expense underwriting profit, equivalent to approximately $13.1 thousand. Total quoted premiums remained close to the benchmark, and the expected claims ratio stayed below 80%. These results were calculated using the simulation's withheld purchase probabilities; they are expected outcomes rather than realised sales or net insurer profit.

The main findings persisted when the experiment was repeated. Competitor information improved conversion prediction in all ten structural sensitivity scenarios and all ten independently regenerated Monte Carlo runs. In each case, targeted repricing increased expected conversion and preserved at least benchmark expected pre-expense underwriting profit. The benefits were smaller when customers were assumed to respond less strongly to price. The Market-position GLM and vehicle-value targeting structure were selected in all ten Monte Carlo replications. These results support the stability of the findings under the assumptions tested.

The conclusions are limited by the assumptions used to generate customer purchases and competing premiums. These behaviours were simulated, not measured in Australian insurance markets. The profit calculation also excludes separate allowances for operating expenses, acquisition costs, reinsurance and capital. The reported improvements should

therefore be treated as evidence about the performance of the framework within the experiment, not as forecasts of results an Australian insurer could achieve.

The main contribution is a reproducible method for studying competitive insurance pricing when quotation and conversion data are missing. It connects claims-cost modelling, competing quotes, purchase probabilities and constrained pricing in one experiment. By hiding the generating purchase probabilities during model fitting and price selection, then using them only for independent evaluation, the framework can assess whether recommended premiums perform as intended under the simulated customer-choice process.

6.1 Further Work

The most important extension would be to calibrate and validate the framework using observed competitor quotations and customer conversion outcomes. Such data would allow the assumed relationships between premiums, competitive position and purchasing behaviour to be assessed against actual market experience.

A further study should compare targeted strategies based on the Own-price GLM and Market-position model using the same pricing objective and financial constraints. This would show how much extra conversion or expected profit is attributable specifically to competitor information, rather than only showing that the Market-position model predicts conversion more accurately.

Future research could also develop a more detailed representation of insurance competition. Rather than generating competing quotations from a common technical-premium reference, each competitor could be assigned an independent risk model, pricing strategy and financial objective. Allowing insurers to respond to one another's premium changes would provide a more realistic representation of strategic competition. Incorporating differences in coverage, service quality and insurer reputation could further improve the representation of customer purchasing decisions.

An additional extension would be to examine renewal business alongside new-business conversion. Existing policyholders may respond differently to premium changes because of customer loyalty, switching costs and previous experience with their insurer. Modelling these effects would allow the framework to investigate pricing decisions across a broader range of insurance customer relationships.

Finally, the robustness and financial evaluation could be extended. Multi-factor sensitivity analysis would allow interactions between competitive conditions, customer demand and pricing constraints to be examined, while additional Monte Carlo replications could provide more precise estimates of simulation uncertainty. Incorporating acquisition and operating expenses, reinsurance, capital requirements and broader profitability measures would also support a more comprehensive evaluation of pricing decisions. These developments, together with assessment of regulatory requirements and customer fairness, would improve the framework's suitability for practical insurance pricing applications.

References

Allianz Australia (2026). Understanding your car insurance premium. Online consumer information. Accessed 28 October 2026.

Allianz Australia Insurance Limited (2015). Allianz car insurance: Premium, excess, discount and options guide.

Australian Bureau of Statistics (2020). Survey of motor vehicle use, Australia, 12 months ended 30 June 2020.

Australian Bureau of Statistics (2021a). Australian statistical geography standard (ASGS), edition 3, July 2021–June 2026. Australian Bureau of Statistics. Includes Remoteness Areas, Postal Areas, Mesh Blocks and other geographic structures; accessed 30 September 2026.

Australian Bureau of Statistics (2021b). Electric passenger vehicle use, experimental estimates. Australian Bureau of Statistics. Published 26 November 2021; estimates for the 12 months ended 30 June 2020.

Australian Bureau of Statistics (2022). 2021 Census DataPacks. Australian Bureau of Statistics. General Community Profile and geographic DataPacks; released 12 April 2022; accessed 30 September 2026.

Australian Bureau of Statistics (2026). Consumer price index, Australia, December 2025. Australian Bureau of Statistics. Reference period December 2025; includes historical CPI expenditure-class series; released 28 January 2026.

Australian Competition and Consumer Commission (2014). The comparator website industry in Australia. Technical report, Australian Competition and Consumer Commission.

Australian Competition and Consumer Commission (2024). Insurance monitoring report 2024. Technical report, Australian Competition and Consumer Commission.

Australian Government (2026). Green vehicle guide. Online vehicle search and comparison database. Vehicle make, model, variant, body, engine, transmission, drive and fuel-type information; accessed 30 September 2026.

Australian Human Rights Commission and Actuaries Institute (2022). Artificial intelligence and discrimination in insurance pricing and underwriting. Technical report, Australian Human Rights Commission and Actuaries Institute, Australia.

Australian Institute of Health and Welfare (2005). Rural, regional and remote health: Indicators of health. Technical Report PHE 59, Australian Institute of Health and Welfare.

Australian Securities and Investments Commission (2024). Beware the gap: Governance arrangements in the face of AI innovation. Report 798, Australian Securities and Investments Commission, Australia.

Avanzi, B., Taylor, G. C., Wang, M., and Wong, B. (2021). SynthETIC: An individual insurance claim simulator with feature control. *Insurance: Mathematics and Economics*, 100:296–308.

Berry, S., Levinsohn, J., and Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890.

Boonen, T. J., Pantelous, A. A., and Wu, R. (2018). Non-cooperative dynamic games for general insurance markets. *Insurance: Mathematics and Economics*, 78:123–135.

Bureau of Infrastructure and Transport Research Economics (2025a). Australian infrastructure and transport statistics—yearbook 2025. Statistics, Bureau of Infrastructure and Transport Research Economics. Released 12 December 2025.

Bureau of Infrastructure and Transport Research Economics (2025b). Road vehicles, Australia, January 2025. Statistics 2025-108, Bureau of Infrastructure and Transport Research Economics. Released 26 September 2025.

CarExpert (2026). Price and specs. CarExpert. Australian vehicle price and specification database; accessed 30 September 2026.

Charlton, J. L., Koppel, S., D'Elia, A., Hua, P., St. Louis, R., Darzins, P., Di Stefano, M., Odell, M., Porter, M., Myers, A., Tuokko, H., and Marshall, S. (2019). Changes in driving patterns of older Australians: Findings from the Candrive/Ozcandrive cohort study. *Safety Science*, 119:219–226.

Compare the Market (2026). One tweak could save hundreds: Car insurance tip revealed.

de Jong, P. and Heller, G. Z. (2008). *Generalized linear models for insurance data*. Cambridge University Press.

Denuit, M., Maréchal, X., Pitrebois, S., and Walhin, J.-F. (2007). *Actuarial modelling of claim counts: Risk classification, credibility and bonus-malus systems*. John Wiley & Sons.

Dionne, G. and Vanasse, C. (1989). A generalization of automobile insurance rating models: The negative binomial distribution with a regression component. *ASTIN Bulletin*, 19(2):199–212.

Efron, B. and Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

El Emam, K., Mosquera, L., and Hoptroff, R. (2020). *Practical synthetic data generation: Balancing privacy and the broad availability of data*. O'Reilly Media.

Elmachtoub, A. N. and Grigas, P. (2022). Smart "predict, then optimize". *Management Science*, 68(1):9–26.

Emms, P., Haberman, S., and Savoulli, I. (2007). Optimal strategies for pricing general insurance. *Insurance: Mathematics and Economics*, 40(1):15–34.

Federal Chamber of Automotive Industries (2024). VFACTS motor vehicle classifications and definitions: Segmentation criteria. Federal Chamber of Automotive Industries. Accessed 30 September 2026.

Ferreira, J. and Minikel, E. (2012). Measuring per mile risk for pay-as-you-drive automobile insurance. *Transportation Research Record*, 2297(1).

Financial Conduct Authority (2014). Price comparison websites in the general insurance sector. Thematic Review TR14/11, Financial Conduct Authority, London, United Kingdom.

Frees, E. W. (2010). *Regression modeling with actuarial and financial applications*. Cambridge University Press.

Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.

Gschlößl, S. and Czado, C. (2007). Spatial modelling of claim frequency and claim size in non-life insurance. *Scandinavian Actuarial Journal*, 2007(3):202–225.

Guelman, L. and Guillén, M. (2014). A causal inference approach to measure price elasticity in automobile insurance. *Expert Systems with Applications*, 41(2):387–396.

Guven, S. and McPhail, M. (2013). Beyond the cost model: Understanding price elasticity and its applications. *Casualty Actuarial Society E-Forum*, 1(Spring):1–29.

Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction*. Springer, 2 edition.

Henckaerts, R., Côté, M.-P., Antonio, K., and Verbelen, R. (2021). Boosting insights in insurance tariff plans with tree-based machine learning methods. *North American Actuarial Journal*, 25(2):255–285.

Insurance Council of Australia (2025). Motor insurance policy paper: A roadmap for reducing rising premiums. Technical report, Insurance Council of Australia.

Jones, H., Lim, B., and Walsh, T. (2025). Not real, but useful: The actuary's guide to synthetic data. Canadian Institute of Actuaries. Accessed 2026-04-16.

Jordon, J., Szpruch, Ł., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S. N., and Weller, A. (2024). Synthetic data: What, why and how? Technical report, The Royal Society. Commissioned by the Royal Society.

Kleijnen, J. P. C. (1988). Analyzing simulation experiments with common random numbers. *Management Science*, 34(1):65–74.

Kleijnen, J. P. C. (2015). *Design and Analysis of Simulation Experiments*. Springer, 2 edition.

Klugman, S. A., Panjer, H. H., and Willmot, G. E. (2012). *Loss Models: From Data to Decisions*. Wiley Series in Probability and Statistics. John Wiley & Sons, Hoboken, NJ, 4 edition.

Mayorga, W. and Torres, D. (2017). A practical model for pricing optimization in car insurance. In *ASTIN and AFIR/ERM Colloquia*, Panama City, Panama. Conference paper.

McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*. Chapman and Hall, London, 2 edition.

Mourdoukoutas, F., Boonen, T. J., Koo, B., and Pantelous, A. A. (2021). Pricing in a competitive stochastic insurance market. *Insurance: Mathematics and Economics*, 97:44–56.

Mourdoukoutas, F., Boonen, T. J., Pantelous, A. A., and Taylor, G. C. (2025). Competitive insurance pricing strategies for multiple lines of business: A game-theoretic approach. *Scandinavian Actuarial Journal*, 2025(8):741–767.

Mozo (2024). Mozo car insurance report 2024. Online report.

Nowok, B., Raab, G. M., and Dibben, C. (2016). synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software*, 74(11):1–26.

Ohlsson, E. and Johansson, B. (2010). *Non-life insurance pricing with generalized linear models*. Springer.

Omari, C., Nyambura, S., and Mwangi, J. (2018). Modeling the frequency and severity of auto insurance claims using statistical distributions. *Journal of Mathematical Finance*, 8:137–160.

Pantelous, A. A. and Passalidou, E. (2013). Optimal premium pricing policy in a competitive insurance market environment. *Annals of Actuarial Science*, 7(2):175–191.

Peng, R. D. (2011). Reproducible research in computational science. *Science*, 334(6060):1226–1227.

Robert, C. P. and Casella, G. (2004). *Monte Carlo Statistical Methods*. Springer, 2 edition.

Royal Automobile Club of Queensland (2025). Running costs 2025. RACQ. Vehicle purchase-price, operating-cost and residual-value information; accessed 30 September 2026.

Ryan, G. A., Legge, M., and Rosman, D. (1998). Age related changes in drivers' crash risk and crash type. *Accident Analysis and Prevention*, 30(3):379–387.

Saltelli, A. and Annoni, P. (2010). How to avoid a perfunctory sensitivity analysis. *Environmental Modelling & Software*, 25(12):1508–1517.

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., and Tarantola, S. (2008). *Global sensitivity analysis: The primer*. John Wiley & Sons, Chichester, UK.

Spedicato, G., Dutang, C., and Petrini, L. (2018). Machine learning methods to perform pricing optimization: A comparison with standard generalized linear models. *Variance*, 12(1):69–89.

Taylor, G. C. (1986). Underwriting strategy in a competitive insurance environment. *Insurance: Mathematics and Economics*, 5(1):59–77.

Train, K. E. (2009). *Discrete choice methods with simulation*. Cambridge University Press, 2 edition.

Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., and Steyerberg, E. W. (2019). Calibration: The Achilles heel of predictive analytics. *BMC Medicine*, 17:230.

Ver Hoef, J. M. and Boveng, P. L. (2007). Quasi-Poisson vs. negative binomial regression: How should we model overdispersed count data? *Ecology*, 88(11):2766–2772.

Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R*. Chapman and Hall/CRC, 2 edition.

Xie, Y., Allaire, J. J., and Grolemund, G. (2018). *R Markdown: The Definitive Guide*. Chapman and Hall/CRC, Boca Raton, Florida.

Yeo, A. C., Smith, K. A., Willis, R. J., and Brooks, M. (2001). Modelling the effect of premium changes on motor insurance customer retention rates using neural networks. In Alexandrov, V. N., Dongarra, J. J., Juliano, B. A., Renner, R. S., and Tan, C. J. K., editors, *Computational Science – ICCS 2001*, volume 2074 of *Lecture Notes in Computer Science*, pages 390–399. Springer, Berlin, Heidelberg.

A Document Conventions

The tables below define the principal abbreviations and mathematical symbols used in the dissertation. They distinguish the underlying relationships used to generate synthetic claims and customer decisions from the fitted models used to estimate expected claims costs, predict conversion and select premiums.

A.1 Acronyms and Abbreviations

Table A.1: Acronyms and abbreviations

Abbreviation	Definition
ABS	Australian Bureau of Statistics
AUC	Area under the receiver operating characteristic curve
BITRE	Bureau of Infrastructure and Transport Research Economics
CV	Coefficient of variation
DGP	Data-generating process
FCAI	Federal Chamber of Automotive Industries
GAM	Generalised additive model
GLM	Generalised linear model
MAE	Mean absolute error
MC	Monte Carlo
O/E	Observed-to-expected ratio
RMSE	Root mean squared error

Two distinct data splits are used. Technical-pricing candidates are fitted on the 2021–2023 cohorts, selected on the 2024 cohort, refitted on 2021–2024, and evaluated on the 2025 cohort. The 2025 quotation cohort is separately divided into a **training sample** (30,000 profiles), **validation sample** (10,000 profiles) and **test sample** (10,000 profiles) for conversion modelling and pricing evaluation. The terms training, validation and test refer to this quotation-level split unless the technical-pricing cohort is specifically identified.

A.2 Summary of Notation

Table A.2: Principal claims and technical-pricing notation

Symbol	Definition
i	Policy or quotation-profile index
j	Reported-claim index
N_i	Number of **reported** claims for policy i in the synthetic claims DGP
C_i	Number of **paid claims** for policy i: reported claims with insurer payment greater than zero
λ_i	Underlying annual **reported-claim** frequency
R_i^A, R_i^L, R_i^K	Driver-age, licence-experience and annual-kilometres frequency relativities
$G_{\ell(i)}$	Shared geographic frequency effect for synthetic locality $\ell(i)$
U_i	Policy-specific Gamma frequency frailty
T_{ij}	Indicator that reported claim j is a **write-off**
X_{ij}	Gross claim cost before deducting voluntary excess
e_i	Policyholder voluntary excess
$Y_{ij} = \max(X_{ij} - e_i, 0)$	Insurer payment for a reported claim, possibly zero
B_i	Reference new-vehicle price on the 2025 price basis
V_i, V_0	Simulated depreciated vehicle value and reference vehicle value on the 2025 price basis

Symbol	Definition
s_i	Underlying expected insurer payment **per reported claim**, including zero-payment claims
$\mu_i = \lambda_i s_i$	Underlying expected annual insurer claims cost
$\widehat{\lambda}_i, \widehat{s}_i$	Fitted annual paid-claim frequency and mean insurer payment per paid claim
$\widehat{\mu}_i = \widehat{\lambda}_i \widehat{s}_i$	Fitted expected annual claims cost
$\pi_i = 1.30\widehat{\mu}_i$	Fitted technical premium, including the 30% loading
E_i	Policy exposure; one year per simulated profile
A_i, L_i, K_i	Driver age, licence experience in years and annual kilometres

Table A.3: Principal quote, demand and optimisation notation

Symbol	Definition
k, h	Competitor index and insurer-alternative index, respectively
P_i	Initial reference insurer quotation; also used for a proposed premium when the pricing context is explicit
Q_{ik}	Quotation returned by competitor k
$Q_i^{\min}$	Cheapest **competing** quotation returned for profile i
ω_i	Initial reference insurer commercial-price multiplier, Uniform$(0.95, 1.05)$
$\alpha_k, \delta_{ik}, \varepsilon_{ik}$	Competitor log-price stance, segment adjustment and stochastic quote variation
$\mathcal{K}_i$	Set of competitors returning a quote to customer i
$\mathcal{J}_i$	Complete available insurer choice set, including the reference insurer
$RP_i = P_i/Q_i^{\min}$	Initial reference insurer premium relative to the cheapest competitor
R_i	Initial reference insurer quote rank; one denotes the cheapest quotation
q_i	Underlying probability of purchasing from any available insurer in the comparison market
w_{ih}	Conditional probability of choosing insurer alternative h after deciding to purchase
w_i	Conditional probability of choosing the reference insurer, given a market purchase
U_{ih}	Customer-choice utility for alternative h
β_i	Customer-specific price sensitivity in the choice DGP
$p_i = q_i w_i$	Underlying reference insurer conversion probability at the initial quotation
D_i	Realised binary reference insurer conversion outcome
$\widehat{p}_i(P_i)$	Fitted conversion probability at proposed reference insurer premium P_i
$p_i(s)$	Underlying reference insurer conversion probability under pricing strategy s
m_i	Multiplier applied to fitted technical premium during repricing
x_{gm}	Binary indicator that price multiplier m is selected for segment g
RL	Aggregate quoted rate level: total proposed premiums divided by total technical premiums
$\widehat{M}$	Model-expected pre-expense underwriting profit, $\sum_i \widehat{p}_i(P_i)(P_i - \widehat{\mu}_i)$
M_s	Expected pre-expense underwriting profit under strategy s, using underlying simulated conversion probabilities and fitted claims costs
CR	Expected claims expenditure divided by expected written premium, both weighted by conversion probability

Claims definitions: A reported claim includes claims whose gross cost is fully absorbed by the policyholder's excess. A paid claim results in a strictly positive insurer payment. The simulation uses reported-claim frequency λ_i and expected payment per reported claim s_i, while the fitted technical-pricing model uses paid-claim frequency $\widehat{\lambda}_i$ and average paid-claim severity $\widehat{s}_i$. Both approaches estimate annual insurer claims expenditure.

Financial definitions: Expected pre-expense underwriting profit is the sum, across customer quotations, of the conversion probability multiplied by the proposed premium less fitted expected claims cost. Absolute expected profit uplift is this amount under the selected strategy minus the technical benchmark; relative uplift is the difference divided

by benchmark profit; expected profit retention is the strategy-to-benchmark ratio. These measures omit acquisition, operating, reinsurance and capital costs and are not net insurer profit. The aggregate quoted rate level instead considers all offered premiums, including quotations that are not accepted. The cheapest competing quotation used for the reference insurer's relative price may differ from the cheapest quotation across the complete choice set used in insurer-choice utility.

B Calibration and Simulation Assumptions

The 250,000-policy synthetic portfolio comprises 50,000 independent one-year new-business policies for each calendar cohort from 2021 to 2025. Australian public data inform selected observable characteristics; unobserved claims heterogeneity, competitor behaviour and customer choice are governed by explicit assumptions. Empirical inputs are not evidence that the joint synthetic population reproduces the insured Australian market.

B.1 Empirical Inputs and Portfolio Calibration

Table B.1: Empirical calibration register

Component	Source or basis	Implementation and qualification
State and registered-vehicle composition	BITRE annual *Road Vehicles Australia* statistics (Bureau of Infrastructure and Transport Research Economics, 2025b)	Annual State and vehicle manufacture-year/make/model calibration; probabilities vary by cohort where annual information is available.
Remoteness conditional on State	BITRE vehicle registration data; ABS ASGS Edition 3 and 2021 Census (Bureau of Infrastructure and Transport Research Economics, 2025b; Australian Bureau of Statistics, 2021a, 2022)	Five remoteness categories; overlapping postal areas allocated with dwelling weights. The empirically derived **2025** $P(\text{Remoteness} \mid \text{State})$ is held constant across all five cohorts, while annual State shares vary.
Driver ages	BITRE *Australian Infrastructure and Transport Statistics Yearbook 2025* (Bureau of Infrastructure and Transport Research Economics, 2025a)	Drawn from the national licence-holder structure. **50%** of the source 15–19 band is retained to represent eligible 18–19-year-olds; this frozen approximation differs from the 40% implied by equal numbers at each single age.
Vehicle segment and motive power	Green Vehicle Guide; FCAI classification; BITRE registrations (Australian Government, 2026; Federal Chamber of Automotive Industries, 2024; Bureau of Infrastructure and Transport Research Economics, 2025b)	Curated make/model/year classification and manufacture-year-conditional motive-power distribution.
Physical vehicle specifications	CarExpert vehicle specifications and compatibility-based donor matching (CarExpert, 2026)	Engine capacity, cylinder count, power, torque, transmission and unladen weight are assigned using a CarExpert-based specification lookup and compatibility-based donor matching. Unmatched vehicle records are handled using the documented fallback rules.
Vehicle value	RACQ *Vehicle Running Costs* (Royal Automobile Club of Queensland, 2025)	Reference new prices and five-year value ratios; interpolated/extrapolated depreciation is a simulation assumption, with a $3,000 floor and new-price ceiling.

Component	Source or basis	Implementation and qualification
Vehicle usage	ABS (2020) *Survey of Motor Vehicle Use* and ABS (2021) *Electric passenger vehicle use* (Australian Bureau of Statistics, 2020, 2021b)	Mean annual distance calibrated to **11,100 km** based on Australian passenger vehicles for the 12 months ended 30 June 2020. Petrol (**10,500 km**), diesel (**13,900 km**) and battery-electric (**11,100 km**) evidence informs motive-power relativities. Hybrid and other-fuel relativities are neutral assumptions; the empirical battery-electric value is applied to the combined battery/fuel-cell category. Driver-age/remoteness effects and lognormal CV **0.45** are assumed.
Calendar-year monetary levels	ABS Consumer Price Index motor-vehicle and maintenance/repair series (Australian Bureau of Statistics, 2026)	Write-off values and partial-claim costs respectively converted to the common 2025 pricing basis.
Underwriting characteristics	Explicit simulation distributions	Voluntary excess, parking, financing, modification, pre-existing damage and regular-driver status; not observed insurer prevalences.

Approximately 450 **synthetic localities** receive a shared random frequency effect. These are simulation groupings, not observed postcode-level risk relativities. The national annual-kilometre benchmark is taken from the ABS *Survey of Motor Vehicle Use* for 2019–20 (Australian Bureau of Statistics, 2020) and applied to each synthetic cohort; it is not an estimate of average driving in every year from 2021 to 2025.

Table B.2: Selected scalar generation and claims assumptions

Parameter	Baseline	Interpretation
Policy count / cohorts	250,000; 2021–2025	50,000 independent policies per year
Annual policy exposure	1 year	Applied to each profile
Synthetic geographic groups	About 450	Shared local frequency heterogeneity
Driver-age effect	$\min\{2.20, \exp[0.00055(A_i - 50)^2]\}$	Minimum at age 50; cap 2.20
Licence effect	$\exp[-0.020 \min(L_i, 15)]$	Reduction capped at 15 years licensed
kilometre effect	$(K_i/11{,}100)^{0.55}$	Diminishing frequency effect of annual kilometres
Geographic frequency effect	Lognormal mean 1; log-SD 0.15	Shared across each synthetic locality
Individual frequency frailty	Gamma mean 1; variance 0.20	Unobserved policy risk
2025 claimant target	14%	Probability of at least one **reported** claim
Write-off claim cost share	25%	Share of total **insurer-paid claim expenditure**, not share of claim counts
Average insurer payment per reported claim	Approximately \$5,202	**Includes zero-payment claims absorbed by excess**; 2025 basis
Partial-claim gross severity	Lognormal; CV 1.00	Conditional on a claim not being a write-off
Partial-claim vehicle-value exponent	0.35	Relativity $(V_i/V_0)^{0.35}$
Mean annual kilometres	Approximately 11,100	Based on the ABS 2019–20 passenger-vehicle average; imposed in each synthetic policy year, subject to rounding and bounds
Annual-kilometres CV	0.45	Initial lognormal simulation assumption
Vehicle-age cap / value floor	30 years / \$3,000	Synthetic portfolio/value constraints
Technical-premium loading	30%	$\pi_i = 1.30\hat{\mu}_i$; implies fitted-basis claims ratio 76.92%

Claims calibration evidence: The Insurance Council of Australia's *Motor Insurance Policy Paper* (2025, pp. 7, 12 and 17) reports approximately one in seven policyholders making a claim annually, an average claim size of $5,202 as at June 2024, and write-offs accounting for approximately one-quarter of claims expenditure. These figures inform the simulation but do not determine its parameters uniquely. Module 01 applies the $5,202 target on a **2025 price basis** to the average insurer payment per **reported claim**, including claims with zero insurer payment. The denominator and price basis do not necessarily match the published industry claim-size statistic. Accordingly, the study uses the industry figure as an indicative calibration target rather than claiming an exact like-for-like match.

Voluntary excess and claim calibration: Voluntary excesses are $400 (10%), $600 (25%), $800 (35%), $1,000 (20%) and $1,500 (10%). The insurer payment for a claim is $Y_{ij} = \max(X_{ij} - e_i, 0)$. The write-off probability and partial-claim severity scale are calibrated jointly to the average-payment and write-off-expenditure-share targets after allowing for excesses. Claims fully absorbed by excess remain in the reported-claim denominator but are excluded from the fitted paid-claim frequency and severity models.

Distinction between simulation and fitted premiums: Module 01 uses a separate internal scale target, the mean target technical premium = $1,052, for its underlying claims calibration. This is not the technical premium used to construct competing quotations or select pricing adjustments. The operative technical premium is $\pi_i = 1.30\hat{\mu}_i$, fitted in Module 02. Its reported mean is $1,028.85 across the full synthetic portfolio and $1,041.48 in the 2025 quotation cohort (Section 4.1).

B.2 Baseline Competitor Assumptions

Six stylised competitors are priced relative to the reference insurer's **fitted** technical premium, using $Q_{ik} = \pi_i \exp(\alpha_k + \delta_{ik} + \varepsilon_{ik})$. The log stance α_k is applied before segment adjustments and quote-specific variation.

Australian motor-insurance quotation research documents variation in premiums across providers, while insurers identify driver age, claims history, vehicle characteristics and vehicle value as factors influencing quoted prices (Mozo, 2024; Allianz Australia, 2026). The Australian Competition and Consumer Commission (2024) also highlighted that insurers may differ in their risk appetite, premium adjustments and willingness to return a quotation, although the detailed evidence on quotation availability primarily concerns property insurance. These findings motivate pricing stances, customer-segment adjustments and incomplete quotation availability. The archetypes, exact log-price adjustments, quote-return probabilities and quotation-variation parameters are intended to reproduce plausible differences between insurers' pricing approaches but remain explicit simulation assumptions.

Table B.3: Competitor stance and quote availability

ID	Archetype	Log stance α_k	Probability of returning a quote
C01	Value Direct	−0.055	96%
C02	Mainstream Alpha	−0.015	98%
C03	Mainstream Beta	+0.010	96%
C04	Premium Brand	+0.065	90%
C05	Young Driver Specialist	+0.025	92%
C06	Low-Risk Specialist	+0.015	91%

Table B.4: Competitor segment-appetite rules

Competitor	Condition	Log-price adjustment δ_{ik}
Mainstream Beta	Small Passenger segment	−0.020
Mainstream Beta	Sports/Performance segment	+0.020
Premium Brand	Vehicle value ≥ $45,000 and driver age 35–69	−0.020
Premium Brand	Driver age ≤ 24	+0.025
Young Driver Specialist	Driver age ≤ 29	−0.055
Young Driver Specialist	Driver age ≥ 55	+0.018
Low-Risk Specialist	No prior claims and at least three claim-free years	−0.045
Low-Risk Specialist	At least two prior claims	+0.030

Note: The specialist price adjustments represent relative deviations from the common fitted technical premium, not claims that young drivers or expensive vehicles have lower absolute premiums. For example, a young-driver specialist may apply a lower multiplier than other competitors to young drivers while still quoting a higher absolute premium for those drivers because of their greater estimated claims costs.

These conditions are evaluated in the order specified in Module 03 using an inline conditional expression; **if multiple conditions for the same competitor overlap, the first matching condition applies**, rather than summing the adjustments. The baseline adjustment scale is 1.00. Residual log-price noise has standard deviation **0.045**. Competitor premium multipliers are limited to **0.75**–**1.35** times reference technical premium, and at least **two** competitor offers are retained. The reference insurer's initial commercial multiplier is drawn independently from Uniform $(0.95, 1.05)$.

B.3 Baseline Customer-Demand Assumptions

The two-stage DGP first determines whether a customer purchases insurance through the simulated market and then which insurer is selected, conditional on purchase. The calibration targets are **78%** mean market participation (purchase from any available insurer) and **18%** mean reference insurer conversion.

Table B.5: Parameters of the baseline demand DGP

Parameter	Baseline value
Mean probability of market purchase	78%
Mean reference conversion probability	18%
Market-participation price sensitivity	0.75
Mean individual conditional-choice price sensitivity	7.00
Conditional-choice price-sensitivity CV	0.35
Rank-salience scale	1.00
Relative-price penalty threshold	1.08
Relative-price threshold penalty strength	0.65
Rank penalty begins beyond	Rank 4
Rank-threshold penalty strength	0.10
Latent taste-shock SD	0.35

Customer-specific price-sensitivity coefficients follow a lognormal distribution. The utility model ranks the premiums of all available insurers, including the reference insurer. Its relative-price ratio uses the cheapest premium across this **complete choice set**. This differs from the reference insurer's market-position variable RP_i, whose denominator is the cheapest **competing** quotation. A calibrated reference insurer utility constant establishes the 18% mean conversion target.

The market-participation linear predictor includes $-0.75 \log(\text{customer median quotation}/\text{portfolio median quotation})$ and a claims-history adjustment of $0.035 \min(\text{prior five-year claim count}, 3)$. Driver-age adjustments are -0.12 for ages up to 24, $+0.02$ for ages 25–34, $+0.10$ for ages 35–69, $+0.02$ for ages 70–79 and -0.06 for ages 80 and above. The intercept is calibrated to the 78% mean market-participation target.

Conditional insurer choice assigns baseline rank-utility bonuses of **0.55**, **0.26** and **0.10** at ranks one, two and three (multiplied by the scenario rank-salience scale). For an alternative with relative price r_{ih} and rank R_{ih}, the nonlinear penalties are

$$Q(r_{ih}) = c_Q \left[\frac{\max\{r_{ih} - r_0, 0\}}{0.05} \right]^2, \qquad K(R_{ih}) = c_K [\max\{R_{ih} - R_0, 0\}]^2,$$

where r_0, R_0, c_Q and c_K are the thresholds and penalty strengths in Table B.5. The baseline non-price utility is also insurer-specific, as shown below.

Table B.6: Baseline non-price insurer-choice utility adjustments

Insurer archetype	Customer condition	Non-price utility
Reference insurer	Before reference-brand calibration	0.00
Value Direct	All	−0.04
Mainstream Alpha	All	+0.08
Mainstream Beta	All	+0.04
Premium Brand	Vehicle value $\geq$ \$45,000 and age 35–69	+0.18
Premium Brand	Otherwise	+0.09
Young Driver Specialist	Age $\leq$ 29	+0.20
Young Driver Specialist	Otherwise	−0.04
Low-Risk Specialist	Zero prior claims and at least three claim-free years	+0.18
Low-Risk Specialist	At least two prior claims	−0.08
Low-Risk Specialist	Otherwise	+0.03

As in the competitor-appetite implementation, the choice utilities are evaluated in order, with the first applicable condition taking precedence. Normal latent preference shocks have baseline standard deviation 0.35. These utility adjustments are stylised behavioural assumptions, not estimated brand preferences from Australian consumers.

B.4 Structural Sensitivity Assumptions

Ten cases were considered in total: the baseline and nine alternatives. Each scenario changes a defined group of assumptions while retaining the remaining baseline structure.

Table B.7: Alternative customer-demand parameters

Parameter	Baseline	Low price sensitivity	Agent-mediated demand	Stronger price thresholds
Participation price sensitivity	0.75	0.50	0.90	0.70
Mean choice price sensitivity	7.0	4.0	10.0	6.0
Choice-sensitivity CV	0.35	0.40	0.25	0.30
Rank-salience scale	1.00	0.60	1.35	0.80
Relative-price threshold	1.08	1.10	1.06	1.05
Price-threshold penalty strength	0.65	0.35	0.95	1.70
Rank threshold	4	5	3	3
Rank-threshold penalty strength	0.10	0.05	0.16	0.34
Latent taste-shock SD	0.35	0.50	0.22	0.30

All demand scenarios retain the calibration targets of 78% market participation and 18% reference insurer conversion. They examine changes in the **shape and strength** of customer price response, rather than changes in the overall starting conversion target. The term *agent-mediated demand* identifies a specified high-price-sensitivity scenario; it does not represent evidence of actual autonomous-agent purchasing behaviour.

Table B.8: Alternative market and pricing assumptions

Scenario	Changes from baseline
High competition	Competitor log-price stances shift **−0.025**; quote-return probabilities increase by **0.04** (subject to probability bounds); log-quote noise SD **0.050**; competitor segment adjustments ×**1.10**.
Low competition	Competitor stances shift **+0.025** and are scaled by **0.90**; quote-return probabilities fall by **0.16**; noise SD **0.035**; segment adjustments ×**0.85**.
Fragmented market	Competitor stance scale **1.55**; availability shift **−0.02**; noise SD **0.080**; segment adjustments ×**1.50**.
Narrow pricing support	Permitted multiplier grid **0.97**–**1.03** instead of 0.95–1.05, in 0.01 increments.
Strict adequacy	Expected claims-ratio ceiling **78%** rather than 80%.
Wide pricing stress	Permitted multiplier grid **0.90**–**1.10**, in **0.025** increments; permits extrapolation beyond the reference-pricing range used to generate baseline quote observations.

The baseline targeted optimiser uses premium multipliers from **0.95 to 1.05** in steps of **0.01**, and imposes the following constraints: aggregate quoted rate level from **99.5% to 100.5%**; expected profit retention across all customers $\geq$ 100%; expected profit retention within each customer group $\geq$ 90%; expected claims ratio $\leq$ 80%; and model-expected conversion retention $\geq$ 100%. At least **50** validation profiles are required per retained segment. Continuous predictors are generally segmented using validation-derived quartiles. If candidate structures differ in validation conversion by no more than **0.0005** in absolute probability (0.05 percentage points), the structure with fewer segments is preferred. Uniform pricing instead maximises model-expected pre-expense underwriting profit, subject to at least **95%** benchmark conversion and its specified financial constraints; it is not subject to the narrow rate-level restriction.

C Supplementary Results and Model Specifications

This appendix provides the complete comparison of candidate segmentation variables, the selected targeting rule and detailed robustness results that support Section 4. Results labelled **validation** were calculated during model or strategy selection. Results labelled **test** were calculated after the relevant decisions were fixed. Where specified, test pricing outcomes use the underlying simulated conversion probabilities rather than fitted predictions.

C.1 Complete Comparison of Targeting Variables

Table C.1: Validation comparison of 18 predefined eligible targeting bases

Targeting basis	Segments	Relative conversion uplift	Aggregate quoted rate level	Expected profit retention	Minimum group retention
Vehicle-value band	4	22.58%	99.54%	102.21%	90.12%
Vehicle-age band	5	18.85%	99.53%	102.34%	90.22%
Vehicle segment	6	16.44%	99.50%	102.65%	90.07%
Unladen-weight band	4	11.94%	99.58%	102.82%	90.18%
Power band	4	11.70%	99.57%	102.15%	90.01%
Torque band	4	11.59%	99.52%	103.20%	90.18%
Annual-kilometres band	4	11.48%	99.63%	102.09%	90.28%
Claim-free-years band	4	10.78%	99.54%	103.75%	90.11%
Prior-claims band	4	10.43%	99.51%	104.07%	90.23%
Engine-capacity band	4	9.93%	99.50%	103.02%	90.17%
Motive power	5	9.77%	99.52%	104.36%	90.39%
Driver age band	8	9.29%	99.50%	102.54%	90.25%
Remoteness area	4	7.70%	99.50%	104.44%	90.69%
State	8	7.46%	99.50%	102.76%	90.31%
Voluntary excess	5	7.37%	99.51%	104.11%	90.17%
Licence-years band	4	7.28%	99.55%	103.89%	90.67%
Cylinder count	5	6.51%	99.50%	103.77%	100.00%
Transmission	2	6.11%	99.65%	102.40%	100.00%

C.2 Selected Vehicle-Value Segments and Multipliers

Table C.2: Baseline validation-selected targeting rule

Vehicle-value group	Validation profiles	Selected multiplier	Model-expected conversion	Segment expected profit retention
Q1 (lowest)	2,504	0.96	30.31%	127.43%
Q2	2,534	0.96	28.97%	127.54%
Q3	2,468	1.01	15.30%	90.16%
Q4 (highest)	2,494	1.01	14.05%	90.12%

The four groups contain all 10,000 validation profiles. Vehicle-value cut-offs were derived on validation and remained fixed when assigning test profiles. Segment expected profit retention depends jointly on the change in premium and the resulting change in predicted purchases. The Q1 and Q2 discounts can increase expected profit where they generate sufficient additional business, whereas a premium increase in the higher-value groups can reduce expected profit if purchasing probabilities decline sufficiently. These effects arise under the simulated demand mechanism, not from empirical evidence of causal differences in price sensitivity by vehicle value.

C.3 Structural Sensitivity Results

Table C.3: Complete ten-case targeted-pricing sensitivity results

Scenario	Selected model	Targeting basis	DGP-relative conversion uplift	DGP expected profit retention	Combined pricing criterion
Baseline	GLM	Vehicle-value band	23.82%	103.27%	Yes
High competition	GLM	Vehicle-value band	19.58%	105.61%	Yes
Low competition	GLM	Vehicle-value band	23.07%	103.49%	Yes
Fragmented market	GAM	Vehicle-value band	17.44%	104.86%	Yes
Low price sensitivity	GLM	Vehicle-value band	6.51%	100.32%	Yes
Agent-mediated demand	GLM	Vehicle-value band	28.75%	109.47%	Yes
Stronger price thresholds	GAM	Vehicle-value band	32.15%	110.67%	Yes
Narrow pricing support	GLM	Vehicle-value band	19.11%	103.79%	Yes
Strict adequacy	GLM	Vehicle-value band	23.82%	103.27%	Yes
Wide pricing stress	GLM	Vehicle-value band	8.96%	102.53%	Yes

The combined criterion requires an increase in **underlying expected reference-insurer conversion** and expected profit retention of at least **100%**, using withheld simulated conversion probabilities and fitted expected claims costs. The Market-position models improved predictive performance relative to the Own-price GLM in all ten scenarios. All ten cases also satisfied the combined pricing criterion, although the low-price-sensitivity case generated only a small profit improvement.

C.4 Monte Carlo Replication Detail

Ten independent runs regenerated Modules 01–06 while retaining Module 00 calibration inputs and baseline structural settings. Model selection and targeted pricing were repeated within each run.

Table C.4: Predictive stability across ten replications

Replication	Selected market-position model	Test log-loss gain over Own-price GLM	DGP Spearman correlation gain
1	GLM	2.50%	0.171
2	GLM	3.02%	0.176
3	GLM	2.54%	0.176
4	GLM	3.13%	0.183
5	GLM	2.80%	0.182

Replication	Selected market-position model	Test log-loss gain over Own-price GLM	DGP Spearman correlation gain
6	GLM	2.70%	0.178
7	GLM	3.03%	0.178
8	GLM	2.29%	0.169
9	GLM	2.89%	0.183
10	GLM	2.44%	0.175

Table C.5: Independent-test pricing stability across ten replications

Replication	DGP-relative conversion uplift	Expected profit retention	Expected claims ratio	Aggregate quoted rate level
1	22.97%	102.67%	77.91%	99.58%
2	24.07%	103.15%	77.95%	99.56%
3	23.52%	102.86%	77.91%	99.58%
4	24.77%	103.11%	78.01%	99.51%
5	15.12%	103.90%	77.54%	99.52%
6	25.00%	103.68%	78.01%	99.47%
7	24.06%	102.96%	77.93%	99.56%
8	22.40%	102.67%	77.88%	99.61%
9	23.38%	103.07%	77.96%	99.52%
10	23.97%	102.90%	77.96%	99.54%

Vehicle-value band and the Market-position GLM were selected in all ten replications. Each produced positive underlying expected conversion uplift and expected profit retention above 100%. The test aggregate quoted rate level fell below the 99.5% lower limit in one replication (replication 6). The constraint was imposed during **validation-based selection**, whereas these quoted rate levels were calculated on independently generated test samples. They therefore do not establish a violation of the optimisation constraint.

C.5 Paired Test Bootstrap

The paired-bootstrap procedure uses 5,000 resamples of the independent test sample, with 10,000 customer profiles drawn with replacement in each resample. Each selected customer contributes the benchmark and targeted-pricing expected outcomes together, preserving the within-customer comparison. The fitted models, pricing strategy and claims-cost predictions remain fixed. Percentile intervals describe uncertainty from test-sample composition, conditional on the selected strategy and specified customer-choice mechanism; they do not capture uncertainty in the DGP or alternative model selections.

The 5,000 resamples generated the following percentile intervals for the fixed independent-test comparison. Values were calculated using the underlying simulated conversion probabilities and fitted expected claims costs.

Table C.6: Paired bootstrap estimates and 95% percentile intervals from the 10,000-profile independent test sample

Measure	Point estimate	95% bootstrap percentile interval
Absolute conversion uplift	4.24 percentage points	4.10–4.38 percentage points
Relative conversion uplift	23.82%	23.03%–24.62%
Expected profit retention	103.27%	102.81%–103.72%
Targeted expected claims ratio	77.94%	77.90%–77.99%
Aggregate quoted rate level	99.53%	99.48%–99.57%

All 5,000 resamples produced positive conversion uplift and retained at least 100% of benchmark expected pre-expense underwriting profit. The rate-level interval overlaps values below the 99.5% lower validation constraint, even though the baseline test estimate satisfies that limit. These are conditional resampling frequencies and intervals, not probabilities of real-world pricing success.

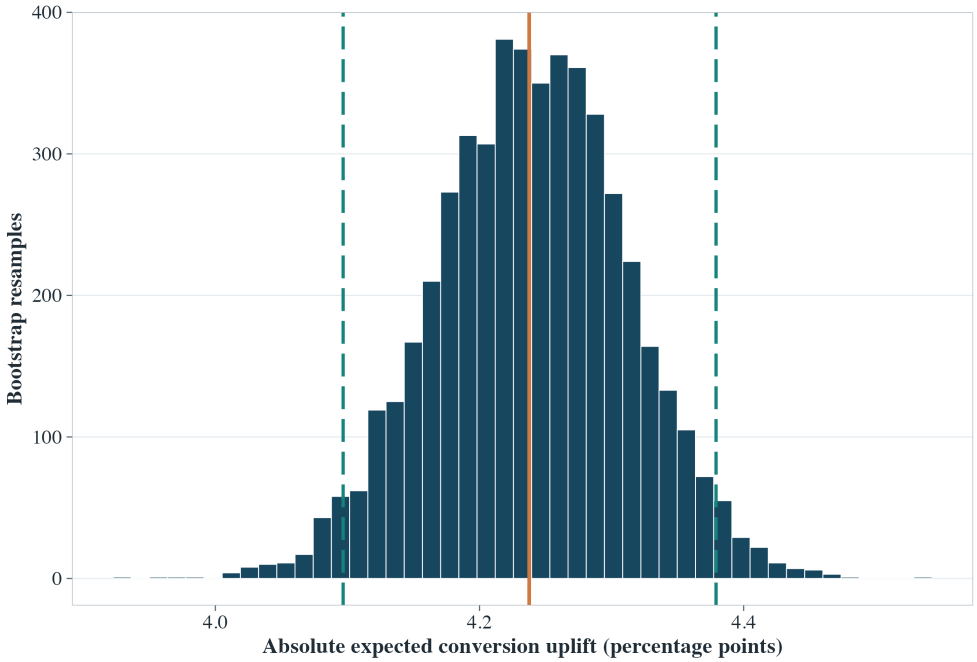


Figure C.1: Distribution of absolute expected conversion uplift across 5,000 paired bootstrap resamples of the independent 10,000-customer test sample. The vertical solid line denotes the original test estimate and dashed lines denote the 95% percentile interval. Pricing decisions and the underlying simulated demand mechanism are fixed during resampling.

C.6 Model-Specification Register

The following registers summarise the implemented candidate models and selection rules from Modules 02 and 05. They supplement the descriptions in Sections 3.3 and 3.5 and are intended to make the predictor sets and model comparisons explicit.

Table C.7: Technical-pricing candidate specifications and selection

Candidate	Frequency (quasi-Poisson, log link)	Severity (Gamma, log link)
Benchmark	Linear age, licence years, log kilometres; prior claims, State, remoteness	Log vehicle value, excess, vehicle segment
Flexible	Natural splines: age (df 4), licence years (df 3), log kilometres (df 3); prior claims, State, remoteness, segment	Natural splines: log vehicle value (df 4), vehicle age (df 3); excess, vehicle segment
Flexible + motive power	Frequency flexible + motive power	Severity flexible + motive power
Flexible + physical specifications	Frequency flexible + motive power, transmission and physical vehicle specifications	Severity flexible + motive power, transmission and physical vehicle specifications
Selected on 2024	**Flexible + physical specifications**	**Benchmark**

Additional physical-vehicle candidates include engine capacity, power, torque and unladen weight (each with a natural spline of three degrees of freedom), together with cylinder count and transmission. Frequency estimation uses all policies and includes the exposure offset $\log(E_i)$. Severity estimation is restricted to policies with at least one paid claim, weighted by the number of such claims. Model specifications are selected using 2024 validation deviance, refitted on 2021–2024 and adjusted by development-period balancing factors before 2025 evaluation.

Table C.8: Conversion-model information sets and flexibility

Candidate	Distinguishing variables relative to preceding information set
Technical-only GLM	Driver age, licence years, prior-claims band, claim-free years, State, remoteness, vehicle segment, motive power, vehicle age, log vehicle value, log annual kilometres, voluntary excess and log technical premium
Own-price GLM	Technical-only + $\log(P_i/\pi_i)$
Relative-price GLM	Own-price + log reference/cheapest-competitor relative price
Rank GLM	Own-price + reference rank percentile (**rather than also inheriting Relative-price GLM**)
Market-position GLM	Own-price + log relative price to cheapest, rank percentile, excess over market-price threshold, log price relative to market median, log competitor-price spread and centred number of competitor quotes
Market-position GAM	Same broad explanatory set as Market-position GLM, with selected continuous terms represented by cubic regression smooths

The Rank and Relative-price GLMs are **parallel extensions** of the Own-price GLM, rather than stages of one sequentially nested model. In the GLMs, natural cubic splines are used for driver age (4 degrees of freedom), licence experience (3), consecutive claim-free years (3) and vehicle age (3); the remaining transformed continuous terms are entered linearly. The GAM replaces selected linear or spline effects with cubic regression smooths: driver age ($k = 5$), licence experience ($k = 4$), claim-free years ($k = 4$), vehicle age ($k = 4$), log vehicle value ($k = 5$), log annual kilometres ($k = 5$), log technical premium ($k = 5$), log reference-to-technical premium ($k = 5$), log relative price to the cheapest competitor ($k = 6$), rank percentile ($k = 5$), log relative price to market median ($k = 5$) and log competitor-price spread ($k = 4$). Categorical and threshold effects remain parametric. The GAM is selected over the Market-position GLM only when validation log loss improves by at least **0.5%** and the calibration slope is between **0.70 and 1.30**.

C.7 Interpretation of Selected Technical-Pricing Relativities

The selected frequency and severity GLMs include nonlinear terms, so their fitted effects can be illustrated through predicted risk relativities. Figure C.2 varies one rating characteristic at a time over the central range of the development sample while holding the other model inputs fixed at a representative policy profile.

Each prediction is divided by the prediction for the reference profile. A relativity of 1.00 therefore represents the reference profile's fitted expected claim outcome, while a value of 1.20 indicates a fitted outcome 20% higher, with the remaining characteristics held constant.

These curves describe the relationships estimated by the fitted models rather than the underlying simulation parameters. However, they should not be interpreted as causal effects. For example, as driver age and licence experience are highly correlated, varying one while holding the other fixed does not reproduce their combined effect; the older-driver increase specified in the simulation is therefore not visible in this panel.

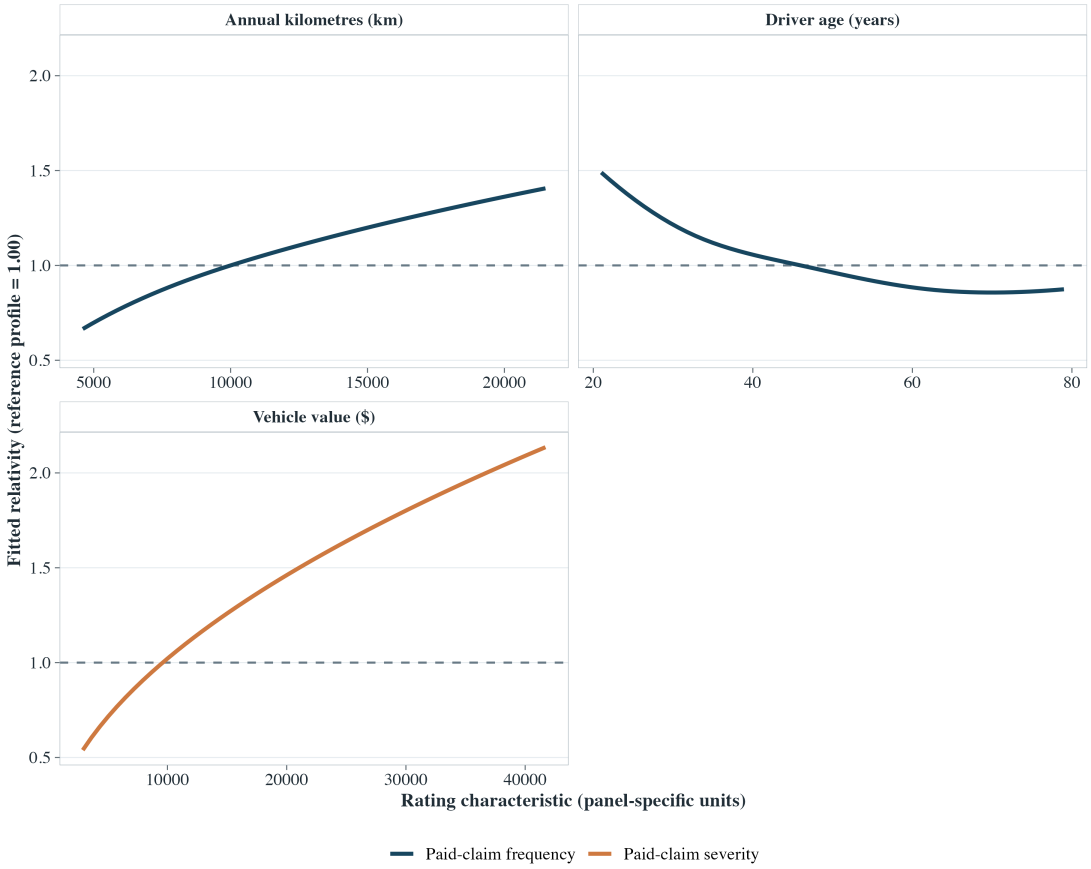


Figure C.2: Fitted frequency and severity risk relativities from the technical-pricing GLMs. Each panel varies one rating characteristic across its central development-sample range while holding other predictors at the representative reference profile. Relativity 1.00 denotes that profile's predicted outcome; lines describe the fitted model responses, not the synthetic generating parameters or Australian empirical risk factors.

D Computational Implementation and Reproducibility

D.1 Module Structure

The experiment is implemented as a sequence of R Markdown modules. Each stage reads the required upstream inputs, applies the specified procedures and writes intermediate datasets, diagnostic summaries and validation checks. This structure separates data generation, model estimation, pricing selection and evaluation.

Table D.1: Computational modules

Module	Main task
00	Prepare and validate Australian empirical calibration inputs
01	Generate the five-cohort synthetic motor portfolio, claim records and withheld latent claims truth
02	Fit/select technical frequency and severity models; estimate 2025 fitted technical premiums
03	Generate reference and competitor quotations, quote availability, competitive-position variables and independent conversion split
04	Generate market participation, insurer choice, observed conversion and withheld DGP demand probabilities
05	Fit/select logistic GLMs and GAM; evaluate validation and test predictions

Module	Main task
06	Select constrained premium adjustments on validation and assess the decision on the test sample against the DGP
06B	Paired bootstrap of the selected test pricing comparison, holding the decision fixed
07	Structural sensitivity analysis, changing one related scenario group at a time
08	Ten full-pipeline Monte Carlo seed replications
09	Dissertation figures and supplementary exhibits

Module 00 prepares the inputs required by Module 01. The baseline experiment then proceeds sequentially through Modules 01–06. Bootstrap, structural-sensitivity and Monte Carlo calculations are conducted after the required baseline outputs are available; Module 09 assembles the dissertation exhibits. Module 06B implements the paired-bootstrap analysis and is provided with the accompanying computational materials.

D.2 Information Controls

The analysis separates the underlying simulation relationships from information used for estimation and selection. The general sequence is: generate claims and demand outcomes, restrict access to the underlying probabilities, fit and validate models, select the pricing decision, hold that decision fixed, and evaluate it using the underlying simulated customer-choice mechanism. The principal information controls are:

1. Technical models are trained on 2021–2023, selected on 2024 and refitted on 2021–2024. The 2025 cohort remains unused during technical model development and selection.
2. A separate 30,000/10,000/10,000 train/validation/test partition is assigned to the 2025 quote profiles before conversion outcomes are generated.
3. Conversion-model selection, probability recalibration and targeted pricing are undertaken without using test conversion outcomes or DGP-expected conversion probabilities for strategy selection.
4. Market-position predictors are reconstructed for each counterfactual reference premium, while competitor quotations remain fixed.
5. The selected targeting variable, segment cut-points and premium multipliers are frozen before independent test evaluation using known DGP conversion probabilities.
6. Validation checks assess required output columns, calibration/consistency conditions, model-prediction reconstruction and pricing constraints.

The technical-pricing cohort split and the conversion-model split serve different purposes. The aggregate quoted rate level may differ between validation and test because the two samples contain different policy characteristics and vehicle-value distributions; a test level outside the validation bounds is not itself evidence that the optimisation constraint failed during selection.

D.3 Code, Inputs and Reproducibility Requirements

The R Markdown modules are available in a dedicated GitHub repository: **Competitive Motor Insurance Pricing with Synthetic Quote Markets**. The repository contains the computational framework for synthetic portfolio generation, technical pricing, competitive quote-market simulation, conversion modelling, pricing optimisation and robustness analysis. It also documents the supporting calibration inputs and dependencies required to reproduce the analysis.

Reproducing the analysis requires the final R Markdown modules, their supporting functions in `R/helpers.R`, and the calibration datasets used by Module 00. The saved outputs and fixed random-number seeds document the executed analyses, while the supplied execution order and session information record the software environment needed to reproduce the reported results. The source files should be run in the dependency order summarised in Table D.1.